

Induction and Recursion

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Many mathematical statements assert that a property is true for all positive integers. Examples of such statements are that for every positive integer n : $n! \leq n^n$, $n^3 - n$ is divisible by 3; a set with n elements has 2^n subsets; and the sum of the first n positive integers is $n(n+1)/2$. A major goal of this chapter, and the book, is to provide a thorough understanding of mathematical induction, which is used to prove results of this kind.

Proofs using mathematical induction have two parts. First, they show that the statement holds for the positive integer 1. Second, they show that if the statement holds for a positive integer then it must also hold for the next larger integer. Mathematical induction is based on the rule of inference that tells us that if $P(1)$ and $\forall k(P(k) \rightarrow P(k+1))$ are true for the domain of positive integers, then $\forall n P(n)$ is true. Mathematical induction can be used to prove a tremendous variety of results. Understanding how to read and construct proofs by mathematical induction is a key goal of learning discrete mathematics.

In Chapter 2 we explicitly defined sets and functions. That is, we described sets by listing their elements or by giving some property that characterizes these elements. We gave formulae for the values of functions. There is another important way to define such objects, based on mathematical induction. To define functions, some initial terms are specified, and a rule is given for finding subsequent values from values already known. (We briefly touched on this sort of definition in Chapter 2 when we showed how sequences can be defined using recurrence relations.) Sets can be defined by listing some of their elements and giving rules for constructing elements from those already known to be in the set. Such definitions, called *recursive definitions*, are used throughout discrete mathematics and computer science. Once we have defined a set recursively, we can use a proof method called structural induction to prove results about this set.

When a procedure is specified for solving a problem, this procedure must *always* solve the problem correctly. Just testing to see that the correct result is obtained for a set of input values does not show that the procedure always works correctly. The correctness of a procedure can be guaranteed only by proving that it always yields the correct result. The final section of this chapter contains an introduction to the techniques of program verification. This is a formal technique to verify that procedures are correct. Program verification serves as the basis for attempts under way to prove in a mechanical fashion that programs are correct.

5.1 Mathematical Induction

5.1.1 Introduction

Suppose that we have an infinite ladder, as shown in Figure 1, and we want to know whether we can reach every step on this ladder. We know two things:

1. We can reach the first rung of the ladder.
2. If we can reach a particular rung of the ladder, then we can reach the next rung.

Can we conclude that we can reach every rung? By (1), we know that we can reach the first rung of the ladder. Moreover, because we can reach the first rung, by (2), we can also reach the second rung; it is the next rung after the first rung. Applying (2) again, because we can reach the second rung, we can also reach the third rung. Continuing in this way, we can show that we can

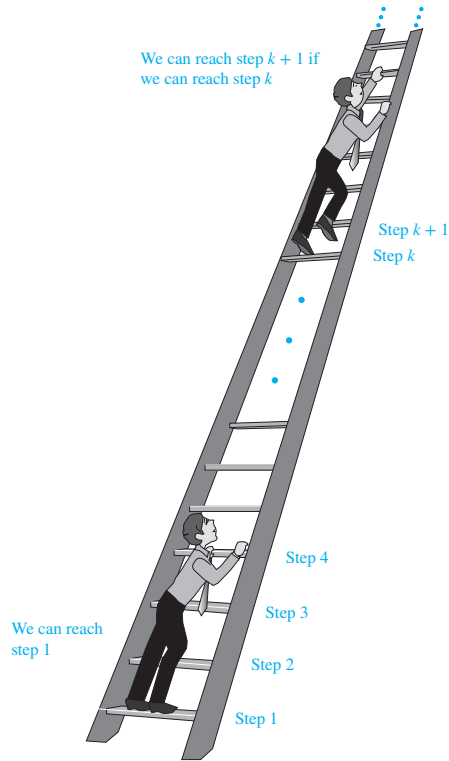


FIGURE 1 Climbing an infinite ladder.

reach the fourth rung, the fifth rung, and so on. For example, after 100 uses of (2), we know that we can reach the 101st rung. But can we conclude that we are able to reach every rung of this infinite ladder? The answer is yes, something we can verify using an important proof technique called **mathematical induction**. That is, we can show that the statement that we can reach the n th rung of the ladder is true for all positive integers n .

Mathematical induction is an extremely important proof technique that can be used to prove assertions of this type. As we will see in this section and in subsequent sections of this chapter and later chapters, mathematical induction is used extensively to prove results about a large variety of discrete objects. For example, it is used to prove results about the complexity of algorithms, the correctness of certain types of computer programs, theorems about graphs and trees, as well as a wide range of identities and inequalities.

In this section, we will describe how mathematical induction can be used and why it is a valid proof technique. It is extremely important to note that mathematical induction can be used only to prove results obtained in some other way. It is *not* a tool for discovering formulae or theorems.

5.1.2 Mathematical Induction

Assessment

In general, mathematical induction* can be used to prove statements that assert that $P(n)$ is true for all positive integers n , where $P(n)$ is a propositional function. A proof by mathematical

*Unfortunately, using the terminology “mathematical induction” clashes with the terminology used to describe different types of reasoning. In logic, **deductive reasoning** uses rules of inference to draw conclusions from premises, whereas **inductive reasoning** makes conclusions only supported, but not ensured, by evidence. Mathematical proofs, including arguments that use mathematical induction, are deductive, not inductive.

induction has two parts, a **basis step**, where we show that $P(1)$ is true, and an **inductive step**, where we show that for all positive integers k , if $P(k)$ is true, then $P(k + 1)$ is true.

PRINCIPLE OF MATHEMATICAL INDUCTION To prove that $P(n)$ is true for all positive integers n , where $P(n)$ is a propositional function, we complete two steps:

BASIS STEP: We verify that $P(1)$ is true.

INDUCTIVE STEP: We show that the conditional statement $P(k) \rightarrow P(k + 1)$ is true for all positive integers k .

To complete the inductive step of a proof using the principle of mathematical induction, we assume that $P(k)$ is true for an arbitrary positive integer k and show that under this assumption, $P(k + 1)$ must also be true. The assumption that $P(k)$ is true is called the **inductive hypothesis**. Once we complete both steps in a proof by mathematical induction, we have shown that $P(n)$ is true for all positive integers n , that is, we have shown that $\forall n P(n)$ is true where the quantification is over the set of positive integers. In the inductive step, we show that $\forall k (P(k) \rightarrow P(k + 1))$ is true, where again, the domain is the set of positive integers.

Expressed as a rule of inference, this proof technique can be stated as

$$(P(1) \wedge \forall k (P(k) \rightarrow P(k + 1))) \rightarrow \forall n P(n),$$

when the domain is the set of positive integers. Because mathematical induction is such an important technique, it is worthwhile to explain in detail the steps of a proof using this technique. The first thing we do to prove that $P(n)$ is true for all positive integers n is to show that $P(1)$ is true. This amounts to showing that the particular statement obtained when n is replaced by 1 in $P(n)$ is true. Then we must show that $P(k) \rightarrow P(k + 1)$ is true for every positive integer k . To prove that this conditional statement is true for every positive integer k , we need to show that $P(k + 1)$ cannot be false when $P(k)$ is true. This can be accomplished by assuming that $P(k)$ is true and showing that *under this hypothesis* $P(k + 1)$ must also be true.

Remark: In a proof by mathematical induction it is *not* assumed that $P(k)$ is true for all positive integers! It is only shown that *if it is assumed* that $P(k)$ is true, then $P(k + 1)$ is also true. Thus, a proof by mathematical induction is not a case of begging the question, or circular reasoning.

After completing the basis and inductive steps of a proof that $P(n)$ is true for all positive integers n , we know that $P(1)$ is true. That is what is shown in the basis step. We can conclude that $P(2)$ is true, because we know that $P(1)$ is true and from the inductive step we know that $P(1) \rightarrow P(2)$. Furthermore, we know that $P(3)$ is true because $P(2)$ is true and we know that $P(2) \rightarrow P(3)$ from the inductive step. Continuing along these lines using a finite number of implications, we can show that $P(n)$ is true for any particular positive integer n .

Links

HISTORICAL NOTE The first known use of mathematical induction is in the work of the sixteenth-century mathematician Francesco Maurolico (1494 – 1575). Maurolico wrote extensively on the works of classical mathematics and made many contributions to geometry and optics. In his book *Arithmeticonum Libri Duo*, Maurolico presented a variety of properties of the integers together with proofs of these properties. To prove some of these properties, he devised the method of mathematical induction. His first use of mathematical induction in this book was to prove that the sum of the first n odd positive integers equals n^2 . Augustus De Morgan is credited with the first presentation in 1838 of formal proofs using mathematical induction, as well as introducing the terminology “mathematical induction.” Maurolico’s proofs were informal and he never used the word “induction.” See [Gu10] to learn more about the history of the method of mathematical induction.

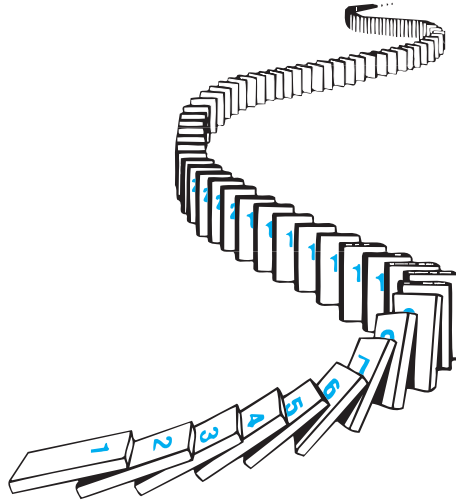


FIGURE 2 Illustrating how mathematical induction works using dominoes.

WAYS TO REMEMBER HOW MATHEMATICAL INDUCTION WORKS Thinking of the infinite ladder and the rules for reaching steps can help you remember how mathematical induction works. Note that statements (1) and (2) for the infinite ladder are exactly the basis step and inductive step, respectively, of the proof that $P(n)$ is true for all positive integers n , where $P(n)$ is the statement that we can reach the n th rung of the ladder. Consequently, we can invoke mathematical induction to conclude that we can reach every rung.

Another way to illustrate the principle of mathematical induction is to consider an infinite row of dominoes, labeled $1, 2, 3, \dots, n, \dots$, where each domino is standing up. Let $P(n)$ be the proposition that domino n is knocked over. If the first domino is knocked over—that is, if $P(1)$ is true—and if, whenever the k th domino is knocked over, it also knocks the $(k + 1)$ st domino over—that is, if $P(k) \rightarrow P(k + 1)$ is true for all positive integers k —then all the dominoes are knocked over. This is illustrated in Figure 2.

5.1.3 Why Mathematical Induction is Valid

Why is mathematical induction a valid proof technique? The reason comes from the well-ordering property, listed in Appendix 1 as an axiom for the set of positive integers, which states that every nonempty subset of the set of positive integers has a least element. So, suppose we know that $P(1)$ is true and that the proposition $P(k) \rightarrow P(k + 1)$ is true for all positive integers k . To show that $P(n)$ must be true for all positive integers n , assume that there is at least one positive integer n for which $P(n)$ is false. Then the set S of positive integers n for which $P(n)$ is false is nonempty. Thus, by the well-ordering property, S has a least element, which will be denoted by m . We know that m cannot be 1, because $P(1)$ is true. Because m is positive and greater than 1, $m - 1$ is a positive integer. Furthermore, because $m - 1$ is less than m , it is not in S , so $P(m - 1)$ must be true. Because the conditional statement $P(m - 1) \rightarrow P(m)$ is also true, it must be the case that $P(m)$ is true. This contradicts the choice of m . Hence, $P(n)$ must be true for every positive integer n .

Remark: In this book we take the well-ordering property for the positive integers as an axiom. We proved that mathematical induction is a valid proof technique. Instead, we could have taken the principle of mathematical induction as an axiom and proved that the positive integers are well ordered. That is, the well-ordering property for positive integers and the principle of mathematical induction are equivalent. (In Section 5.2 we will present examples of proofs that use the well-ordering

property directly. Also, Exercise 41 in that section asks for a proof that the well-ordering property for positive integers is a consequence of the principle of mathematical induction.)

5.1.4 Choosing the Correct Basis Step

Mathematical induction can be used to prove theorems other than those of the form “ $P(n)$ is true for all positive integers n .” Often, we will need to show that $P(n)$ is true for $n = b, b + 1, b + 2, \dots$, where b is an integer other than 1. We can use mathematical induction to accomplish this, as long as we change the basis step by replacing $P(1)$ with $P(b)$. In other words, to use mathematical induction to show that $P(n)$ is true for $n = b, b + 1, b + 2, \dots$, where b is an integer other than 1, we show that $P(b)$ is true in the basis step. In the inductive step, we show that the conditional statement $P(k) \rightarrow P(k + 1)$ is true for $k = b, b + 1, b + 2, \dots$. Note that b can be negative, zero, or positive. Following the domino analogy we used earlier, imagine that we begin by knocking down the b th domino (the basis step), and as each domino falls, it knocks down the next domino (the inductive step). We leave it to the reader to show that this form of induction is valid (see Exercise 85).

We will illustrate this notion in Example 3, which states that a summation formula is valid for all nonnegative integers. In this example, we need to prove that $P(n)$ is true for $n = 0, 1, 2, \dots$. So, the basis step in Example 3 will show that $P(0)$ is true.

5.1.5 Guidelines for Proofs by Mathematical Induction

Examples 1–14 will illustrate how to use mathematical induction to prove a diverse collection of theorems. Each of these examples includes all the elements needed in a proof by mathematical induction. We will also present an example of an invalid proof by mathematical induction. Before we give these proofs, we will provide some useful guidelines for constructing correct proofs by mathematical induction.

Template for Proofs by Mathematical Induction

1. Express the statement that is to be proved in the form “for all $n \geq b$, $P(n)$ ” for a fixed integer b . For statements of the form “ $P(n)$ for all positive integers n ,” let $b = 1$, and for statements of the form “ $P(n)$ for all nonnegative integers n ,” let $b = 0$. For some statements of the form $P(n)$, such as inequalities, you may need to determine the appropriate value of b by checking the truth values of $P(n)$ for small values of n , as is done in Example 6.
2. Write out the words “Basis Step.” Then show that $P(b)$ is true, taking care that the correct value of b is used. This completes the first part of the proof.
3. Write out the words “Inductive Step” and state, and clearly identify, the inductive hypothesis, in the form “Assume that $P(k)$ is true for an arbitrary fixed integer $k \geq b$.”
4. State what needs to be proved under the assumption that the inductive hypothesis is true. That is, write out what $P(k + 1)$ says.
5. Prove the statement $P(k + 1)$ making use of the assumption $P(k)$. (Generally, this is the most difficult part of a mathematical induction proof. Decide on the most promising proof strategy and look ahead to see how to use the induction hypothesis to build your proof of the inductive step. Also, be sure that your proof is valid for all integers k with $k \geq b$, taking care that the proof works for small values of k , including $k = b$.)
6. Clearly identify the conclusion of the inductive step, such as by saying “This completes the inductive step.”
7. After completing the basis step and the inductive step, state the conclusion, namely, “By mathematical induction, $P(n)$ is true for all integers n with $n \geq b$ ”.

Readers will find it worthwhile to see how the steps described in the template are completed in each of the 14 examples. It will also be useful to follow these guidelines in the solutions of the exercises that ask for proofs by mathematical induction. The guidelines that we presented can be adapted for each of the variants of mathematical induction that we introduce in the exercises and later in this chapter.

5.1.6 The Good and the Bad of Mathematical Induction

An important point needs to be made about mathematical induction before we commence a study of its use. The good thing about mathematical induction is that it can be used to prove a conjecture once it has been made (and is true). The bad thing about it is that it cannot be used to find new theorems. Mathematicians sometimes find proofs by mathematical induction unsatisfying because they do not provide insights as to why theorems are true. Many theorems can be proved in many ways, including by mathematical induction. Proofs of these theorems by methods other than mathematical induction are often preferred because of the insights they bring. (See Example 8 and the subsequent remark for an example of this.)

You can prove a theorem by mathematical induction even if you do not have the slightest idea why it is true!

5.1.7 Examples of Proofs by Mathematical Induction

Many theorems assert that $P(n)$ is true for all positive integers n , where $P(n)$ is a propositional function. Mathematical induction is a technique for proving theorems of this kind. In other words, mathematical induction can be used to prove statements of the form $\forall n P(n)$, where the domain is the set of positive integers. Mathematical induction can be used to prove an extremely wide variety of theorems, each of which is a statement of this form. (Remember, many mathematical assertions include an implicit universal quantifier. The statement “if n is a positive integer, then $n^3 - n$ is divisible by 3” is an example of this. Making the implicit universal quantifier explicit yields the statement “for every positive integer n , $n^3 - n$ is divisible by 3.”)

Links ➤

We will use a variety of examples to illustrate how theorems are proved using mathematical induction. The theorems we will prove include summation formulae, inequalities, identities for combinations of sets, divisibility results, theorems about algorithms, and some other creative results. In this section, and in later sections, we will employ mathematical induction to prove many other types of results, including the correctness of computer programs and algorithms. Mathematical induction can be used to prove a wide variety of theorems both similar to, and also quite different from, the examples here. (For proofs by mathematical induction of many more interesting and diverse results, see the *Handbook of Mathematical Induction* by David Gunderson [Gu11].)

There are many opportunities for errors in induction proofs. We will describe some incorrect proofs by mathematical induction at the end of this section and in the exercises. To avoid making errors in proofs by mathematical induction, try to follow the guidelines for such proofs provided previously in Section 5.1.5.

SEEING WHERE THE INDUCTIVE HYPOTHESIS IS USED To help the reader understand each of the mathematical induction proofs in this section, we will note where the inductive hypothesis is used. We indicate this use in three different ways: by explicit mention in the text, by inserting the acronym IH (for inductive hypothesis) over an equals sign or a sign for an inequality, or by specifying the inductive hypothesis as the reason for a step in a multi-line display.

Look for the ^{IH} = symbol to see where the inductive hypothesis is used.

PROVING SUMMATION FORMULAE We begin by using mathematical induction to prove several summation formulae. As we will see, mathematical induction is particularly well suited for proving that such formulae are valid. However, summation formulae can be proven in other ways. This is not surprising because there are often different ways to prove a theorem. The major

disadvantage of using mathematical induction to prove a summation formula is that you cannot use it to derive this formula. That is, you must already have the formula before you attempt to prove it by mathematical induction.

Examples 1–4 illustrate how to use mathematical induction to prove summation formulae. The first summation formula we will prove by mathematical induction, in Example 1, is a closed formula for the sum of the smallest n positive integers.

EXAMPLE 1 Show that if n is a positive integer, then

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

**Extra
Examples** 

Solution: Let $P(n)$ be the proposition that the sum of the first n positive integers, $1 + 2 + \cdots + n = \frac{n(n+1)}{2}$, is $n(n+1)/2$. We must do two things to prove that $P(n)$ is true for $n = 1, 2, 3, \dots$. Namely, we must show that $P(1)$ is true and that the conditional statement $P(k)$ implies $P(k+1)$ is true for $k = 1, 2, 3, \dots$.

BASIS STEP: $P(1)$ is true, because $1 = \frac{1(1+1)}{2}$. (The left-hand side of this equation is 1 because 1 is the sum of the first positive integer. The right-hand side is found by substituting 1 for n in $n(n+1)/2$.)

INDUCTIVE STEP: For the inductive hypothesis we assume that $P(k)$ holds for an arbitrary positive integer k . That is, we assume that

$$1 + 2 + \cdots + k = \frac{k(k+1)}{2}.$$

Under this assumption, it must be shown that $P(k+1)$ is true, namely, that

$$1 + 2 + \cdots + k + (k+1) = \frac{(k+1)[(k+1)+1]}{2} = \frac{(k+1)(k+2)}{2}$$

is also true.

We now look ahead to see how we might be able to prove that $P(k+1)$ holds under the assumption that $P(k)$ is true. We observe that the summation in the left-hand side of $P(k+1)$ is $k+1$ more than the summation in the left-hand side of $P(k)$. Our strategy will be to add $k+1$ to both sides of the equation in $P(k)$ and simplify the result algebraically to complete the inductive step.

We now return to the proof of the inductive step. When we add $k+1$ to both sides of the equation in $P(k)$, we obtain

$$\begin{aligned} 1 + 2 + \cdots + k + (k+1) &\stackrel{\text{IH}}{=} \frac{k(k+1)}{2} + (k+1) \\ &= \frac{k(k+1) + 2(k+1)}{2} \\ &= \frac{(k+1)(k+2)}{2}. \end{aligned}$$

This last equation shows that $P(k+1)$ is true under the assumption that $P(k)$ is true. This completes the inductive step.

We have completed the basis step and the inductive step, so by mathematical induction we know that $P(n)$ is true for all positive integers n . That is, we have proven that $1 + 2 + \cdots + n = n(n+1)/2$ for all positive integers n . 

If you are rusty simplifying algebraic expressions, this is the time to do some reviewing!

As we noted, mathematical induction is not a tool for finding theorems about all positive integers. Rather, it is a proof method for proving such results once they are conjectured. In Example 2, using mathematical induction to prove a summation formula, we will both formulate and then prove a conjecture.

EXAMPLE 2 Conjecture a formula for the sum of the first n positive odd integers. Then prove your conjecture using mathematical induction.

Solution: The sums of the first n positive odd integers for $n = 1, 2, 3, 4, 5$ are

$$\begin{array}{lll} 1 = 1, & 1 + 3 = 4, & 1 + 3 + 5 = 9, \\ 1 + 3 + 5 + 7 = 16, & 1 + 3 + 5 + 7 + 9 = 25. & \end{array}$$

From these values it is reasonable to conjecture that the sum of the first n positive odd integers is n^2 , that is, $1 + 3 + 5 + \cdots + (2n - 1) = n^2$. We need a method to *prove* that this *conjecture* is correct, if in fact it is.

Let $P(n)$ denote the proposition that the sum of the first n odd positive integers is n^2 . Our conjecture is that $P(n)$ is true for all positive integers n . To use mathematical induction to prove this conjecture, we must first complete the basis step; that is, we must show that $P(1)$ is true. Then we must carry out the inductive step; that is, we must show that $P(k + 1)$ is true when $P(k)$ is assumed to be true. We now attempt to complete these two steps.

BASIS STEP: $P(1)$ states that the sum of the first one odd positive integer is 1^2 . This is true because the sum of the first odd positive integer is 1. The basis step is complete.

INDUCTIVE STEP: To complete the inductive step we must show that the proposition $P(k) \rightarrow P(k + 1)$ is true for every positive integer k . To do this, we first assume the inductive hypothesis. The inductive hypothesis is the statement that $P(k)$ is true for an arbitrary positive integer k , that is,

$$1 + 3 + 5 + \cdots + (2k - 1) = k^2.$$

(Note that the k th odd positive integer is $(2k - 1)$, because this integer is obtained by adding 2 a total of $k - 1$ times to 1.)

To show that $\forall k(P(k) \rightarrow P(k + 1))$ is true, we must show that if $P(k)$ is true (the inductive hypothesis), then $P(k + 1)$ is true. Note that $P(k + 1)$ is the statement that

$$1 + 3 + 5 + \cdots + (2k - 1) + (2k + 1) = (k + 1)^2.$$

Before we complete the inductive step, we will take a time out to figure out a strategy. At this stage of a mathematical induction proof it is time to look for a way to use the inductive hypothesis to show that $P(k + 1)$ is true. Here we note that $1 + 3 + 5 + \cdots + (2k - 1) + (2k + 1)$ is the sum of its first k terms $1 + 3 + 5 + \cdots + (2k - 1)$ and its last term $2k + 1$. So, we can use our inductive hypothesis to replace $1 + 3 + 5 + \cdots + (2k - 1)$ by k^2 .

We now return to our proof. We find that

$$\begin{aligned} 1 + 3 + 5 + \cdots + (2k - 1) + (2k + 1) &= [1 + 3 + \cdots + (2k - 1)] + (2k + 1) \\ &\stackrel{\text{IH}}{=} k^2 + (2k + 1) \\ &= k^2 + 2k + 1 \\ &= (k + 1)^2. \end{aligned}$$

This shows that $P(k + 1)$ follows from $P(k)$. Note that we used the inductive hypothesis $P(k)$ in the second equality to replace the sum of the first k odd positive integers by k^2 .

We have now completed both the basis step and the inductive step. That is, we have shown that $P(1)$ is true and the conditional statement $P(k) \rightarrow P(k+1)$ is true for all positive integers k . Consequently, by the principle of mathematical induction we can conclude that $P(n)$ is true for all positive integers n . That is, we know that $1 + 3 + 5 + \cdots + (2n-1) = n^2$ for all positive integers n . 

EXAMPLE 3 Use mathematical induction to show that

$$1 + 2 + 2^2 + \cdots + 2^n = 2^{n+1} - 1$$

for all nonnegative integers n .

Solution: Let $P(n)$ be the proposition that $1 + 2 + 2^2 + \cdots + 2^n = 2^{n+1} - 1$ for the integer n .

BASIS STEP: $P(0)$ is true because $2^0 = 1 = 2^1 - 1$. This completes the basis step.

INDUCTIVE STEP: For the inductive hypothesis, we assume that $P(k)$ is true for an arbitrary nonnegative integer k . That is, we assume that

$$1 + 2 + 2^2 + \cdots + 2^k = 2^{k+1} - 1.$$

To carry out the inductive step using this assumption, we must show that when we assume that $P(k)$ is true, then $P(k+1)$ is also true. That is, we must show that

$$1 + 2 + 2^2 + \cdots + 2^k + 2^{k+1} = 2^{(k+1)+1} - 1 = 2^{k+2} - 1$$

assuming the inductive hypothesis $P(k)$. Under the assumption of $P(k)$, we see that

$$\begin{aligned} 1 + 2 + 2^2 + \cdots + 2^k + 2^{k+1} &= (1 + 2 + 2^2 + \cdots + 2^k) + 2^{k+1} \\ &\stackrel{\text{IH}}{=} (2^{k+1} - 1) + 2^{k+1} \\ &= 2 \cdot 2^{k+1} - 1 \\ &= 2^{k+2} - 1. \end{aligned}$$

Note that we used the inductive hypothesis in the second equation in this string of equalities to replace $1 + 2 + 2^2 + \cdots + 2^k$ by $2^{k+1} - 1$. We have completed the inductive step.

Because we have completed the basis step and the inductive step, by mathematical induction we know that $P(n)$ is true for all nonnegative integers n . That is, $1 + 2 + \cdots + 2^n = 2^{n+1} - 1$ for all nonnegative integers n . 

The formula given in Example 3 is a special case of a general result for the sum of terms of a geometric progression (Theorem 1 in Section 2.4). We will use mathematical induction to provide an alternative proof of this formula.

EXAMPLE 4 Sums of Geometric Progressions Use mathematical induction to prove this formula for the sum of a finite number of terms of a geometric progression with initial term a and common ratio r :

$$\sum_{j=0}^n ar^j = a + ar + ar^2 + \cdots + ar^n = \frac{ar^{n+1} - a}{r - 1} \quad \text{when } r \neq 1,$$

where n is a nonnegative integer.

Solution: To prove this formula using mathematical induction, let $P(n)$ be the statement that the sum of the first $n + 1$ terms of a geometric progression in this formula is correct.

BASIS STEP: $P(0)$ is true, because

$$\frac{ar^{0+1} - a}{r - 1} = \frac{ar - a}{r - 1} = \frac{a(r - 1)}{r - 1} = a.$$

INDUCTIVE STEP: The inductive hypothesis is the statement that $P(k)$ is true, where k is an arbitrary nonnegative integer. That is, $P(k)$ is the statement that

$$a + ar + ar^2 + \cdots + ar^k = \frac{ar^{k+1} - a}{r - 1}.$$

To complete the inductive step we must show that if $P(k)$ is true, then $P(k + 1)$ is also true. To show that this is the case, we first add ar^{k+1} to both sides of the equality asserted by $P(k)$. We find that

$$a + ar + ar^2 + \cdots + ar^k + ar^{k+1} \stackrel{\text{IH}}{=} \frac{ar^{k+1} - a}{r - 1} + ar^{k+1}.$$

Rewriting the right-hand side of this equation shows that

$$\begin{aligned} \frac{ar^{k+1} - a}{r - 1} + ar^{k+1} &= \frac{ar^{k+1} - a}{r - 1} + \frac{ar^{k+2} - ar^{k+1}}{r - 1} \\ &= \frac{ar^{k+2} - a}{r - 1}. \end{aligned}$$

Combining these last two equations gives

$$a + ar + ar^2 + \cdots + ar^k + ar^{k+1} = \frac{ar^{k+2} - a}{r - 1}.$$

This shows that if the inductive hypothesis $P(k)$ is true, then $P(k + 1)$ must also be true. This completes the inductive argument.

We have completed the basis step and the inductive step, so by mathematical induction $P(n)$ is true for all nonnegative integers n . This shows that the formula for the sum of the terms of a geometric series is correct. 

As previously mentioned, the formula in Example 3 is the case of the formula in Example 4 with $a = 1$ and $r = 2$. The reader should verify that putting these values for a and r into the general formula gives the same formula as in Example 3.

PROVING INEQUALITIES Mathematical induction can be used to prove a variety of inequalities that hold for all positive integers greater than a particular positive integer, as Examples 5–7 illustrate.

EXAMPLE 5 Use mathematical induction to prove the inequality

$$n < 2^n$$

for all positive integers n .

Solution: Let $P(n)$ be the proposition that $n < 2^n$.

**Extra
Examples**

BASIS STEP: $P(1)$ is true, because $1 < 2^1 = 2$. This completes the basis step.

INDUCTIVE STEP: We first assume the inductive hypothesis that $P(k)$ is true for an arbitrary positive integer k . That is, the inductive hypothesis $P(k)$ is the statement that $k < 2^k$. To complete the inductive step, we need to show that if $P(k)$ is true, then $P(k+1)$, which is the statement that $k+1 < 2^{k+1}$ is true. That is, we need to show that if $k < 2^k$, then $k+1 < 2^{k+1}$. To show that this conditional statement is true for the positive integer k , we first add 1 to both sides of $k < 2^k$, and then note that $1 \leq 2^k$. This tells us that

$$k+1 < \overset{\text{IH}}{2^k} + 1 \leq 2^k + 2^k = 2 \cdot 2^k = 2^{k+1}.$$

This shows that $P(k+1)$ is true, namely, that $k+1 < 2^{k+1}$, based on the assumption that $P(k)$ is true. The induction step is complete.

Therefore, because we have completed both the basis step and the inductive step, by the principle of mathematical induction we have shown that $n < 2^n$ is true for all positive integers n . 

EXAMPLE 6 Use mathematical induction to prove that $2^n < n!$ for every integer n with $n \geq 4$. (Note that this inequality is false for $n = 1, 2$, and 3 .)

Solution: Let $P(n)$ be the proposition that $2^n < n!$.

BASIS STEP: To prove the inequality for $n \geq 4$ requires that the basis step be $P(4)$. Note that $P(4)$ is true, because $2^4 = 16 < 24 = 4!$

INDUCTIVE STEP: For the inductive step, we assume that $P(k)$ is true for an arbitrary integer k with $k \geq 4$. That is, we assume that $2^k < k!$ for the positive integer k with $k \geq 4$. We must show that under this hypothesis, $P(k+1)$ is also true. That is, we must show that if $2^k < k!$ for an arbitrary positive integer k where $k \geq 4$, then $2^{k+1} < (k+1)!$. We have

$$\begin{aligned} 2^{k+1} &= 2 \cdot 2^k && \text{by definition of exponent} \\ &\overset{\text{IH}}{<} 2 \cdot k! && \text{by the inductive hypothesis} \\ &< (k+1)k! && \text{because } 2 < k+1 \\ &= (k+1)! && \text{by definition of factorial function.} \end{aligned}$$

This shows that $P(k+1)$ is true when $P(k)$ is true. This completes the inductive step of the proof.

We have completed the basis step and the inductive step. Hence, by mathematical induction $P(n)$ is true for all integers n with $n \geq 4$. That is, we have proved that $2^n < n!$ is true for all integers n with $n \geq 4$. 

An important inequality for the sum of the reciprocals of a set of positive integers will be proved in Example 7.

EXAMPLE 7 **An Inequality for Harmonic Numbers** The **harmonic numbers** $H_j, j = 1, 2, 3, \dots$, are defined by

$$H_j = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{j}.$$

For instance,

$$H_4 = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} = \frac{25}{12}.$$

Use mathematical induction to show that

$$H_{2^n} \geq 1 + \frac{n}{2},$$

whenever n is a nonnegative integer.

Solution: To carry out the proof, let $P(n)$ be the proposition that $H_{2^n} \geq 1 + \frac{n}{2}$.

BASIS STEP: $P(0)$ is true, because $H_{2^0} = H_1 = 1 \geq 1 + \frac{0}{2}$.

INDUCTIVE STEP: The inductive hypothesis is the statement that $P(k)$ is true, that is, $H_{2^k} \geq 1 + \frac{k}{2}$, where k is an arbitrary nonnegative integer. We must show that if $P(k)$ is true, then $P(k+1)$, which states that $H_{2^{k+1}} \geq 1 + \frac{k+1}{2}$, is also true. So, assuming the inductive hypothesis, it follows that

$$\begin{aligned} H_{2^{k+1}} &= 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{2^k} + \frac{1}{2^k + 1} + \cdots + \frac{1}{2^{k+1}} && \text{by the definition of harmonic number} \\ &= H_{2^k} + \frac{1}{2^k + 1} + \cdots + \frac{1}{2^{k+1}} && \text{by the definition of } 2^k\text{th harmonic number} \\ &\stackrel{\text{IH}}{\geq} \left(1 + \frac{k}{2}\right) + \frac{1}{2^k + 1} + \cdots + \frac{1}{2^{k+1}} && \text{by the inductive hypothesis} \\ &\geq \left(1 + \frac{k}{2}\right) + 2^k \cdot \frac{1}{2^{k+1}} && \text{because there are } 2^k \text{ terms each } \geq 1/2^{k+1} \\ &\geq \left(1 + \frac{k}{2}\right) + \frac{1}{2} && \text{canceling a common factor of } 2^k \text{ in second term} \\ &= 1 + \frac{k+1}{2}. \end{aligned}$$

This establishes the inductive step of the proof.

We have completed the basis step and the inductive step. Thus, by mathematical induction $P(n)$ is true for all nonnegative integers n . That is, the inequality $H_{2^n} \geq 1 + \frac{n}{2}$ for the harmonic numbers holds for all nonnegative integers n . 

Remark: The inequality established here shows that the **harmonic series**

$$1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n} + \cdots$$

is a divergent infinite series. This is an important example in the study of infinite series.

PROVING DIVISIBILITY RESULTS Mathematical induction can be used to prove divisibility results about integers. Although such results are often easier to prove using basic results in

number theory, it is instructive to see how to prove such results using mathematical induction, as Examples 8 and 9 illustrate.

EXAMPLE 8 Use mathematical induction to prove that $n^3 - n$ is divisible by 3 whenever n is a positive integer. (Note that this is the statement with $p = 3$ of Fermat's little theorem, which is Theorem 3 of Section 4.4.)

Extra
Examples

Solution: To construct the proof, let $P(n)$ denote the proposition: “ $n^3 - n$ is divisible by 3.”

BASIS STEP: The statement $P(1)$ is true because $1^3 - 1 = 0$ is divisible by 3. This completes the basis step.

INDUCTIVE STEP: For the inductive hypothesis we assume that $P(k)$ is true; that is, we assume that $k^3 - k$ is divisible by 3 for an arbitrary positive integer k . To complete the inductive step, we must show that when we assume the inductive hypothesis, it follows that $P(k + 1)$, the statement that $(k + 1)^3 - (k + 1)$ is divisible by 3, is also true. That is, we must show that $(k + 1)^3 - (k + 1)$ is divisible by 3. Note that

$$\begin{aligned}(k + 1)^3 - (k + 1) &= (k^3 + 3k^2 + 3k + 1) - (k + 1) \\ &= (k^3 - k) + 3(k^2 + k).\end{aligned}$$

Using the inductive hypothesis, we conclude that the first term $k^3 - k$ is divisible by 3. The second term is divisible by 3 because it is 3 times an integer. So, by part (i) of Theorem 1 in Section 4.1, we know that $(k + 1)^3 - (k + 1)$ is also divisible by 3. This completes the inductive step.

Because we have completed both the basis step and the inductive step, by the principle of mathematical induction we know that $n^3 - n$ is divisible by 3 whenever n is a positive integer. 

Remark: We have included Example 8 as an illustration how a divisibility result can be proved by mathematical induction. However, there are simpler proofs. For example, we can prove that $n^3 - n$ is divisible by 3 for all positive integers n using the factorization $n^3 - n = n(n^2 - 1) = n(n - 1)(n + 1) = (n - 1)n(n + 1)$. Hence, $n^3 - 1$ is divisible by 3 because it is the product of three consecutive integers, one of which is divisible by 3.

Example 9 presents a more challenging proof by mathematical induction of a divisibility result.

EXAMPLE 9 Use mathematical induction to prove that $7^{n+2} + 8^{2n+1}$ is divisible by 57 for every nonnegative integer n .

Extra
Examples

Solution: To construct the proof, let $P(n)$ denote the proposition: “ $7^{n+2} + 8^{2n+1}$ is divisible by 57.”

BASIS STEP: To complete the basis step, we must show that $P(0)$ is true, because we want to prove that $P(n)$ is true for every nonnegative integer n . We see that $P(0)$ is true because $7^{0+2} + 8^{2 \cdot 0 + 1} = 7^2 + 8^1 = 57$ is divisible by 57. This completes the basis step.

INDUCTIVE STEP: For the inductive hypothesis we assume that $P(k)$ is true for an arbitrary nonnegative integer k ; that is, we assume that $7^{k+2} + 8^{2k+1}$ is divisible by 57. To complete the inductive step, we must show that when we assume that the inductive hypothesis $P(k)$ is true, then $P(k + 1)$, the statement that $7^{(k+1)+2} + 8^{2(k+1)+1}$ is divisible by 57, is also true.

The difficult part of the proof is to see how to use the inductive hypothesis. To take advantage of the inductive hypothesis, we use these steps:

$$\begin{aligned}
 7^{(k+1)+2} + 8^{2(k+1)+1} &= 7^{k+3} + 8^{2k+3} \\
 &= 7 \cdot 7^{k+2} + 8^2 \cdot 8^{2k+1} \\
 &= 7 \cdot 7^{k+2} + 64 \cdot 8^{2k+1} \\
 &= 7(7^{k+2} + 8^{2k+1}) + 57 \cdot 8^{2k+1}.
 \end{aligned}$$

We can now use the inductive hypothesis, which states that $7^{k+2} + 8^{2k+1}$ is divisible by 57. We will use parts (i) and (ii) of Theorem 1 in Section 4.1. By part (ii) of this theorem, and the inductive hypothesis, we conclude that the first term in this last sum, $7(7^{k+2} + 8^{2k+1})$, is divisible by 57. By part (ii) of this theorem, the second term in this sum, $57 \cdot 8^{2k+1}$, is divisible by 57. Hence, by part (i) of this theorem, we conclude that $7(7^{k+2} + 8^{2k+1}) + 57 \cdot 8^{2k+1} = 7^{k+3} + 8^{2k+3}$ is divisible by 57. This completes the inductive step.

Because we have completed both the basis step and the inductive step, by the principle of mathematical induction we know that $7^{n+2} + 8^{2n+1}$ is divisible by 57 for every nonnegative integer n . 

PROVING RESULTS ABOUT SETS Mathematical induction can be used to prove many results about sets. In particular, in Example 10 we prove a formula for the number of subsets of a finite set and in Example 11 we establish a set identity.

EXAMPLE 10 The Number of Subsets of a Finite Set Use mathematical induction to show that if S is a finite set with n elements, where n is a nonnegative integer, then S has 2^n subsets. (We will prove this result directly in several ways in Chapter 6.)

Solution: Let $P(n)$ be the proposition that a set with n elements has 2^n subsets.

BASIS STEP: $P(0)$ is true, because a set with zero elements, the empty set, has exactly $2^0 = 1$ subset, namely, itself.

INDUCTIVE STEP: For the inductive hypothesis we assume that $P(k)$ is true for an arbitrary nonnegative integer k , that is, we assume that every set with k elements has 2^k subsets. It must be shown that under this assumption, $P(k+1)$, which is the statement that every set with $k+1$ elements has 2^{k+1} subsets, must also be true. To show this, let T be a set with $k+1$ elements. Then, it is possible to write $T = S \cup \{a\}$, where a is one of the elements of T and $S = T - \{a\}$ (and hence $|S| = k$). The subsets of T can be obtained in the following way. For each subset X of S there are exactly two subsets of T , namely, X and $X \cup \{a\}$. (This is illustrated in Figure 3.) These constitute all the subsets of T and are all distinct. We now use the inductive hypothesis to conclude that S has 2^k subsets, because it has k elements. We also know that there are two subsets of T for each subset of S . Therefore, there are $2 \cdot 2^k = 2^{k+1}$ subsets of T . This finishes the inductive argument.

Because we have completed the basis step and the inductive step, by mathematical induction it follows that $P(n)$ is true for all nonnegative integers n . That is, we have proved that a set with n elements has 2^n subsets whenever n is a nonnegative integer. 

EXAMPLE 11 Use mathematical induction to prove the following generalization of one of De Morgan's laws:

$$\overline{\bigcap_{j=1}^n A_j} = \bigcup_{j=1}^n \overline{A_j}$$

whenever $A_1, A_2, \dots, A_n$ are subsets of a universal set U and $n \geq 2$.

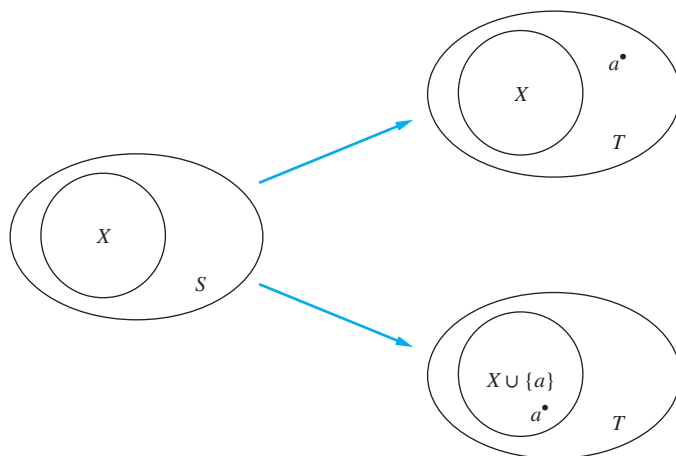


FIGURE 3 Generating subsets of a set with $k + 1$ elements. Here $T = S \cup \{a\}$.

Solution: Let $P(n)$ be the identity for n sets.

BASIS STEP: The statement $P(2)$ asserts that $\overline{A_1 \cap A_2} = \overline{A_1} \cup \overline{A_2}$. This is one of De Morgan's laws; it was proved in Example 11 of Section 2.2.

INDUCTIVE STEP: The inductive hypothesis is the statement that $P(k)$ is true, where k is an arbitrary integer with $k \geq 2$; that is, it is the statement that

$$\overline{\bigcap_{j=1}^k A_j} = \bigcup_{j=1}^k \overline{A_j}$$

whenever $A_1, A_2, \dots, A_k$ are subsets of the universal set U . To carry out the inductive step, we need to show that this assumption implies that $P(k + 1)$ is true. That is, we need to show that if this equality holds for every collection of k subsets of U , then it must also hold for every collection of $k + 1$ subsets of U . Suppose that $A_1, A_2, \dots, A_k, A_{k+1}$ are subsets of U . When the inductive hypothesis is assumed to hold, it follows that

$$\begin{aligned} \overline{\bigcap_{j=1}^{k+1} A_j} &= \overline{\left(\bigcap_{j=1}^k A_j \right) \cap A_{k+1}} && \text{by the definition of intersection} \\ &= \overline{\left(\bigcap_{j=1}^k A_j \right) \cup \overline{A_{k+1}}} && \text{by De Morgan's law (where the two sets are } \bigcap_{j=1}^k A_j \text{ and } A_{k+1}) \\ &\stackrel{\text{IH}}{=} \overline{\left(\bigcup_{j=1}^k \overline{A_j} \right) \cup \overline{A_{k+1}}} && \text{by the inductive hypothesis} \\ &= \bigcup_{j=1}^{k+1} \overline{A_j} && \text{by the definition of union.} \end{aligned}$$

This completes the inductive step.

Because we have completed both the basis step and the inductive step, by mathematical induction we know that $P(n)$ is true whenever n is a positive integer, $n \geq 2$. That is, we know that

$$\overline{\bigcap_{j=1}^n A_j} = \bigcup_{j=1}^n \overline{A_j}$$

whenever $A_1, A_2, \dots, A_n$ are subsets of a universal set U and $n \geq 2$. 

PROVING RESULTS ABOUT ALGORITHMS Next, we provide an example (somewhat more difficult than previous examples) that illustrates one of many ways mathematical induction is used in the study of algorithms. We will show how mathematical induction can be used to prove that a greedy algorithm we introduced in Section 3.1 always yields an optimal solution.

EXAMPLE 12 Recall the algorithm for scheduling talks discussed in Example 7 of Section 3.1. The input to this algorithm is a group of m proposed talks with preset starting and ending times. The goal is to schedule as many of these lectures as possible in the main lecture hall so that no two talks overlap. Suppose that talk t_j begins at time s_j and ends at time e_j . (No two lectures can proceed in the main lecture hall at the same time, but a lecture in this hall can begin at the same time another one ends.)

Without loss of generality, we assume that the talks are listed in order of nondecreasing ending time, so that $e_1 \leq e_2 \leq \dots \leq e_m$. The greedy algorithm proceeds by selecting at each stage a talk with the earliest ending time among all those talks that begin no sooner than when the last talk scheduled in the main lecture hall has ended. Note that a talk with the earliest end time is always selected first by the algorithm. We will show that this greedy algorithm is optimal in the sense that it always schedules the most talks possible in the main lecture hall. To prove the optimality of this algorithm we use mathematical induction on the variable n , the number of talks scheduled by the algorithm. We let $P(n)$ be the proposition that if the greedy algorithm schedules n talks in the main lecture hall, then it is not possible to schedule more than n talks in this hall.



BASIS STEP: Suppose that the greedy algorithm managed to schedule just one talk, t_1 , in the main lecture hall. This means that no other talk can start at or after e_1 , the end time of t_1 . Otherwise, the first such talk we come to as we go through the talks in order of nondecreasing end times could be added. Hence, at time e_1 each of the remaining talks needs to use the main lecture hall because they all start before e_1 and end after e_1 . It follows that no two talks can be scheduled because both need to use the main lecture hall at time e_1 . This shows that $P(1)$ is true and completes the basis step.

INDUCTIVE STEP: The inductive hypothesis is that $P(k)$ is true, where k is an arbitrary positive integer, that is, that the greedy algorithm always schedules the most possible talks when it selects k talks, where k is a positive integer, given any set of talks, no matter how many. We must show that $P(k+1)$ follows from the assumption that $P(k)$ is true, that is, we must show that under the assumption of $P(k)$, the greedy algorithm always schedules the most possible talks when it selects $k+1$ talks.

Now suppose that the greedy algorithm has selected $k+1$ talks. Our first step in completing the inductive step is to show there is a schedule including the most talks possible that contains talk t_1 , a talk with the earliest end time. This is easy to see because a schedule that begins with the talk t_i in the list, where $i > 1$, can be changed so that talk t_1 replaces talk t_i . To see this, note that because $e_1 \leq e_i$, all talks that were scheduled to follow talk t_i can still be scheduled.

Once we included talk t_1 , scheduling the talks so that as many as possible are scheduled is reduced to scheduling as many talks as possible that begin at or after time e_1 . So, if we have scheduled as many talks as possible, the schedule of talks other than talk t_1 is an optimal

schedule of the original talks that begin once talk t_1 has ended. Because the greedy algorithm schedules k talks when it creates this schedule, we can apply the inductive hypothesis to conclude that it has scheduled the most possible talks. It follows that the greedy algorithm has scheduled the most possible talks, $k + 1$, when it produced a schedule with $k + 1$ talks, so $P(k + 1)$ is true. This completes the inductive step.

We have completed the basis step and the inductive step. So, by mathematical induction we know that $P(n)$ is true for all positive integers n . This completes the proof of optimality. That is, we have proved that when the greedy algorithm schedules n talks, when n is a positive integer, then it is not possible to schedule more than n talks. ◀

CREATIVE USES OF MATHEMATICAL INDUCTION Mathematical induction can often be used in unexpected ways. We will illustrate two particularly clever uses of mathematical induction here, the first relating to survivors in a pie fight and the second relating to tilings with regular triominoes of checkerboards with one square missing.

EXAMPLE 13 **Odd Pie Fights** An odd number of people stand in a yard at mutually distinct distances. At the same time each person throws a pie at their nearest neighbor, hitting this person. Use mathematical induction to show that there is at least one survivor, that is, at least one person who is not hit by a pie. (This problem was introduced by Carmony [Ca79]. Note that this result is false when there are an even number of people; see Exercise 77.)

Extra
Examples ▶

Solution: Let $P(n)$ be the statement that there is a survivor whenever $2n + 1$ people stand in a yard at distinct mutual distances and each person throws a pie at their nearest neighbor. To prove this result, we will show that $P(n)$ is true for all positive integers n . This follows because as n runs through all positive integers, $2n + 1$ runs through all odd integers greater than or equal to 3. Note that one person cannot engage in a pie fight because there is no one else to throw the pie at.

BASIS STEP: When $n = 1$, there are $2n + 1 = 3$ people in the pie fight. Of the three people, suppose that the closest pair are A and B , and C is the third person. Because distances between pairs of people are different, the distance between A and C and the distance between B and C are both different from, and greater than, the distance between A and B . It follows that A and B throw pies at each other, while C throws a pie at either A or B , whichever is closer. Hence, C is not hit by a pie. This shows that at least one of the three people is not hit by a pie, completing the basis step.

INDUCTIVE STEP: For the inductive step, assume that $P(k)$ is true for an arbitrary odd integer k with $k \geq 3$. That is, assume that there is at least one survivor whenever $2k + 1$ people stand in a yard at distinct mutual distances and each throws a pie at their nearest neighbor. We must show that if the inductive hypothesis $P(k)$ is true, then $P(k + 1)$, the statement that there is at least one survivor whenever $2(k + 1) + 1 = 2k + 3$ people stand in a yard at distinct mutual distances and each throws a pie at their nearest neighbor, is also true.

So suppose that we have $2(k + 1) + 1 = 2k + 3$ people in a yard with distinct distances between pairs of people. Let A and B be the closest pair of people in this group of $2k + 3$ people. When each person throws a pie at the nearest person, A and B throw pies at each other. We have two cases to consider, (i) when someone else throws a pie at either A or B and (ii) when no one else throws a pie at either A or B .

Case (i): Because A and B throw pies at each other and someone else throws a pie at either A and B , at least three pies are thrown at A and B , and at most $(2k + 3) - 3 = 2k$ pies are thrown at the remaining $2k + 1$ people. This guarantees that at least one person is a survivor, for if each

of these $2k + 1$ people was hit by at least one pie, a total of at least $2k + 1$ pies would have to be thrown at them. (The reasoning used in this last step is an example of the pigeonhole principle discussed further in Section 6.2.)

Case (ii): No one else throws a pie at either A and B . Besides A and B , there are $2k + 1$ people. Because the distances between pairs of these people are all different, we can use the inductive hypothesis to conclude that there is at least one survivor S when these $2k + 1$ people each throws a pie at their nearest neighbor. Furthermore, S is also not hit by either the pie thrown by A or the pie thrown by B because A and B throw their pies at each other, so S is a survivor because S is not hit by any of the pies thrown by these $2k + 3$ people.

We have completed both the basis step and the inductive step, using a proof by cases. So by mathematical induction it follows that $P(n)$ is true for all positive integers n . We conclude that whenever an odd number of people located in a yard at distinct mutual distances each throws a pie at their nearest neighbor, there is at least one survivor. ◀

Links ▶

In Section 1.8 we discussed the tiling of checkerboards by polyominoes. Example 14 illustrates how mathematical induction can be used to prove a result about covering checkerboards with right triominoes, pieces shaped like the letter “L.”

EXAMPLE 14



FIGURE 4 A right triomino.

Let n be a positive integer. Show that every $2^n \times 2^n$ checkerboard with one square removed can be tiled using right triominoes, where these pieces cover three squares at a time, as shown in Figure 4.

Solution: Let $P(n)$ be the proposition that every $2^n \times 2^n$ checkerboard with one square removed can be tiled using right triominoes. We can use mathematical induction to prove that $P(n)$ is true for all positive integers n .

BASIS STEP: $P(1)$ is true, because each of the four 2×2 checkerboards with one square removed can be tiled using one right triomino, as shown in Figure 5.



FIGURE 5 Tiling 2×2 checkerboards with one square removed.

INDUCTIVE STEP: The inductive hypothesis is the assumption that $P(k)$ is true for the positive integer k ; that is, it is the assumption that every $2^k \times 2^k$ checkerboard with one square removed can be tiled using right triominoes. It must be shown that under the assumption of the inductive hypothesis, $P(k + 1)$ must also be true; that is, any $2^{k+1} \times 2^{k+1}$ checkerboard with one square removed can be tiled using right triominoes.

To see this, consider a $2^{k+1} \times 2^{k+1}$ checkerboard with one square removed. Split this checkerboard into four checkerboards of size $2^k \times 2^k$, by dividing it in half in both directions. This is illustrated in Figure 6. No square has been removed from three of these four checkerboards. The fourth $2^k \times 2^k$ checkerboard has one square removed, so we now use the inductive hypothesis to conclude that it can be covered by right triominoes. Now temporarily remove the square from each of the other three $2^k \times 2^k$ checkerboards that has the center of the original, larger checkerboard as one of its corners, as shown in Figure 7. By the inductive hypothesis, each of these three $2^k \times 2^k$ checkerboards with a square removed can be tiled by right triominoes. Furthermore, the three squares that were temporarily removed can be covered by one right triomino. Hence, the entire $2^{k+1} \times 2^{k+1}$ checkerboard can be tiled with right triominoes.

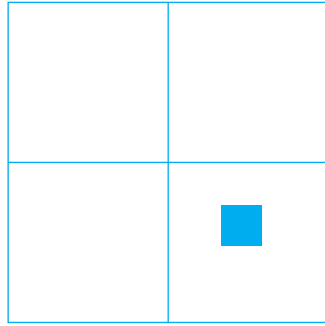


FIGURE 6 Dividing a $2^{k+1} \times 2^{k+1}$ checkerboard into four $2^k \times 2^k$ checkerboards.

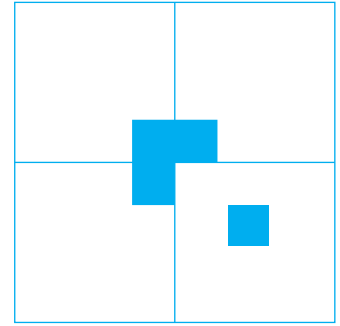


FIGURE 7 Tiling the $2^{k+1} \times 2^{k+1}$ checkerboard with one square removed.

We have completed the basis step and the inductive step. Therefore, by mathematical induction $P(n)$ is true for all positive integers n . This shows that we can tile every $2^n \times 2^n$ checkerboard, where n is a positive integer, with one square removed, using right triominoes. ◀

5.1.8 Mistaken Proofs By Mathematical Induction

Consult *Common Errors in Discrete Mathematics* on this book's website for more basic mistakes.

As with every proof method, there are many opportunities for making errors when using mathematical induction. Many well-known mistaken, and often entertaining, proofs by mathematical induction of clearly false statements have been devised, as exemplified by Example 15 and Exercises 49–51. Often, it is not easy to find where the error in reasoning occurs in such mistaken proofs.

To uncover errors in proofs by mathematical induction, remember that in every such proof, both the basis step and the inductive step must be done correctly. Not completing the basis step in a supposed proof by mathematical induction can lead to mistaken proofs of clearly ridiculous statements such as “ $n = n + 1$ whenever n is a positive integer.” (We leave it to the reader to show that it is easy to construct a correct inductive step in an attempted proof of this statement.) Locating the error in a faulty proof by mathematical induction, as Example 15 illustrates, can be quite tricky, especially when the error is hidden in the basis step.

EXAMPLE 15 Find the error in this “proof” of the clearly false claim that every set of lines in the plane, no two of which are parallel, meet in a common point.

“Proof:” Let $P(n)$ be the statement that every set of n lines in the plane, no two of which are parallel, meet in a common point. We will attempt to prove that $P(n)$ is true for all positive integers $n \geq 2$.

BASIS STEP: The statement $P(2)$ is true because any two lines in the plane that are not parallel meet in a common point (by the definition of parallel lines).

INDUCTIVE STEP: The inductive hypothesis is the statement that $P(k)$ is true for the positive integer k , that is, it is the assumption that every set of k lines in the plane, no two of which are parallel, meet in a common point. To complete the inductive step, we must show that if $P(k)$ is true, then $P(k + 1)$ must also be true. That is, we must show that if every set of k lines in the plane, no two of which are parallel, meet in a common point, then every set of $k + 1$ lines in the plane, no two of which are parallel, meet in a common point. So, consider a set of $k + 1$ distinct lines in the plane. By the inductive hypothesis, the first k of these lines meet in a common point

p_1 . Moreover, by the inductive hypothesis, the last k of these lines meet in a common point p_2 . We will show that p_1 and p_2 must be the same point. If p_1 and p_2 were different points, all lines containing both of them must be the same line because two points determine a line. This contradicts our assumption that all these lines are distinct. Thus, p_1 and p_2 are the same point. We conclude that the point $p_1 = p_2$ lies on all $k + 1$ lines. We have shown that $P(k + 1)$ is true assuming that $P(k)$ is true. That is, we have shown that if we assume that every k , $k \geq 2$, distinct lines meet in a common point, then every $k + 1$ distinct lines meet in a common point. This completes the inductive step.

We have completed the basis step and the inductive step, and supposedly we have a correct proof by mathematical induction.



Solution: Examining this supposed proof by mathematical induction it appears that everything is in order. However, there is an error, as there must be. The error is rather subtle. Carefully looking at the inductive step shows that this step requires that $k \geq 3$. We cannot show that $P(2)$ implies $P(3)$. When $k = 2$, our goal is to show that every three distinct lines meet in a common point. The first two lines must meet in a common point p_1 and the last two lines must meet in a common point p_2 . But in this case, p_1 and p_2 do not have to be the same, because only the second line is common to both sets of lines. Here is where the inductive step fails. ◀

Exercises

- There are infinitely many stations on a train route. Suppose that the train stops at the first station and suppose that if the train stops at a station, then it stops at the next station. Show that the train stops at all stations.
- Suppose that you know that a golfer plays the first hole of a golf course with an infinite number of holes and that if this golfer plays one hole, then the golfer goes on to play the next hole. Prove that this golfer plays every hole on the course.

Use mathematical induction in Exercises 3–17 to prove summation formulae. Be sure to identify where you use the inductive hypothesis.

- Let $P(n)$ be the statement that $1^2 + 2^2 + \cdots + n^2 = n(n + 1)(2n + 1)/6$ for the positive integer n .
 - What is the statement $P(1)$?
 - Show that $P(1)$ is true, completing the basis step of a proof that $P(n)$ is true for all positive integers n .
 - What is the inductive hypothesis of a proof that $P(n)$ is true for all positive integers n ?
 - What do you need to prove in the inductive step of a proof that $P(n)$ is true for all positive integers n ?
 - Complete the inductive step of a proof that $P(n)$ is true for all positive integers n , identifying where you use the inductive hypothesis.
 - Explain why these steps show that this formula is true whenever n is a positive integer.
- Let $P(n)$ be the statement that $1^3 + 2^3 + \cdots + n^3 = (n(n + 1)/2)^2$ for the positive integer n .
 - What is the statement $P(1)$?
 - Show that $P(1)$ is true, completing the basis step of the proof of $P(n)$ for all positive integers n .
 - What is the inductive hypothesis of a proof that $P(n)$ is true for all positive integers n ?
 - What do you need to prove in the inductive step of a proof that $P(n)$ is true for all positive integers n ?
 - Complete the inductive step of a proof that $P(n)$ is true for all positive integers n , identifying where you use the inductive hypothesis.
 - Explain why these steps show that this formula is true whenever n is a positive integer.
- What is the inductive hypothesis of a proof that $P(n)$ is true for all positive integers n ?
- What do you need to prove in the inductive step of a proof that $P(n)$ is true for all positive integers n ?
- Complete the inductive step of a proof that $P(n)$ is true for all positive integers n , identifying where you use the inductive hypothesis.
- Explain why these steps show that this formula is true whenever n is a positive integer.
- Prove that $1^2 + 3^2 + 5^2 + \cdots + (2n + 1)^2 = (n + 1)(2n + 1)(2n + 3)/3$ whenever n is a nonnegative integer.
- Prove that $1 \cdot 1! + 2 \cdot 2! + \cdots + n \cdot n! = (n + 1)! - 1$ whenever n is a positive integer.
- Prove that $3 + 3 \cdot 5 + 3 \cdot 5^2 + \cdots + 3 \cdot 5^n = 3(5^{n+1} - 1)/4$ whenever n is a nonnegative integer.
- Prove that $2 - 2 \cdot 7 + 2 \cdot 7^2 - \cdots + 2(-7)^n = (1 - (-7)^{n+1})/4$ whenever n is a nonnegative integer.
- Find a formula for the sum of the first n even positive integers.
 - Prove the formula that you conjectured in part (a).
- Find a formula for
$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \cdots + \frac{1}{n(n + 1)}$$
 by examining the values of this expression for small values of n .
 - Prove the formula you conjectured in part (a).
- Find a formula for
$$\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots + \frac{1}{2^n}$$
 by examining the values of this expression for small values of n .
 - Prove the formula you conjectured in part (a).

12. Prove that

$$\sum_{j=0}^n \left(-\frac{1}{2}\right)^j = \frac{2^{n+1} + (-1)^n}{3 \cdot 2^n}$$

whenever n is a nonnegative integer.

13. Prove that $1^2 - 2^2 + 3^2 - \dots + (-1)^{n-1} n^2 = (-1)^{n-1} n(n+1)/2$ whenever n is a positive integer.

14. Prove that for every positive integer n , $\sum_{k=1}^n k2^k = (n-1)2^{n+1} + 2$.

15. Prove that for every positive integer n ,

$$1 \cdot 2 + 2 \cdot 3 + \dots + n(n+1) = n(n+1)(n+2)/3.$$

16. Prove that for every positive integer n ,

$$1 \cdot 2 \cdot 3 + 2 \cdot 3 \cdot 4 + \dots + n(n+1)(n+2) = n(n+1)(n+2)(n+3)/4.$$

17. Prove that $\sum_{j=1}^n j^4 = n(n+1)(2n+1)(3n^2+3n-1)/30$ whenever n is a positive integer.

Use mathematical induction to prove the inequalities in Exercises 18–30.

18. Let $P(n)$ be the statement that $n! < n^n$, where n is an integer greater than 1.

- What is the statement $P(2)$?
- Show that $P(2)$ is true, completing the basis step of a proof by mathematical induction that $P(n)$ is true for all integers n greater than 1.
- What is the inductive hypothesis of a proof by mathematical induction that $P(n)$ is true for all integers n greater than 1?
- What do you need to prove in the inductive step of a proof by mathematical induction that $P(n)$ is true for all integers n greater than 1?
- Complete the inductive step of a proof by mathematical induction that $P(n)$ is true for all integers n greater than 1.
- Explain why these steps show that this inequality is true whenever n is an integer greater than 1.

19. Let $P(n)$ be the statement that

$$1 + \frac{1}{4} + \frac{1}{9} + \dots + \frac{1}{n^2} < 2 - \frac{1}{n},$$

where n is an integer greater than 1.

- What is the statement $P(2)$?
- Show that $P(2)$ is true, completing the basis step of a proof by mathematical induction that $P(n)$ is true for all integers n greater than 1.
- What is the inductive hypothesis of a proof by mathematical induction that $P(n)$ is true for all integers n greater than 1?
- What do you need to prove in the inductive step of a proof by mathematical induction that $P(n)$ is true for all integers n greater than 1?
- Complete the inductive step of a proof by mathematical induction that $P(n)$ is true for all integers n greater than 1.
- Explain why these steps show that this inequality is true whenever n is an integer greater than 1.

20. Prove that $3^n < n!$ if n is an integer greater than 6.

21. Prove that $2^n > n^2$ if n is an integer greater than 4.

22. For which nonnegative integers n is $n^2 \leq n!$? Prove your answer.

23. For which nonnegative integers n is $2n + 3 \leq 2^n$? Prove your answer.

24. Prove that $1/(2n) \leq [1 \cdot 3 \cdot 5 \cdot \dots \cdot (2n-1)]/(2 \cdot 4 \cdot \dots \cdot 2n)$ whenever n is a positive integer.

*25. Prove that if $h > -1$, then $1 + nh \leq (1 + h)^n$ for all nonnegative integers n . This is called **Bernoulli's inequality**.

*26. Suppose that a and b are real numbers with $0 < b < a$. Prove that if n is a positive integer, then $a^n - b^n \leq na^{n-1}(a - b)$.

*27. Prove that for every positive integer n ,

$$1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \dots + \frac{1}{\sqrt{n}} > 2(\sqrt{n+1} - 1).$$

28. Prove that $n^2 - 7n + 12$ is nonnegative whenever n is an integer with $n \geq 3$.

In Exercises 29 and 30, H_n denotes the n th harmonic number.

*29. Prove that $H_{2^n} \leq 1 + n$ whenever n is a nonnegative integer.

*30. Prove that

$$H_1 + H_2 + \dots + H_n = (n+1)H_n - n.$$

Use mathematical induction in Exercises 31–37 to prove divisibility facts.

31. Prove that 2 divides $n^2 + n$ whenever n is a positive integer.

32. Prove that 3 divides $n^3 + 2n$ whenever n is a positive integer.

33. Prove that 5 divides $n^5 - n$ whenever n is a nonnegative integer.

34. Prove that 6 divides $n^3 - n$ whenever n is a nonnegative integer.

*35. Prove that $n^2 - 1$ is divisible by 8 whenever n is an odd positive integer.

*36. Prove that 21 divides $4^{n+1} + 5^{2n-1}$ whenever n is a positive integer.

*37. Prove that if n is a positive integer, then 133 divides $11^{n+1} + 12^{2n-1}$.

Use mathematical induction in Exercises 38–46 to prove results about sets.

38. Prove that if $A_1, A_2, \dots, A_n$ and $B_1, B_2, \dots, B_n$ are sets such that $A_j \subseteq B_j$ for $j = 1, 2, \dots, n$, then

$$\bigcup_{j=1}^n A_j \subseteq \bigcup_{j=1}^n B_j.$$

39. Prove that if $A_1, A_2, \dots, A_n$ and $B_1, B_2, \dots, B_n$ are sets such that $A_j \subseteq B_j$ for $j = 1, 2, \dots, n$, then

$$\bigcap_{j=1}^n A_j \subseteq \bigcap_{j=1}^n B_j.$$

40. Prove that if $A_1, A_2, \dots, A_n$ and B are sets, then

$$(A_1 \cap A_2 \cap \dots \cap A_n) \cup B \\ = (A_1 \cup B) \cap (A_2 \cup B) \cap \dots \cap (A_n \cup B).$$

41. Prove that if $A_1, A_2, \dots, A_n$ and B are sets, then

$$(A_1 \cup A_2 \cup \dots \cup A_n) \cap B \\ = (A_1 \cap B) \cup (A_2 \cap B) \cup \dots \cup (A_n \cap B).$$

42. Prove that if $A_1, A_2, \dots, A_n$ and B are sets, then

$$(A_1 - B) \cap (A_2 - B) \cap \dots \cap (A_n - B) \\ = (A_1 \cap A_2 \cap \dots \cap A_n) - B.$$

43. Prove that if $A_1, A_2, \dots, A_n$ are subsets of a universal set U , then

$$\overline{\bigcup_{k=1}^n A_k} = \bigcap_{k=1}^n \overline{A_k}.$$

44. Prove that if $A_1, A_2, \dots, A_n$ and B are sets, then

$$(A_1 - B) \cup (A_2 - B) \cup \dots \cup (A_n - B) \\ = (A_1 \cup A_2 \cup \dots \cup A_n) - B.$$

45. Prove that a set with n elements has $n(n-1)/2$ subsets containing exactly two elements whenever n is an integer greater than or equal to 2.

- *46. Prove that a set with n elements has $n(n-1)(n-2)/6$ subsets containing exactly three elements whenever n is an integer greater than or equal to 3.

In Exercises 47 and 48 we consider the problem of placing towers along a straight road, so that every building on the road receives cellular service. Assume that a building receives cellular service if it is within one mile of a tower.

47. Devise a greedy algorithm that uses the minimum number of towers possible to provide cell service to d buildings located at positions $x_1, x_2, \dots, x_d$ from the start of the road. [Hint: At each step, go as far as possible along the road before adding a tower so as not to leave any buildings without coverage.]

- *48. Use mathematical induction to prove that the algorithm you devised in Exercise 47 produces an optimal solution, that is, that it uses the fewest towers possible to provide cellular service to all buildings.

Exercises 49–51 present incorrect proofs using mathematical induction. You will need to identify an error in reasoning in each exercise.

49. What is wrong with this “proof” that all horses are the same color?
Let $P(n)$ be the proposition that all the horses in a set of n horses are the same color.

Basis Step: Clearly, $P(1)$ is true.

Inductive Step: Assume that $P(k)$ is true, so that all the horses in any set of k horses are the same color. Consider any $k+1$ horses; number these as horses 1, 2, 3, ..., $k, k+1$. Now the first k of these horses all must have the same color, and the last k of these must

also have the same color. Because the set of the first k horses and the set of the last k horses overlap, all $k+1$ must be the same color. This shows that $P(k+1)$ is true and finishes the proof by induction.

50. What is wrong with this “proof”?

“Theorem” For every positive integer n , $\sum_{i=1}^n i = (n + \frac{1}{2})^2/2$.

Basis Step: The formula is true for $n = 1$.

Inductive Step: Suppose that $\sum_{i=1}^n i = (n + \frac{1}{2})^2/2$. Then $\sum_{i=1}^{n+1} i = (\sum_{i=1}^n i) + (n+1)$. By the inductive hypothesis, we have $\sum_{i=1}^{n+1} i = (n + \frac{1}{2})^2/2 + n + 1 = (n^2 + n + \frac{1}{4})/2 + n + 1 = (n^2 + 3n + \frac{9}{4})/2 = (n + \frac{3}{2})^2/2 = [(n+1) + \frac{1}{2}]^2/2$, completing the inductive step.

51. What is wrong with this “proof”?

“Theorem” For every positive integer n , if x and y are positive integers with $\max(x, y) = n$, then $x = y$.

Basis Step: Suppose that $n = 1$. If $\max(x, y) = 1$ and x and y are positive integers, we have $x = 1$ and $y = 1$.

Inductive Step: Let k be a positive integer. Assume that whenever $\max(x, y) = k$ and x and y are positive integers, then $x = y$. Now let $\max(x, y) = k+1$, where x and y are positive integers. Then $\max(x-1, y-1) = k$, so by the inductive hypothesis, $x-1 = y-1$. It follows that $x = y$, completing the inductive step.

52. Suppose that m and n are positive integers with $m > n$ and f is a function from $\{1, 2, \dots, m\}$ to $\{1, 2, \dots, n\}$. Use mathematical induction on the variable n to show that f is not one-to-one.

- *53. Use mathematical induction to show that n people can divide a cake (where each person gets one or more separate pieces of the cake) so that the cake is divided fairly, that is, in the sense that each person thinks he or she got at least $(1/n)$ th of the cake. [Hint: For the inductive step, take a fair division of the cake among the first k people, have each person divide their share into what this person thinks are $k+1$ equal portions, and then have the $(k+1)$ st person select a portion from each of the k people. When showing this produces a fair division for $k+1$ people, suppose that person $k+1$ thinks that person i got p_i of the cake, where $\sum_{i=1}^k p_i = 1$.]

54. Use mathematical induction to show that given a set of $n+1$ positive integers, none exceeding $2n$, there is at least one integer in this set that divides another integer in the set.

- *55. A knight on a chessboard can move one space horizontally (in either direction) and two spaces vertically (in either direction) or two spaces horizontally (in either direction) and one space vertically (in either direction). Suppose that we have an infinite chessboard, made up of all squares (m, n) , where m and n are nonnegative integers that denote the row number and the column number of the square, respectively. Use mathematical induction to show that a knight starting at $(0, 0)$ can visit every square using

a finite sequence of moves. [Hint: Use induction on the variable $s = m + n$.]

56. Suppose that

$$\mathbf{A} = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix},$$

where a and b are real numbers. Show that

$$\mathbf{A}^n = \begin{bmatrix} a^n & 0 \\ 0 & b^n \end{bmatrix}$$

for every positive integer n .

57. (Requires calculus) Use mathematical induction to prove that the derivative of $f(x) = x^n$ equals nx^{n-1} whenever n is a positive integer. (For the inductive step, use the product rule for derivatives.)
58. Suppose that $\mathbf{A}$ and $\mathbf{B}$ are square matrices with the property $\mathbf{AB} = \mathbf{BA}$. Show that $\mathbf{AB}^n = \mathbf{B}^n\mathbf{A}$ for every positive integer n .
59. Suppose that m is a positive integer. Use mathematical induction to prove that if a and b are integers with $a \equiv b \pmod{m}$, then $a^k \equiv b^k \pmod{m}$ whenever k is a nonnegative integer.
60. Use mathematical induction to show that $\neg(p_1 \vee p_2 \vee \cdots \vee p_n)$ is equivalent to $\neg p_1 \wedge \neg p_2 \wedge \cdots \wedge \neg p_n$ whenever $p_1, p_2, \dots, p_n$ are propositions.

- *61. Show that

$$\begin{aligned} [(p_1 \rightarrow p_2) \wedge (p_2 \rightarrow p_3) \wedge \cdots \wedge (p_{n-1} \rightarrow p_n)] \\ \rightarrow [(p_1 \wedge p_2 \wedge \cdots \wedge p_{n-1}) \rightarrow p_n] \end{aligned}$$

is a tautology whenever $p_1, p_2, \dots, p_n$ are propositions, where $n \geq 2$.

- *62. Show that n lines separate the plane into $(n^2 + n + 2)/2$ regions if no two of these lines are parallel and no three pass through a common point.
- **63. Let $a_1, a_2, \dots, a_n$ be positive real numbers. The **arithmetic mean** of these numbers is defined by

$$A = (a_1 + a_2 + \cdots + a_n)/n,$$

and the **geometric mean** of these numbers is defined by

$$G = (a_1 a_2 \cdots a_n)^{1/n}.$$

Use mathematical induction to prove that $A \geq G$.

64. Use mathematical induction to prove Lemma 3 of Section 4.3, which states that if p is a prime and $p \mid a_1 a_2 \cdots a_n$, where a_i is an integer for $i = 1, 2, 3, \dots, n$, then $p \mid a_i$ for some integer i .
65. Show that if n is a positive integer, then

$$\sum_{\{a_1, \dots, a_k\} \subseteq \{1, 2, \dots, n\}} \frac{1}{a_1 a_2 \cdots a_k} = n.$$

(Here the sum is over all nonempty subsets of the set of the n smallest positive integers.)

- *66. Use the well-ordering property to show that the following form of mathematical induction is a valid method to prove that $P(n)$ is true for all positive integers n .

Basis Step: $P(1)$ and $P(2)$ are true.

Inductive Step: For each positive integer k , if $P(k)$ and $P(k+1)$ are both true, then $P(k+2)$ is true.

67. Show that if $A_1, A_2, \dots, A_n$ are sets where $n \geq 2$, and for all pairs of integers i and j with $1 \leq i < j \leq n$, either A_i is a subset of A_j or A_j is a subset of A_i , then there is an integer i , $1 \leq i \leq n$, such that A_i is a subset of A_j for all integers j with $1 \leq j \leq n$.

- *68. A guest at a party is a **celebrity** if this person is known by every other guest, but knows none of them. There is at most one celebrity at a party, for if there were two, they would know each other. A particular party may have no celebrity. Your assignment is to find the celebrity, if one exists, at a party, by asking only one type of question—asking a guest whether they know a second guest. Everyone must answer your questions truthfully. That is, if Alice and Bob are two people at the party, you can ask Alice whether she knows Bob; she must answer correctly. Use mathematical induction to show that if there are n people at the party, then you can find the celebrity, if there is one, with $3(n-1)$ questions. [Hint: First ask a question to eliminate one person as a celebrity. Then use the inductive hypothesis to identify a potential celebrity. Finally, ask two more questions to determine whether that person is actually a celebrity.]

Suppose there are n people in a group, each aware of a scandal no one else in the group knows about. These people communicate by telephone; when two people in the group talk, they share information about all scandals each knows about. For example, on the first call, two people share information, so by the end of the call, each of these people knows about two scandals. The **gossip problem** asks for $G(n)$, the minimum number of telephone calls that are needed for all n people to learn about all the scandals. Exercises 69–71 deal with the gossip problem.

69. Find $G(1)$, $G(2)$, $G(3)$, and $G(4)$.

70. Use mathematical induction to prove that $G(n) \leq 2n - 4$ for $n \geq 4$. [Hint: In the inductive step, have a new person call a particular person at the start and at the end.]

- **71. Prove that $G(n) = 2n - 4$ for $n \geq 4$.

- *72. Show that it is possible to arrange the numbers $1, 2, \dots, n$ in a row so that the average of any two of these numbers never appears between them. [Hint: Show that it suffices to prove this fact when n is a power of 2. Then use mathematical induction to prove the result when n is a power of 2.]

- *73. Show that if $I_1, I_2, \dots, I_n$ is a collection of open intervals on the real number line, $n \geq 2$, and every pair of these intervals has a nonempty intersection, that is, $I_i \cap I_j \neq \emptyset$ whenever $1 \leq i \leq n$ and $1 \leq j \leq n$, then the intersection of all these sets is nonempty, that is, $I_1 \cap I_2 \cap \cdots \cap I_n \neq \emptyset$. (Recall that an **open interval** is the set of real numbers x with $a < x < b$, where a and b are real numbers with $a < b$.)

Sometimes we cannot use mathematical induction to prove a result we believe to be true, but we can use mathematical induction to prove a stronger result. Because the inductive hypothesis of the stronger result provides more to work with, this process is called **inductive loading**. We use inductive loading in Exercise 74–76.

74. Show that we cannot use mathematical induction to prove that $\sum_{j=1}^n 1/j^2 < 2$ for all positive integers n , but that this inequality is a consequence of the inequality proved by mathematical induction in Exercise 19.
75. Suppose that we want to prove that

$$\sum_{j=1}^n j/(j+1)! < 1$$

for all positive integers n .

- a) Show that if we try to prove this inequality using mathematical induction, the basis step works, but the inductive step fails.
- b) Show that mathematical induction can be used to prove the stronger inequality

$$\sum_{j=1}^n j/(j+1)! \leq 1 - 1/(n+1)!$$

for all positive integers n , implying that the weaker inequality is also true.

76. Suppose that we want to prove that

$$\frac{1}{2} \cdot \frac{3}{4} \cdots \frac{2n-1}{2n} < \frac{1}{\sqrt{3n}}$$

for all positive integers n .

- a) Show that if we try to prove this inequality using mathematical induction, the basis step works, but the inductive step fails.
- b) Show that mathematical induction can be used to prove the stronger inequality

$$\frac{1}{2} \cdot \frac{3}{4} \cdots \frac{2n-1}{2n} < \frac{1}{\sqrt{3n+1}}$$

for all integers n greater than 1, which, together with a verification for the case where $n = 1$, establishes the

weaker inequality we originally tried to prove using mathematical induction.

77. Let n be an even integer. Show that it is possible for n people to stand in a yard at mutually distinct distances so that when each person throws a pie at their nearest neighbor, everyone is hit by a pie.
78. Construct a tiling using right triominoes of the 4×4 checkerboard with the square in the upper left corner removed.
79. Construct a tiling using right triominoes of the 8×8 checkerboard with the square in the upper left corner removed.
80. Prove or disprove that all checkerboards of these shapes can be completely covered using right triominoes whenever n is a positive integer.
- a) 3×2^n b) 6×2^n
c) $3^n \times 3^n$ d) $6^n \times 6^n$
- *81. Show that a three-dimensional $2^n \times 2^n \times 2^n$ checkerboard with one $1 \times 1 \times 1$ cube missing can be completely covered by $2 \times 2 \times 2$ cubes with one $1 \times 1 \times 1$ cube removed.
- *82. Show that an $n \times n$ checkerboard with one square removed can be completely covered using right triominoes if $n > 5$, n is odd, and $3 \nmid n$.
83. Show that a 5×5 checkerboard with a corner square removed can be tiled using right triominoes.
- *84. Find a 5×5 checkerboard with a square removed that cannot be tiled using right triominoes. Prove that such a tiling does not exist for this board.
85. Use the principle of mathematical induction to show that $P(n)$ is true for $n = b, b+1, b+2, \dots$, where b is an integer, if $P(b)$ is true and the conditional statement $P(k) \rightarrow P(k+1)$ is true for all integers k with $k \geq b$.

5.2 Strong Induction and Well-Ordering

5.2.1 Introduction

In Section 5.1 we introduced mathematical induction and we showed how to use it to prove a variety of theorems. In this section we will introduce another form of mathematical induction, called **strong induction**, which can often be used when we cannot easily prove a result using mathematical induction. The basis step of a proof by strong induction is the same as a proof of the same result using mathematical induction. That is, in a strong induction proof that $P(n)$ is true for all positive integers n , the basis step shows that $P(1)$ is true. However, the inductive steps in these two proof methods are different. In a proof by mathematical induction, the inductive step shows that if the inductive hypothesis $P(k)$ is true, then $P(k+1)$ is also true. In a proof by strong induction, the inductive step shows that if $P(j)$ is true for all positive integers j not exceeding k , then $P(k+1)$ is true. That is, for the inductive hypothesis we assume that $P(j)$ is true for $j = 1, 2, \dots, k$.

The validity of both mathematical induction and strong induction follow from the well-ordering property in Appendix 1. In fact, mathematical induction, strong induction, and well-ordering are all equivalent principles (as shown in Exercises 41, 42, and 43). That is, the validity

of each can be proved from either of the other two. This means that a proof using one of these two principles can be rewritten as a proof using either of the other two principles. Just as it is sometimes the case that it is much easier to see how to prove a result using strong induction rather than mathematical induction, it is sometimes easier to use well-ordering than one of the two forms of mathematical induction. In this section we will give some examples of how the well-ordering property can be used to prove theorems.

5.2.2 Strong Induction

Before we illustrate how to use strong induction, we state this principle again.

STRONG INDUCTION To prove that $P(n)$ is true for all positive integers n , where $P(n)$ is a propositional function, we complete two steps:

BASIS STEP: We verify that the proposition $P(1)$ is true.

INDUCTIVE STEP: We show that the conditional statement $[P(1) \wedge P(2) \wedge \cdots \wedge P(k)] \rightarrow P(k+1)$ is true for all positive integers k .

Note that when we use strong induction to prove that $P(n)$ is true for all positive integers n , our inductive hypothesis is the assumption that $P(j)$ is true for $j = 1, 2, \dots, k$. That is, the inductive hypothesis includes all k statements $P(1), P(2), \dots, P(k)$. Because we can use all k statements $P(1), P(2), \dots, P(k)$ to prove $P(k+1)$, rather than just the statement $P(k)$ as in a proof by mathematical induction, strong induction is a more flexible proof technique. Because of this, some mathematicians prefer to always use strong induction instead of mathematical induction, even when a proof by mathematical induction is easy to find.

You may be surprised that mathematical induction and strong induction are equivalent. That is, each can be shown to be a valid proof technique assuming that the other is valid. In particular, any proof using mathematical induction can also be considered to be a proof by strong induction because the inductive hypothesis of a proof by mathematical induction is part of the inductive hypothesis in a proof by strong induction. That is, if we can complete the inductive step of a proof using mathematical induction by showing that $P(k+1)$ follows from $P(k)$ for every positive integer k , then it also follows that $P(k+1)$ follows from all the statements $P(1), P(2), \dots, P(k)$, because we are assuming that not only $P(k)$ is true, but also more, namely, that the $k-1$ statements $P(1), P(2), \dots, P(k-1)$ are true. However, it is much more awkward to convert a proof by strong induction into a proof using the principle of mathematical induction. (See Exercise 42.)

Strong induction is sometimes called the **second principle of mathematical induction** or **complete induction**. When the terminology “complete induction” is used, the principle of mathematical induction is called **incomplete induction**, a technical term that is a somewhat unfortunate choice because there is nothing incomplete about the principle of mathematical induction; after all, it is a valid proof technique.

STRONG INDUCTION AND THE INFINITE LADDER To better understand strong induction, consider the infinite ladder in Section 5.1. Strong induction tells us that we can reach all rungs if

1. we can reach the first rung, and
2. for every positive integer k , if we can reach all the first k rungs, then we can reach the $(k+1)$ st rung.

That is, if $P(n)$ is the statement that we can reach the n th rung of the ladder, by strong induction we know that $P(n)$ is true for all positive integers n , because (1) tells us $P(1)$ is true, completing

the basis step and (2) tells us that $P(1) \wedge P(2) \wedge \cdots \wedge P(k)$ implies $P(k + 1)$, completing the inductive step.

Example 1 illustrates how strong induction can help us prove a result that cannot easily be proved using the principle of mathematical induction.

EXAMPLE 1 Suppose we can reach the first and second rungs of an infinite ladder, and we know that if we can reach a rung, then we can reach two rungs higher. Can we prove that we can reach every rung using the principle of mathematical induction? Can we prove that we can reach every rung using strong induction?

Solution: We first try to prove this result using the principle of mathematical induction.

BASIS STEP: The basis step of such a proof holds; here it simply verifies that we can reach the first rung.

ATTEMPTED INDUCTIVE STEP: The inductive hypothesis is the statement that we can reach the k th rung of the ladder. To complete the inductive step, we need to show that if we assume the inductive hypothesis for the positive integer k , namely, if we assume that we can reach the k th rung of the ladder, then we can show that we can reach the $(k + 1)$ st rung of the ladder. However, there is no obvious way to complete this inductive step because we do not know from the given information that we can reach the $(k + 1)$ st rung from the k th rung. After all, we only know that if we can reach a rung we can reach the rung two higher.

Now consider a proof using strong induction.

BASIS STEP: The basis step is the same as before; it simply verifies that we can reach the first rung.

INDUCTIVE STEP: The inductive hypothesis states that we can reach each of the first k rungs. To complete the inductive step, we need to show that if we assume that the inductive hypothesis is true, that is, if we can reach each of the first k rungs, then we can reach the $(k + 1)$ st rung. We already know that we can reach the second rung. We can complete the inductive step by noting that as long as $k \geq 2$, we can reach the $(k + 1)$ st rung from the $(k - 1)$ st rung because we know we can climb two rungs from a rung we can already reach, and because $k - 1 \leq k$, by the inductive hypothesis we can reach the $(k - 1)$ st rung. This completes the inductive step and finishes the proof by strong induction.

We have proved that if we can reach the first two rungs of an infinite ladder and for every positive integer k if we can reach all the first k rungs then we can reach the $(k + 1)$ st rung, then we can reach all rungs of the ladder. ◀

5.2.3 Examples of Proofs Using Strong Induction

Now that we have both mathematical induction and strong induction, how do we decide which method to apply in a particular situation? Although there is no cut-and-dried answer, we can supply some useful pointers. In practice, you should use mathematical induction when it is straightforward to prove that $P(k) \rightarrow P(k + 1)$ is true for all positive integers k . This is the case for all the proofs in the examples in Section 5.1. In general, you should restrict your use of the principle of mathematical induction to such scenarios. Unless you can clearly see that the inductive step of a proof by mathematical induction goes through, you should attempt a proof by strong induction. That is, use strong induction and not mathematical induction when you see how to prove that $P(k + 1)$ is true from the assumption that $P(j)$ is true for all positive integers j not exceeding k , but you cannot see how to prove that $P(k + 1)$ follows from just $P(k)$. Keep this in mind as you examine the proofs in this section. For each of these proofs, consider why strong induction works better than mathematical induction.

We will illustrate how strong induction is employed in Examples 2–4. In these examples, we will prove a diverse collection of results. Pay particular attention to the inductive step in each of these examples, where we show that a result $P(k + 1)$ follows under the assumption that $P(j)$ holds for all positive integers j not exceeding k , where $P(n)$ is a propositional function.

Before we present these examples, note that we can slightly modify strong induction to handle a wider variety of situations. In particular, we can adapt strong induction to handle cases where the inductive step is valid only for integers greater than a particular integer. Let b be a fixed integer and j a fixed positive integer. The form of strong induction we need tells us that $P(n)$ is true for all integers n with $n \geq b$ if we can complete these two steps:

BASIS STEP: We verify that the propositions $P(b)$, $P(b + 1)$, $\dots$, $P(b + j)$ are true.

INDUCTIVE STEP: We show that $[P(b) \wedge P(b + 1) \wedge \dots \wedge P(k)] \rightarrow P(k + 1)$ is true for every integer $k \geq b + j$.

That this alternative form is equivalent to strong induction is left as Exercise 28.

We begin with one of the most prominent uses of strong induction, the part of the fundamental theorem of arithmetic that tells us that every positive integer can be written as the product of primes.

EXAMPLE 2 Show that if n is an integer greater than 1, then n can be written as the product of primes.

**Extra
Examples** 

Solution: Let $P(n)$ be the proposition that n can be written as the product of primes.

BASIS STEP: $P(2)$ is true, because 2 can be written as the product of one prime, itself. (Note that $P(2)$ is the first case we need to establish.)

INDUCTIVE STEP: The inductive hypothesis is the assumption that $P(j)$ is true for all integers j with $2 \leq j \leq k$, that is, the assumption that j can be written as the product of primes whenever j is a positive integer at least 2 and not exceeding k . To complete the inductive step, it must be shown that $P(k + 1)$ is true under this assumption, that is, that $k + 1$ is the product of primes.

There are two cases to consider, namely, when $k + 1$ is prime and when $k + 1$ is composite. If $k + 1$ is prime, we immediately see that $P(k + 1)$ is true. Otherwise, $k + 1$ is composite and can be written as the product of two positive integers a and b with $2 \leq a \leq b < k + 1$. Because both a and b are integers at least 2 and not exceeding k , we can use the inductive hypothesis to write both a and b as the product of primes. Thus, if $k + 1$ is composite, it can be written as the product of primes, namely, those primes in the factorization of a and those in the factorization of b . 

Remark: Because 1 can be thought of as an *empty* product of primes, that is, the product of no primes, we could have started the proof in Example 2 with $P(1)$ as the basis step. We chose not to do so because many people find this confusing.

Example 2 completes the proof of the fundamental theorem of arithmetic, which asserts that every nonnegative integer can be written uniquely as the product of primes in nondecreasing order. We showed in Section 4.3 that an integer has at most one such factorization into primes. Example 2 shows there is at least one such factorization.

Next, we show how strong induction can be used to prove that a player has a winning strategy in a game.

EXAMPLE 3 Consider a game in which two players take turns removing any positive number of matches they want from one of two piles of matches. The player who removes the last match wins the game. Show that if the two piles contain the same number of matches initially, the second player can always guarantee a win.

Solution: Let n be the number of matches in each pile. We will use strong induction to prove $P(n)$, the statement that the second player can win when there are initially n matches in each pile.

BASIS STEP: When $n = 1$, the first player has only one choice, removing one match from one of the piles, leaving a single pile with a single match, which the second player can remove to win the game.

INDUCTIVE STEP: The inductive hypothesis is the statement that $P(j)$ is true for all j with $1 \leq j \leq k$, that is, the assumption that the second player can always win whenever there are j matches, where $1 \leq j \leq k$ in each of the two piles at the start of the game. We need to show that $P(k + 1)$ is true, that is, that the second player can win when there are initially $k + 1$ matches in each pile, under the assumption that $P(j)$ is true for $j = 1, 2, \dots, k$. So suppose that there are $k + 1$ matches in each of the two piles at the start of the game and suppose that the first player removes r matches ($1 \leq r \leq k$) from one of the piles, leaving $k + 1 - r$ matches in this pile. By removing the same number of matches from the other pile, the second player creates the situation where there are two piles each with $k + 1 - r$ matches. Because $1 \leq k + 1 - r \leq k$, we can now use the inductive hypothesis to conclude that the second player can always win. We complete the proof by noting that if the first player removes all $k + 1$ matches from one of the piles, the second player can win by removing all the remaining matches. 

Using the principle of mathematical induction, instead of strong induction, to prove the results in Examples 2 and 3 is difficult. However, as Example 4 shows, some results can be readily proved using either the principle of mathematical induction or strong induction.

EXAMPLE 4 Prove that every amount of postage of 12 cents or more can be formed using just 4-cent and 5-cent stamps.

Solution: We will prove this result using the principle of mathematical induction. Then we will present a proof using strong induction. Let $P(n)$ be the statement that postage of n cents can be formed using 4-cent and 5-cent stamps.

We begin by using the principle of mathematical induction.

BASIS STEP: Postage of 12 cents can be formed using three 4-cent stamps.

INDUCTIVE STEP: The inductive hypothesis is the statement that $P(k)$ is true. That is, under this hypothesis, postage of k cents can be formed using 4-cent and 5-cent stamps. To complete the inductive step, we need to show that when we assume $P(k)$ is true, then $P(k + 1)$ is also true where $k \geq 12$. That is, we need to show that if we can form postage of k cents, then we can form postage of $k + 1$ cents. So, assume the inductive hypothesis is true; that is, assume that we can form postage of k cents using 4-cent and 5-cent stamps. We consider two cases, when at least one 4-cent stamp has been used and when no 4-cent stamps have been used. First, suppose that at least one 4-cent stamp was used to form postage of k cents. Then we can replace this stamp with a 5-cent stamp to form postage of $k + 1$ cents. But if no 4-cent stamps were used, we can form postage of k cents using only 5-cent stamps. Moreover, because $k \geq 12$, we needed at least three 5-cent stamps to form postage of k cents. So, we can replace three 5-cent stamps with four 4-cent stamps to form postage of $k + 1$ cents. This completes the inductive step.

Because we have completed the basis step and the inductive step, we know that $P(n)$ is true for all $n \geq 12$. That is, we can form postage of n cents, where $n \geq 12$ using just 4-cent and 5-cent stamps. This completes the proof by mathematical induction.

Next, we will use strong induction to prove the same result. In this proof, in the basis step we show that $P(12)$, $P(13)$, $P(14)$, and $P(15)$ are true, that is, that postage of 12, 13, 14, or 15 cents can be formed using just 4-cent and 5-cent stamps. In the inductive step we show how to get postage of $k + 1$ cents for $k \geq 15$ from postage of $k - 3$ cents.

BASIS STEP: We can form postage of 12, 13, 14, and 15 cents using three 4-cent stamps, two 4-cent stamps and one 5-cent stamp, one 4-cent stamp and two 5-cent stamps, and three 5-cent stamps, respectively. This shows that $P(12)$, $P(13)$, $P(14)$, and $P(15)$ are true. This completes the basis step.

INDUCTIVE STEP: The inductive hypothesis is the statement that $P(j)$ is true for $12 \leq j \leq k$, where k is an integer with $k \geq 15$. To complete the inductive step, we assume that we can form postage of j cents, where $12 \leq j \leq k$. We need to show that under the assumption that $P(k+1)$ is true, we can also form postage of $k+1$ cents. Using the inductive hypothesis, we can assume that $P(k-3)$ is true because $k-3 \geq 12$, that is, we can form postage of $k-3$ cents using just 4-cent and 5-cent stamps. To form postage of $k+1$ cents, we need only add another 4-cent stamp to the stamps we used to form postage of $k-3$ cents. That is, we have shown that if the inductive hypothesis is true, then $P(k+1)$ is also true. This completes the inductive step.

Because we have completed the basis step and the inductive step of a strong induction proof, we know by strong induction that $P(n)$ is true for all integers n with $n \geq 12$. That is, we know that every postage of n cents, where n is at least 12, can be formed using 4-cent and 5-cent stamps. This finishes the proof by strong induction.

(There are other ways to approach this problem besides those described here. Can you find a solution that does not use mathematical induction?) ◀

5.2.4 Using Strong Induction in Computational Geometry

Our next example of strong induction will come from **computational geometry**, the part of discrete mathematics that studies computational problems involving geometric objects. Computational geometry is used extensively in computer graphics, computer games, robotics, scientific calculations, and a vast array of other areas. Before we can present this result, we introduce some terminology, possibly familiar from earlier studies in geometry.

A **polygon** is a closed geometric figure consisting of a sequence of line segments $s_1, s_2, \dots, s_n$, called **sides**. Each pair of consecutive sides, s_i and s_{i+1} , $i = 1, 2, \dots, n-1$, as well as the last side s_n and the first side s_1 , of the polygon meet at a common endpoint, called a **vertex**. A polygon is called **simple** if no two nonconsecutive sides intersect. Every simple polygon divides the plane into two regions: its **interior**, consisting of the points inside the curve, and its **exterior**, consisting of the points outside the curve. This last fact is surprisingly complicated to prove. It is a special case of the deceptively simple Jordan curve theorem, an important result with a rich history, which tells us that every simple curve divides the plane into two regions; see [Or00], for example.

A polygon is called **convex** if every line segment connecting two points in the interior of the polygon lies entirely inside the polygon. (A polygon that is not convex is said to be **nonconvex**.) Figure 1 displays some polygons; polygons (a) and (b) are convex, but polygons (c) and (d) are not. A **diagonal** of a simple polygon is a line segment connecting two nonconsecutive vertices of the polygon, and a diagonal is called an **interior diagonal** if it lies entirely inside the polygon, except for its endpoints. For example, in polygon (d), the line segment connecting a and f is an interior diagonal, but the line segment connecting a and d is a diagonal that is not an interior diagonal.

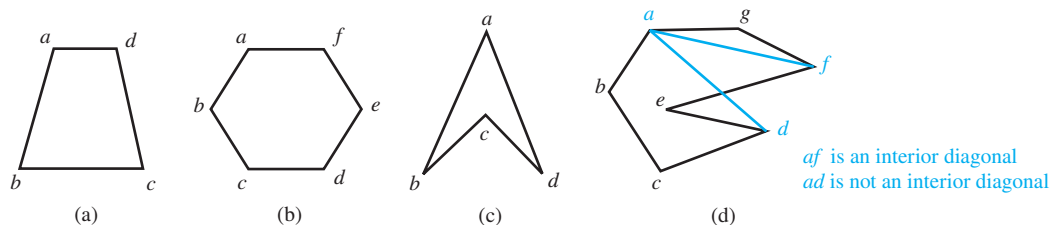
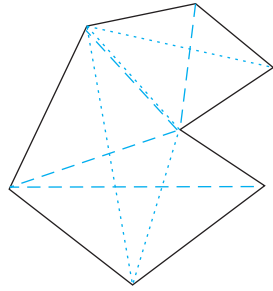


FIGURE 1 Convex and nonconvex polygons.



Two different triangulations of a simple polygon with seven sides into five triangles, shown with dotted lines and with dashed lines, respectively

FIGURE 2 Triangulations of a polygon.

One of the most basic operations of computational geometry involves dividing a simple polygon into triangles by adding nonintersecting diagonals. This process is called **triangulation**. Note that a simple polygon can have many different triangulations, as shown in Figure 2. Perhaps the most basic fact in computational geometry is that it is possible to triangulate every simple polygon, as we state in Theorem 1. Furthermore, this theorem tells us that every triangulation of a simple polygon with n sides includes $n - 2$ triangles.

THEOREM 1

A simple polygon with n sides, where n is an integer with $n \geq 3$, can be triangulated into $n - 2$ triangles.

It seems obvious that we should be able to triangulate a simple polygon by successively adding interior diagonals. Consequently, a proof by strong induction seems promising. However, such a proof requires this crucial lemma.

LEMMA 1

Every simple polygon with at least four sides has an interior diagonal.

Although Lemma 1 seems particularly simple, it is surprisingly tricky to prove. In fact, as recently as 30 years ago, a variety of incorrect proofs thought to be correct were commonly seen in books and articles. We defer the proof of Lemma 1 until after we prove Theorem 1. It is not uncommon to prove a theorem pending the later proof of an important lemma.

Proof (of Theorem 1): We will prove this result using strong induction. Let $T(n)$ be the statement that every simple polygon with n sides can be triangulated into $n - 2$ triangles.

BASIS STEP: $T(3)$ is true because a simple polygon with three sides is a triangle. We do not need to add any diagonals to triangulate a triangle; it is already triangulated into one triangle, itself. Consequently, every simple polygon with $n = 3$ has can be triangulated into $n - 2 = 3 - 2 = 1$ triangle.

INDUCTIVE STEP: For the inductive hypothesis, we assume that $T(j)$ is true for all integers j with $3 \leq j \leq k$. That is, we assume that we can triangulate a simple polygon with j sides into $j - 2$ triangles whenever $3 \leq j \leq k$. To complete the inductive step, we must show that when we assume the inductive hypothesis, $P(k + 1)$ is true, that is, that every simple polygon with $k + 1$ sides can be triangulated into $(k + 1) - 2 = k - 1$ triangles.

So, suppose that we have a simple polygon P with $k + 1$ sides. Because $k + 1 \geq 4$, Lemma 1 tells us that P has an interior diagonal ab . Now, ab splits P into two simple polygons Q , with s sides, and R , with t sides. The sides of Q and R are the sides of P , together with the side ab , which is a side of both Q and R . Note that $3 \leq s \leq k$ and $3 \leq t \leq k$ because both Q and R have at least one fewer side than P does (after all, each of these is formed from P by deleting at least two sides and replacing these sides by the diagonal ab). Furthermore, the number of sides of P is two less than the sum of the numbers of sides of Q and the number of sides

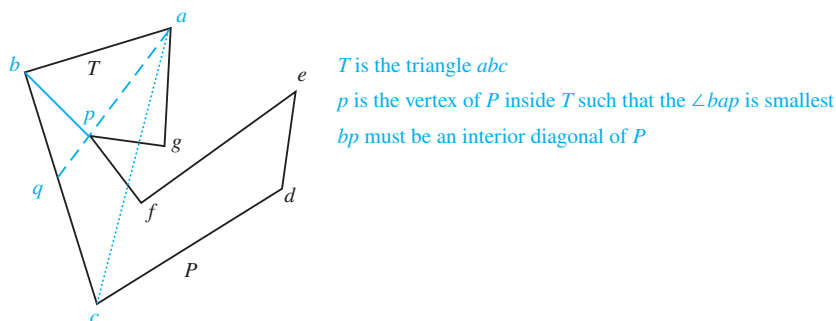


FIGURE 3 Constructing an interior diagonal of a simple polygon.

of R , because each side of P is a side of either Q or of R , but not both, and the diagonal ab is a side of both Q and R , but not P . That is, $k + 1 = s + t - 2$.

We now use the inductive hypothesis. Because both $3 \leq s \leq k$ and $3 \leq t \leq k$, by the inductive hypothesis we can triangulate Q and R into $s - 2$ and $t - 2$ triangles, respectively. Next, note that these triangulations together produce a triangulation of P . (Each diagonal added to triangulate one of these smaller polygons is also a diagonal of P .) Consequently, we can triangulate P into a total of $(s - 2) + (t - 2) = s + t - 4 = (k + 1) - 2$ triangles. This completes the proof by strong induction. That is, we have shown that every simple polygon with n sides, where $n \geq 3$, can be triangulated into $n - 2$ triangles. $\triangleleft$

We now return to our proof of Lemma 1. We present a proof published by Chung-Wu Ho [Ho75]. Note that although this proof may be omitted without loss of continuity, it does provide a correct proof of a result proved incorrectly by many mathematicians.



Proof: Suppose that P is a simple polygon drawn in the plane. Furthermore, suppose that b is the point of P or in the interior of P with the least y -coordinate among the vertices with the smallest x -coordinate. Then b must be a vertex of P , for if it is an interior point, there would have to be a vertex of P with a smaller x -coordinate. Two other vertices each share an edge with b , say a and c . It follows that the angle in the interior of P formed by ab and bc must be less than 180 degrees (otherwise, there would be points of P with smaller x -coordinates than b).

Now let T be the triangle $\triangle abc$. If there are no vertices of P on or inside T , we can connect a and c to obtain an interior diagonal. On the other hand, if there are vertices of P inside T , we will find a vertex p of P on or inside T such that bp is an interior diagonal. (This is the tricky part. Ho noted that in many published proofs of this lemma a vertex p was found such that bp was not necessarily an interior diagonal of P . See Exercise 21.) The key is to select a vertex p such that the angle $\angle bap$ is smallest. To see this, note that the ray starting at a and passing through p hits the line segment bc at a point, say q . It then follows that the triangle $\triangle baq$ cannot contain any vertices of P in its interior. Hence, we can connect b and p to produce an interior diagonal of P . Locating this vertex p is illustrated in Figure 3. $\triangleleft$

5.2.5 Proofs Using the Well-Ordering Property

The validity of both the principle of mathematical induction and strong induction follows from a fundamental axiom of the set of integers, the **well-ordering property** (see Appendix 1). The well-ordering property states that every nonempty set of nonnegative integers has a least element. We will show how the well-ordering property can be used directly in proofs. Furthermore, it can be shown (see Exercises 41, 42, and 43) that the well-ordering property, the principle of mathematical induction, and strong induction are all equivalent. That is, the validity of each of these three proof techniques implies the validity of the other two techniques. In Section 5.1 we

showed that the principle of mathematical induction follows from the well-ordering property. The other parts of this equivalence are left as Exercises 31, 42, and 43.

THE WELL-ORDERING PROPERTY Every nonempty set of nonnegative integers has a least element.

The well-ordering property can be used directly in proofs, as Example 5 illustrates.

EXAMPLE 5 Use the well-ordering property to prove the division algorithm. Recall that the division algorithm states that if a is an integer and d is a positive integer, then there are unique integers q and r with $0 \leq r < d$ and $a = dq + r$.

*Extra
Examples* 

Solution: Let S be the set of nonnegative integers of the form $a - dq$, where q is an integer. This set is nonempty because $-dq$ can be made as large as desired (taking q to be a negative integer with large absolute value). By the well-ordering property, S has a least element $r = a - dq_0$.

The integer r is nonnegative. It is also the case that $r < d$. If it were not, then there would be a smaller nonnegative element in S , namely, $a - d(q_0 + 1)$. To see this, suppose that $r \geq d$. Because $a = dq_0 + r$, it follows that $a - d(q_0 + 1) = (a - dq_0) - d = r - d \geq 0$. Consequently, there are integers q and r with $0 \leq r < d$. The proof that q and r are unique is left as Exercise 37. 

EXAMPLE 6 In a round-robin tournament every player plays every other player exactly once and each match has a winner and a loser. We say that the players $p_1, p_2, \dots, p_m$ form a *cycle* if p_1 beats p_2 , p_2 beats p_3 , $\dots$, p_{m-1} beats p_m , and p_m beats p_1 . Use the well-ordering property to show that if there is a cycle of length m ($m \geq 3$) among the players in a round-robin tournament, there must be a cycle of three of these players.

Solution: We assume that there is no cycle of three players. Because there is at least one cycle in the round-robin tournament, the set of all positive integers n for which there is a cycle of length n is nonempty. By the well-ordering property, this set of positive integers has a least element k , which by assumption must be greater than three. Consequently, there exists a cycle of players $p_1, p_2, p_3, \dots, p_k$ and no shorter cycle exists.

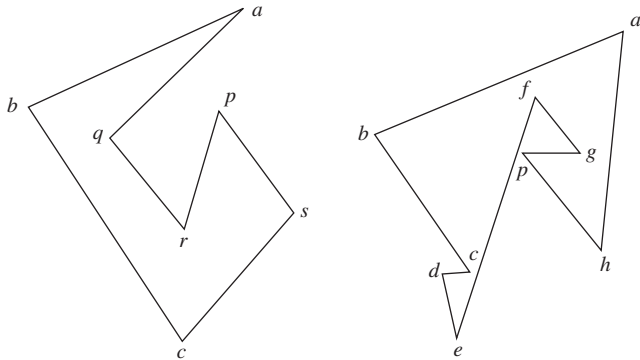
Because there is no cycle of three players, we know that $k > 3$. Consider the first three elements of this cycle, p_1, p_2 , and p_3 . There are two possible outcomes of the match between p_1 and p_3 . If p_3 beats p_1 , it follows that p_1, p_2, p_3 is a cycle of length three, contradicting our assumption that there is no cycle of three players. Consequently, it must be the case that p_1 beats p_3 . This means that we can omit p_2 from the cycle $p_1, p_2, p_3, \dots, p_k$ to obtain the cycle $p_1, p_3, p_4, \dots, p_k$ of length $k - 1$, contradicting the assumption that the smallest cycle has length k . We conclude that there must be a cycle of length three. 

Exercises

- Use strong induction to show that if you can run one mile or two miles, and if you can always run two more miles once you have run a specified number of miles, then you can run any number of miles.
- Use strong induction to show that all dominoes fall in an infinite arrangement of dominoes if you know that the first three dominoes fall, and that when a domino falls, the domino three farther down in the arrangement also falls.
- Let $P(n)$ be the statement that a postage of n cents can be formed using just 3-cent stamps and 5-cent stamps. The parts of this exercise outline a strong induction proof that $P(n)$ is true for all integers $n \geq 8$.
 - Show that the statements $P(8)$, $P(9)$, and $P(10)$ are true, completing the basis step of a proof by strong induction that $P(n)$ is true for all integers $n \geq 8$.
 - What is the inductive hypothesis of a proof by strong induction that $P(n)$ is true for all integers $n \geq 8$?
 - What do you need to prove in the inductive step of a proof by strong induction that $P(n)$ is true for all integers $n \geq 8$?
 - Complete the inductive step for $k \geq 10$.
 - Explain why these steps show that $P(n)$ is true whenever $n \geq 8$.

4. Let $P(n)$ be the statement that a postage of n cents can be formed using just 4-cent stamps and 7-cent stamps. The parts of this exercise outline a strong induction proof that $P(n)$ is true for all integers $n \geq 18$.
 - a) Show that the statements $P(18)$, $P(19)$, $P(20)$, and $P(21)$ are true, completing the basis step of a proof by strong induction that $P(n)$ is true for all integers $n \geq 18$.
 - b) What is the inductive hypothesis of a proof by strong induction that $P(n)$ is true for all integers $n \geq 18$?
 - c) What do you need to prove in the inductive step of a proof that $P(n)$ is true for all integers $n \geq 18$?
 - d) Complete the inductive step for $k \geq 21$.
 - e) Explain why these steps show that $P(n)$ is true for all integers $n \geq 18$.
5. a) Determine which amounts of postage can be formed using just 4-cent and 11-cent stamps.
 b) Prove your answer to (a) using the principle of mathematical induction. Be sure to state explicitly your inductive hypothesis in the inductive step.
 c) Prove your answer to (a) using strong induction. How does the inductive hypothesis in this proof differ from that in the inductive hypothesis for a proof using mathematical induction?
6. a) Determine which amounts of postage can be formed using just 3-cent and 10-cent stamps.
 b) Prove your answer to (a) using the principle of mathematical induction. Be sure to state explicitly your inductive hypothesis in the inductive step.
 c) Prove your answer to (a) using strong induction. How does the inductive hypothesis in this proof differ from that in the inductive hypothesis for a proof using mathematical induction?
7. Which amounts of money can be formed using just two-dollar bills and five-dollar bills? Prove your answer using strong induction.
8. Suppose that a store offers gift certificates in denominations of 25 dollars and 40 dollars. Determine the possible total amounts you can form using these gift certificates. Prove your answer using strong induction.
- *9. Use strong induction to prove that $\sqrt{2}$ is irrational. [Hint: Let $P(n)$ be the statement that $\sqrt{2} \neq n/b$ for any positive integer b .]
10. Assume that a chocolate bar consists of n squares arranged in a rectangular pattern. The entire bar, or any smaller rectangular piece of the bar, can be broken along a vertical or a horizontal line separating the squares. Assuming that only one piece can be broken at a time, determine how many breaks you must successively make to break the bar into n separate squares. Use strong induction to prove your answer.
11. Consider this variation of the game of Nim. The game begins with n matches. Two players take turns removing matches, one, two, or three at a time. The player removing the last match loses. Use strong induction to show that if each player plays the best strategy possible, the first player wins if $n = 4j$, $4j + 2$, or $4j + 3$ for some nonnegative integer j and the second player wins in the remaining case when $n = 4j + 1$ for some nonnegative integer j .
12. Use strong induction to show that every positive integer can be written as a sum of distinct powers of two, that is, as a sum of a subset of the integers $2^0 = 1$, $2^1 = 2$, $2^2 = 4$, and so on. [Hint: For the inductive step, separately consider the case where $k + 1$ is even and where it is odd. When it is even, note that $(k + 1)/2$ is an integer.]
- *13. A jigsaw puzzle is put together by successively joining pieces that fit together into blocks. A move is made each time a piece is added to a block, or when two blocks are joined. Use strong induction to prove that no matter how the moves are carried out, exactly $n - 1$ moves are required to assemble a puzzle with n pieces.
14. Suppose you begin with a pile of n stones and split this pile into n piles of one stone each by successively splitting a pile of stones into two smaller piles. Each time you split a pile you multiply the number of stones in each of the two smaller piles you form, so that if these piles have r and s stones in them, respectively, you compute rs . Show that no matter how you split the piles, the sum of the products computed at each step equals $n(n - 1)/2$.
15. Prove that the first player has a winning strategy for the game of Chomp, introduced in Example 12 in Section 1.8, if the initial board is square. [Hint: Use strong induction to show that this strategy works. For the first move, the first player chomps all cookies except those in the left and top edges. On subsequent moves, after the second player has chomped cookies on either the top or left edge, the first player chomps cookies in the same relative positions in the left or top edge, respectively.]
- *16. Prove that the first player has a winning strategy for the game of Chomp, introduced in Example 12 in Section 1.8, if the initial board is two squares wide, that is, a $2 \times n$ board. [Hint: Use strong induction. The first move of the first player should be to chomp the cookie in the bottom row at the far right.]
17. Use strong induction to show that if a simple polygon with at least four sides is triangulated, then at least two of the triangles in the triangulation have two sides that border the exterior of the polygon.
- *18. Use strong induction to show that when a simple polygon P with consecutive vertices $v_1, v_2, \dots, v_n$ is triangulated into $n - 2$ triangles, the $n - 2$ triangles can be numbered $1, 2, \dots, n - 2$ so that v_i is a vertex of triangle i for $i = 1, 2, \dots, n - 2$.
- *19. **Pick's theorem** says that the area of a simple polygon P in the plane with vertices that are all lattice points (that is, points with integer coordinates) equals $I(P) + B(P)/2 - 1$, where $I(P)$ and $B(P)$ are the number of lattice points in the interior of P and on the boundary of P , respectively. Use strong induction on the number of vertices of P to prove Pick's theorem. [Hint: For the basis step, first prove the theorem for rectangles, then for right triangles, and finally for all triangles by noting that the area of a triangle is the area of a larger rectangle containing it with the areas of at most three triangles subtracted. For the inductive step, take advantage of Lemma 1.]

- **20.** Suppose that P is a simple polygon with vertices $v_1, v_2, \dots, v_n$ listed so that consecutive vertices are connected by an edge, and v_1 and v_n are connected by an edge. A vertex v_i is called an **ear** if the line segment connecting the two vertices adjacent to v_i is an interior diagonal of the simple polygon. Two ears v_i and v_j are called **nonoverlapping** if the interiors of the triangles with vertices v_i and its two adjacent vertices and v_j and its two adjacent vertices do not intersect. Prove that every simple polygon with at least four vertices has at least two nonoverlapping ears.
- 21.** In the proof of Lemma 1 we mentioned that many incorrect methods for finding a vertex p such that the line segment bp is an interior diagonal of P have been published. This exercise presents some of the incorrect ways p has been chosen in these proofs. Show, by considering one of the polygons drawn here, that for each of these choices of p , the line segment bp is not necessarily an interior diagonal of P .
- p is the vertex of P such that the angle $\angle abp$ is smallest.
 - p is the vertex of P with the least x -coordinate (other than b).
 - p is the vertex of P that is closest to b .



Exercises 22 and 23 present examples that show inductive loading can be used to prove results in computational geometry.

- *22.** Let $P(n)$ be the statement that when nonintersecting diagonals are drawn inside a convex polygon with n sides, at least two vertices of the polygon are not endpoints of any of these diagonals.
- Show that when we attempt to prove $P(n)$ for all integers n with $n \geq 3$ using strong induction, the inductive step does not go through.
 - Show that we can prove that $P(n)$ is true for all integers n with $n \geq 3$ by proving by strong induction the stronger assertion $Q(n)$, for $n \geq 4$, where $Q(n)$ states that whenever nonintersecting diagonals are drawn inside a convex polygon with n sides, at least two *nonadjacent* vertices are not endpoints of any of these diagonals.
- 23.** Let $E(n)$ be the statement that in a triangulation of a simple polygon with n sides, at least one of the triangles in the triangulation has two sides bordering the exterior of the polygon.
- Explain where a proof using strong induction that $E(n)$ is true for all integers $n \geq 4$ runs into difficulties.
 - Show that we can prove that $E(n)$ is true for all integers $n \geq 4$ by proving by strong induction the stronger statement $T(n)$ for all integers $n \geq 4$, which states that in every triangulation of a simple polygon, at least two of the triangles in the triangulation have two sides bordering the exterior of the polygon.
- *24.** A stable assignment, defined in the preamble to Exercise 64 in Section 3.1, is called **optimal for suitors** if no stable assignment exists in which a suitor is paired with a suitee whom this suitor prefers to the person to whom this suitor is paired in this stable assignment. Use strong induction to show that the deferred acceptance algorithm produces a stable assignment that is optimal for suitors.
- 25.** Suppose that $P(n)$ is a propositional function. Determine for which positive integers n the statement $P(n)$ must be true, and justify your answer, if
- $P(1)$ is true; for all positive integers n , if $P(n)$ is true, then $P(n+2)$ is true.
 - $P(1)$ and $P(2)$ are true; for all positive integers n , if $P(n)$ and $P(n+1)$ are true, then $P(n+2)$ is true.
 - $P(1)$ is true; for all positive integers n , if $P(n)$ is true, then $P(2n)$ is true.
 - $P(1)$ is true; for all positive integers n , if $P(n)$ is true, then $P(n+1)$ is true.
- 26.** Suppose that $P(n)$ is a propositional function. Determine for which nonnegative integers n the statement $P(n)$ must be true if
- $P(0)$ is true; for all nonnegative integers n , if $P(n)$ is true, then $P(n+2)$ is true.
 - $P(0)$ is true; for all nonnegative integers n , if $P(n)$ is true, then $P(n+3)$ is true.
 - $P(0)$ and $P(1)$ are true; for all nonnegative integers n , if $P(n)$ and $P(n+1)$ are true, then $P(n+2)$ is true.
 - $P(0)$ is true; for all nonnegative integers n , if $P(n)$ is true, then $P(n+2)$ and $P(n+3)$ are true.
- 27.** Show that if the statement $P(n)$ is true for infinitely many positive integers n and $P(n+1) \rightarrow P(n)$ is true for all positive integers n , then $P(n)$ is true for all positive integers n .
- 28.** Let b be a fixed integer and j a fixed positive integer. Show that if $P(b), P(b+1), \dots, P(b+j)$ are true and $[P(b) \wedge P(b+1) \wedge \dots \wedge P(k)] \rightarrow P(k+1)$ is true for every integer $k \geq b+j$, then $P(n)$ is true for all integers n with $n \geq b$.
- 29.** What is wrong with this “proof” by strong induction?
- “Theorem” For every nonnegative integer n , $5n = 0$.
- Basis Step: $5 \cdot 0 = 0$.
- Inductive Step: Suppose that $5j = 0$ for all nonnegative integers j with $0 \leq j \leq k$. Write $k+1 = i+j$, where i and j are natural numbers less than $k+1$. By the inductive hypothesis, $5(k+1) = 5(i+j) = 5i + 5j = 0 + 0 = 0$.

- *30. Find the flaw with the following “proof” that $a^n = 1$ for all nonnegative integers n , whenever a is a nonzero real number.

Basis Step: $a^0 = 1$ is true by the definition of a^0 .

Inductive Step: Assume that $a^j = 1$ for all nonnegative integers j with $j \leq k$. Then note that

$$a^{k+1} = \frac{a^k \cdot a^k}{a^{k-1}} = \frac{1 \cdot 1}{1} = 1.$$

- *31. Show that strong induction is a valid method of proof by showing that it follows from the well-ordering property.

32. Find the flaw with the following “proof” that every postage of three cents or more can be formed using just 3-cent and 4-cent stamps.

Basis Step: We can form postage of three cents with a single 3-cent stamp and we can form postage of four cents using a single 4-cent stamp.

Inductive Step: Assume that we can form postage of j cents for all nonnegative integers j with $j \leq k$ using just 3-cent and 4-cent stamps. We can then form postage of $k + 1$ cents by replacing one 3-cent stamp with a 4-cent stamp or by replacing two 4-cent stamps by three 3-cent stamps.

33. Show that we can prove that $P(n, k)$ is true for all pairs of positive integers n and k if we show

- a) $P(1, 1)$ is true and $P(n, k) \rightarrow [P(n + 1, k) \wedge P(n, k + 1)]$ is true for all positive integers n and k .
- b) $P(1, k)$ is true for all positive integers k , and $P(n, k) \rightarrow P(n + 1, k)$ is true for all positive integers n and k .
- c) $P(n, 1)$ is true for all positive integers n , and $P(n, k) \rightarrow P(n, k + 1)$ is true for all positive integers n and k .

34. Prove that $\sum_{j=1}^n j(j+1)(j+2) \cdots (j+k-1) = n(n+1)(n+2) \cdots (n+k)/(k+1)$ for all positive integers k and n . [Hint: Use a technique from Exercise 33.]

- *35. Show that if $a_1, a_2, \dots, a_n$ are n distinct real numbers, exactly $n - 1$ multiplications are used to compute the product of these n numbers no matter how parentheses are inserted into their product. [Hint: Use strong induction and consider the last multiplication.]
- *36. The well-ordering property can be used to show that there is a unique greatest common divisor of two positive integers. Let a and b be positive integers, and let S be

the set of positive integers of the form $as + bt$, where s and t are integers.

- a) Show that S is nonempty.
- b) Use the well-ordering property to show that S has a smallest element c .
- c) Show that if d is a common divisor of a and b , then d is a divisor of c .
- d) Show that $c \mid a$ and $c \mid b$. [Hint: First, assume that $c \nmid a$. Then $a = qc + r$, where $0 < r < c$. Show that $r \in S$, contradicting the choice of c .]
- e) Conclude from (c) and (d) that the greatest common divisor of a and b exists. Finish the proof by showing that this greatest common divisor is unique.

37. Let a be an integer and d be a positive integer. Show that the integers q and r with $a = dq + r$ and $0 \leq r < d$, which were shown to exist in Example 5, are unique.

38. Use mathematical induction to show that a rectangular checkerboard with an even number of cells and two squares missing, one white and one black, can be covered by dominoes.

- **39. Can you use the well-ordering property to prove the statement: “Every positive integer can be described using no more than fifteen English words”? Assume the words come from a particular dictionary of English. [Hint: Suppose that there are positive integers that cannot be described using no more than fifteen English words. By well ordering, the smallest positive integer that cannot be described using no more than fifteen English words would then exist.]

40. Use the well-ordering property to show that if x and y are real numbers with $x < y$, then there is a rational number r with $x < r < y$. [Hint: Use the Archimedean property, given in Appendix 1, to find a positive integer A with $A > 1/(y - x)$. Then show that there is a rational number r with denominator A between x and y by looking at the numbers $\lfloor x \rfloor + j/A$, where j is a positive integer.]

- *41. Show that the well-ordering property can be proved when the principle of mathematical induction is taken as an axiom.
- *42. Show that the principle of mathematical induction and strong induction are equivalent; that is, each can be shown to be valid from the other.
- *43. Show that we can prove the well-ordering property when we take strong induction as an axiom instead of taking the well-ordering property as an axiom.

5.3

Recursive Definitions and Structural Induction

5.3.1 Introduction

Sometimes it is difficult to define an object explicitly. However, it may be easy to define this object in terms of itself. This process is called **recursion**. For instance, the picture shown in Figure 1 is produced recursively. First, an original picture is given. Then a process of successively superimposing centered smaller pictures on top of the previous pictures is carried out.

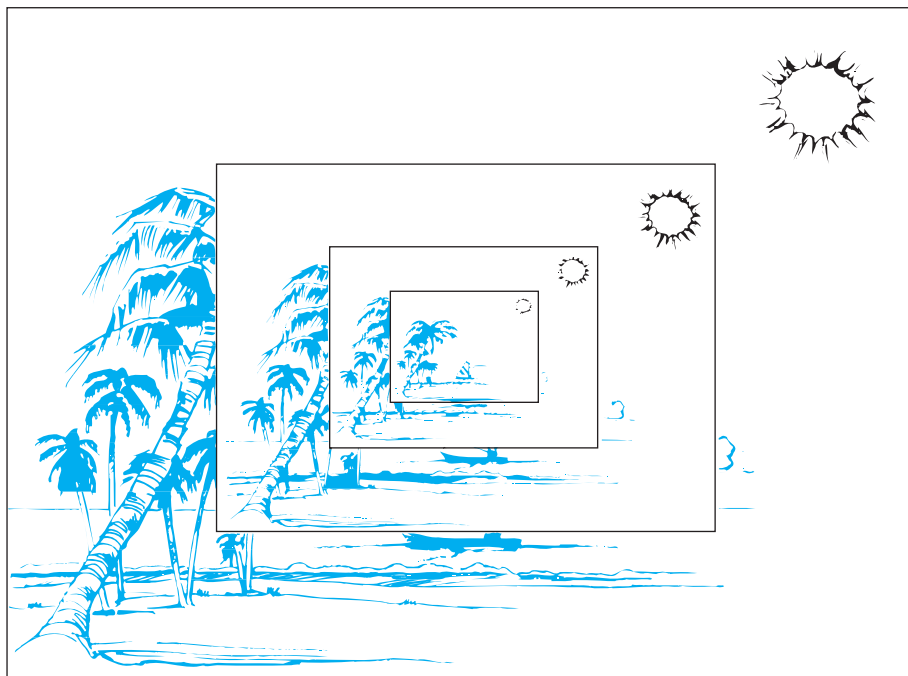


FIGURE 1 A recursively defined picture.

We can use recursion to define sequences, functions, and sets. In Section 2.4, and in most beginning mathematics courses, the terms of a sequence are specified using an explicit formula. For instance, the sequence of powers of 2 is given by $a_n = 2^n$ for $n = 0, 1, 2, \dots$. Recall from Section 2.4 that we can also define a sequence recursively by specifying how terms of the sequence are found from previous terms. The sequence of powers of 2 can also be defined by giving the first term of the sequence, namely, $a_0 = 1$, and a rule for finding a term of the sequence from the previous one, namely, $a_{n+1} = 2a_n$ for $n = 0, 1, 2, \dots$. When we define a sequence recursively by specifying how terms of the sequence are found from previous terms, we can use induction to prove results about the sequence.

When we define a set recursively, we specify some initial elements in a basis step and provide a rule for constructing new elements from those we already have in the recursive step. To prove results about recursively defined sets we use a method called *structural induction*.

5.3.2 Recursively Defined Functions

We use two steps to define a function with the set of nonnegative integers as its domain:

BASIS STEP: Specify the value of the function at zero.



RECURSIVE STEP: Give a rule for finding its value at an integer from its values at smaller integers.

Such a definition is called a **recursive** or **inductive definition**. Note that a function $f(n)$ from the set of nonnegative integers to the set of real numbers is the same as a sequence $a_0, a_1, \dots$, where a_i is a real number for every nonnegative integer i . So, defining a real-valued sequence $a_0, a_1, \dots$ using a recurrence relation, as was done in Section 2.4, is the same as defining a function from the set of nonnegative integers to the set of real numbers.

EXAMPLE 1 Suppose that f is defined recursively by

*Extra
Examples* 

$$\begin{aligned} f(0) &= 3, \\ f(n+1) &= 2f(n) + 3. \end{aligned}$$

Find $f(1)$, $f(2)$, $f(3)$, and $f(4)$.

Solution: From the recursive definition it follows that

$$\begin{aligned} f(1) &= 2f(0) + 3 = 2 \cdot 3 + 3 = 9, \\ f(2) &= 2f(1) + 3 = 2 \cdot 9 + 3 = 21, \\ f(3) &= 2f(2) + 3 = 2 \cdot 21 + 3 = 45, \\ f(4) &= 2f(3) + 3 = 2 \cdot 45 + 3 = 93. \end{aligned}$$


Recursively defined functions are **well defined**. That is, for every positive integer, the value of the function at this integer is determined in an unambiguous way. This means that given any positive integer, we can use the two parts of the definition to find the value of the function at that integer, and that we obtain the same value no matter how we apply the two parts of the definition. This is a consequence of the principle of mathematical induction. (See Exercise 58.) Additional examples of recursive definitions are given in Examples 2 and 3.

EXAMPLE 2 Give a recursive definition of a^n , where a is a nonzero real number and n is a nonnegative integer.

Solution: The recursive definition contains two parts. First a^0 is specified, namely, $a^0 = 1$. Then the rule for finding a^{n+1} from a^n , namely, $a^{n+1} = a \cdot a^n$, for $n = 0, 1, 2, 3, \dots$, is given. These two equations uniquely define a^n for all nonnegative integers n .



EXAMPLE 3 Give a recursive definition of

$$\sum_{k=0}^n a_k.$$

Solution: The first part of the recursive definition is

$$\sum_{k=0}^0 a_k = a_0.$$

The second part is

$$\sum_{k=0}^{n+1} a_k = \left(\sum_{k=0}^n a_k \right) + a_{n+1}.$$


In some recursive definitions of functions, the values of the function at the first k positive integers are specified, and a rule is given for determining the value of the function at larger integers from its values at some or all of the preceding k integers. That recursive definitions defined in this way produce well-defined functions follows from strong induction (see Exercise 59).

Recall from Section 2.4 that the Fibonacci numbers, $f_0, f_1, f_2, \dots$, are defined by the equations $f_0 = 0, f_1 = 1$, and

Links 

$$f_n = f_{n-1} + f_{n-2}$$

for $n = 2, 3, 4, \dots$ [We can think of the Fibonacci number f_n either as the n th term of the sequence of Fibonacci numbers $f_0, f_1, \dots$ or as the value at the integer n of a function $f(n)$.]

We can use the recursive definition of the Fibonacci numbers to prove many properties of these numbers. We give one such property in Example 4.

EXAMPLE 4 Show that whenever $n \geq 3, f_n > \alpha^{n-2}$, where $\alpha = (1 + \sqrt{5})/2$.

Extra Examples 

Solution: We can use strong induction to prove this inequality. Let $P(n)$ be the statement $f_n > \alpha^{n-2}$. We want to show that $P(n)$ is true whenever n is an integer greater than or equal to 3.

BASIS STEP: First, note that

$$\alpha < 2 = f_3, \quad \alpha^2 = (3 + \sqrt{5})/2 < 3 = f_4,$$

so $P(3)$ and $P(4)$ are true.

INDUCTIVE STEP: Assume that $P(j)$ is true, namely, that $f_j > \alpha^{j-2}$, for all integers j with $3 \leq j \leq k$, where $k \geq 4$. We must show that $P(k+1)$ is true, that is, that $f_{k+1} > \alpha^{k-1}$. Because α is a solution of $x^2 - x - 1 = 0$ (as the quadratic formula shows), it follows that $\alpha^2 = \alpha + 1$. Therefore,

$$\alpha^{k-1} = \alpha^2 \cdot \alpha^{k-3} = (\alpha + 1)\alpha^{k-3} = \alpha \cdot \alpha^{k-3} + 1 \cdot \alpha^{k-3} = \alpha^{k-2} + \alpha^{k-3}.$$

By the inductive hypothesis, because $k \geq 4$, we have

$$f_{k-1} > \alpha^{k-3}, \quad f_k > \alpha^{k-2}.$$

Therefore, it follows that

$$f_{k+1} = f_k + f_{k-1} > \alpha^{k-2} + \alpha^{k-3} = \alpha^{k-1}.$$

Hence, $P(k+1)$ is true. This completes the proof. 

Remark: The inductive step of the proof by strong induction in Example 4 shows that whenever $k \geq 4, P(k+1)$ follows from the assumption that $P(j)$ is true for $3 \leq j \leq k$. Hence, the inductive step does *not* show that $P(3) \rightarrow P(4)$. Therefore, we had to show that $P(4)$ is true separately.

We can now show that the Euclidean algorithm, introduced in Section 4.3, uses $O(\log b)$ divisions to find the greatest common divisor of the positive integers a and b , where $a \geq b$.

THEOREM 1

LAMÉ'S THEOREM Let a and b be positive integers with $a \geq b$. Then the number of divisions used by the Euclidean algorithm to find $\gcd(a, b)$ is less than or equal to five times the number of decimal digits in b .

Proof: Recall that when the Euclidean algorithm is applied to find $\gcd(a, b)$ with $a \geq b$, this sequence of equations (where $a = r_0$ and $b = r_1$) is obtained.

$$\begin{aligned} r_0 &= r_1 q_1 + r_2 & 0 \leq r_2 < r_1, \\ r_1 &= r_2 q_2 + r_3 & 0 \leq r_3 < r_2, \\ &\vdots \\ r_{n-2} &= r_{n-1} q_{n-1} + r_n & 0 \leq r_n < r_{n-1}, \\ r_{n-1} &= r_n q_n. \end{aligned}$$

Here n divisions have been used to find $r_n = \gcd(a, b)$. Note that the quotients $q_1, q_2, \dots, q_{n-1}$ are all at least 1. Moreover, $q_n \geq 2$, because $r_n < r_{n-1}$. This implies that

$$\begin{aligned} r_n &\geq 1 = f_2, \\ r_{n-1} &\geq 2r_n \geq 2f_2 = f_3, \\ r_{n-2} &\geq r_{n-1} + r_n \geq f_3 + f_2 = f_4, \\ &\vdots \\ r_2 &\geq r_3 + r_4 \geq f_{n-1} + f_{n-2} = f_n, \\ b = r_1 &\geq r_2 + r_3 \geq f_n + f_{n-1} = f_{n+1}. \end{aligned}$$

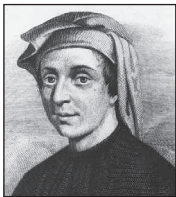
It follows that if n divisions are used by the Euclidean algorithm to find $\gcd(a, b)$ with $a \geq b$, then $b \geq f_{n+1}$. By Example 4 we know that $f_{n+1} > \alpha^{n-1}$ for $n > 2$, where $\alpha = (1 + \sqrt{5})/2$. Therefore, it follows that $b > \alpha^{n-1}$. Furthermore, because $\log_{10} \alpha \approx 0.208 > 1/5$, we see that

$$\log_{10} b > (n-1) \log_{10} \alpha > (n-1)/5.$$

Hence, $n-1 < 5 \cdot \log_{10} b$. Now suppose that b has k decimal digits. Then $b < 10^k$ and $\log_{10} b < k$. It follows that $n-1 < 5k$, and because k is an integer, it follows that $n \leq 5k$. This finishes the proof. $\triangleleft$

Because the number of decimal digits in b , which equals $\lfloor \log_{10} b \rfloor + 1$, is less than or equal to $\log_{10} b + 1$, Theorem 1 tells us that the number of divisions required to find $\gcd(a, b)$ with

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FIBONACCI (1170–1250) Fibonacci (short for *filius Bonacci*, or “son of Bonacci”) was also known as Leonardo of Pisa. He was born in the Italian commercial center of Pisa. Fibonacci was a merchant who traveled extensively throughout the Mideast, where he came into contact with Arabian mathematics. In his book *Liber Abaci*, Fibonacci introduced the European world to Arabic notation for numerals and algorithms for arithmetic. It was in this book that his well known rabbit problem (described in Section 8.1) appeared. Fibonacci also wrote books on geometry and trigonometry and on Diophantine equations, which involve finding integer solutions to equations.

$a > b$ is less than or equal to $5(\log_{10} b + 1)$. Because $5(\log_{10} b + 1)$ is $O(\log b)$, we see that $O(\log b)$ divisions are used by the Euclidean algorithm to find $\gcd(a, b)$ whenever $a > b$.

5.3.3 Recursively Defined Sets and Structures

Assessment

We have explored how functions can be defined recursively. We now turn our attention to how sets can be defined recursively. Just as in the recursive definition of functions, recursive definitions of sets have two parts, a **basis step** and a **recursive step**. In the basis step, an initial collection of elements is specified. In the recursive step, rules for forming new elements in the set from those already known to be in the set are provided. Recursive definitions may also include an **exclusion rule**, which specifies that a recursively defined set contains nothing other than those elements specified in the basis step or generated by applications of the recursive step. In our discussions, we will always tacitly assume that the exclusion rule holds and no element belongs to a recursively defined set unless it is in the initial collection specified in the basis step or can be generated using the recursive step one or more times. Later we will see how we can use a technique known as structural induction to prove results about recursively defined sets.

Examples 5, 6, 8, and 9 illustrate the recursive definition of sets. In each example, we show those elements generated by the first few applications of the recursive step.

EXAMPLE 5 Consider the subset S of the set of integers recursively defined by

BASIS STEP: $3 \in S$.

RECURSIVE STEP: If $x \in S$ and $y \in S$, then $x + y \in S$.

Extra Examples

The new elements found to be in S are 3 by the basis step, $3 + 3 = 6$ at the first application of the recursive step, $3 + 6 = 6 + 3 = 9$ and $6 + 6 = 12$ at the second application of the recursive step, and so on. We will show in Example 10 that S is the set of all positive multiples of 3. ◀

Recursive definitions play an important role in the study of strings. (See Chapter 13 for an introduction to the theory of formal languages, for example.) Recall from Section 2.4 that a string over an alphabet Σ is a finite sequence of symbols from Σ . We can define Σ^* , the set of strings over Σ , recursively, as Definition 1 shows.

Definition 1

The set Σ^* of *strings* over the alphabet Σ is defined recursively by

BASIS STEP: $\lambda \in \Sigma^*$ (where λ is the empty string containing no symbols).

RECURSIVE STEP: If $w \in \Sigma^*$ and $x \in \Sigma$, then $wx \in \Sigma^*$.

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GABRIEL LAMÉ (1795–1870) Gabriel Lamé entered the École Polytechnique in 1813, graduating in 1817. He continued his education at the École des Mines, graduating in 1820.

In 1820 Lamé went to Russia, where he was appointed director of the Schools of Highways and Transportation in St. Petersburg. Not only did he teach, but he also planned roads and bridges while in Russia. He returned to Paris in 1832, where he helped found an engineering firm. However, he soon left the firm, accepting the chair of physics at the École Polytechnique, which he held until 1844. While holding this position, he was active outside academia as an engineering consultant, serving as chief engineer of mines and participating in the building of railways.

Lamé contributed original work to number theory, applied mathematics, and thermodynamics. His best-known work involves the introduction of curvilinear coordinates. His work on number theory includes proving Fermat's last theorem for $n = 7$, as well as providing the upper bound for the number of divisions used by the Euclidean algorithm given in this text.

In the opinion of Gauss, one of the most important mathematicians of all time, Lamé was the foremost French mathematician of his time. However, French mathematicians considered him too practical, whereas French scientists considered him too theoretical.

The basis step of the recursive definition of strings says that the empty string belongs to Σ^* . The recursive step states that new strings are produced by adding a symbol from Σ to the end of strings in Σ^* . At each application of the recursive step, strings containing one additional symbol are generated.

EXAMPLE 6 If $\Sigma = \{0, 1\}$, the strings found to be in Σ^* , the set of all bit strings, are λ , specified to be in Σ^* in the basis step, 0 and 1 formed during the first application of the recursive step, 00, 01, 10, and 11 formed during the second application of the recursive step, and so on. ◀

Recursive definitions can be used to define operations or functions on the elements of recursively defined sets. This is illustrated in Definition 2 of the concatenation of two strings and Example 7 concerning the length of a string.

Definition 2

Two strings can be combined via the operation of *concatenation*. Let Σ be a set of symbols and Σ^* the set of strings formed from symbols in Σ . We can define the concatenation of two strings, denoted by $\cdot$, recursively as follows.

BASIS STEP: If $w \in \Sigma^*$, then $w \cdot \lambda = w$, where λ is the empty string.

RECURSIVE STEP: If $w_1 \in \Sigma^*$ and $w_2 \in \Sigma^*$ and $x \in \Sigma$, then $w_1 \cdot (w_2x) = (w_1 \cdot w_2)x$.

The concatenation of the strings w_1 and w_2 is often written as w_1w_2 rather than $w_1 \cdot w_2$. By repeated application of the recursive definition, it follows that the concatenation of two strings w_1 and w_2 consists of the symbols in w_1 followed by the symbols in w_2 . For instance, the concatenation of $w_1 = abra$ and $w_2 = cadabra$ is $w_1w_2 = abracadabra$.

EXAMPLE 7 Length of a String Give a recursive definition of $l(w)$, the length of the string w .

Solution: The length of a string can be recursively defined by

$$\begin{aligned} l(\lambda) &= 0; \\ l(wx) &= l(w) + 1 \text{ if } w \in \Sigma^* \text{ and } x \in \Sigma. \end{aligned}$$

Another important use of recursive definitions is to define **well-formed formulae** of various types. This is illustrated in Examples 8 and 9.

EXAMPLE 8 Well-Formed Formulae in Propositional Logic We can define the set of well-formed formulae in propositional logic involving **T**, **F**, propositional variables, and operators from the set $\{\neg, \wedge, \vee, \rightarrow, \leftrightarrow\}$.

BASIS STEP: **T**, **F**, and s , where s is a propositional variable, are well-formed formulae.

RECURSIVE STEP: If E and F are well-formed formulae, then $(\neg E)$, $(E \wedge F)$, $(E \vee F)$, $(E \rightarrow F)$, and $(E \leftrightarrow F)$ are well-formed formulae.

For example, by the basis step we know that **T**, **F**, p , and q are well-formed formulae, where p and q are propositional variables. From an initial application of the recursive step, we know that $(p \vee q)$, $(p \rightarrow \mathbf{F})$, $(\mathbf{F} \rightarrow q)$, and $(q \wedge \mathbf{F})$ are well-formed formulae. A second application of the recursive step shows that $((p \vee q) \rightarrow (q \wedge \mathbf{F}))$, $(q \vee (p \vee q))$, and $((p \rightarrow \mathbf{F}) \rightarrow \mathbf{T})$ are well-formed formulae. We leave it to the reader to show that $p\neg \wedge q$, $pq\wedge$, and $\neg \wedge pq$ are *not* well-formed formulae, by showing that none can be obtained using the basis step and one or more applications of the recursive step. ◀

EXAMPLE 9 Well-Formed Formulae of Operators and Operands We can define the set of well-formed formulae consisting of variables, numerals, and operators from the set $\{+, -, *, /, \uparrow\}$ (where $*$ denotes multiplication and $\uparrow$ denotes exponentiation) recursively.

BASIS STEP: x is a well-formed formula if x is a numeral or a variable.

RECURSIVE STEP: If F and G are well-formed formulae, then $(F + G)$, $(F - G)$, $(F * G)$, (F/G) , and $(F \uparrow G)$ are well-formed formulae.

For example, by the basis step we see that x , y , 0 , and 3 are well-formed formulae (as is any variable or numeral). Well-formed formulae generated by applying the recursive step once include $(x + 3)$, $(3 + y)$, $(x - y)$, $(3 - 0)$, $(x * 3)$, $(3 * y)$, $(3/0)$, (x/y) , $(3 \uparrow x)$, and $(0 \uparrow 3)$. Applying the recursive step twice shows that formulae such as $((x + 3) + 3)$ and $(x - (3 * y))$ are well-formed formulae. [Note that $(3/0)$ is a well-formed formula because we are concerned only with syntax matters here.] We leave it to the reader to show that each of the formulae $x3 +$, $y * + x$, and $* x/y$ is *not* a well-formed formula by showing that none of them can be obtained from the basis step and one or more applications of the recursive step. ◀

We will study trees extensively in Chapter 11. A tree is a special type of a graph; a graph is made up of vertices and edges connecting some pairs of vertices. We will study graphs in Chapter 10. We will briefly introduce them here to illustrate how they can be defined recursively.

Definition 3

The set of *rooted trees*, where a rooted tree consists of a set of vertices containing a distinguished vertex called the *root*, and edges connecting these vertices, can be defined recursively by these steps:

BASIS STEP: A single vertex r is a rooted tree.

RECURSIVE STEP: Suppose that $T_1, T_2, \dots, T_n$ are disjoint rooted trees with roots $r_1, r_2, \dots, r_n$, respectively. Then the graph formed by starting with a root r , which is not in any of the rooted trees $T_1, T_2, \dots, T_n$, and adding an edge from r to each of the vertices $r_1, r_2, \dots, r_n$, is also a rooted tree.

In Figure 2 we illustrate some of the rooted trees formed starting with the basis step and applying the recursive step one time and two times. Note that infinitely many rooted trees are formed at each application of the recursive definition.

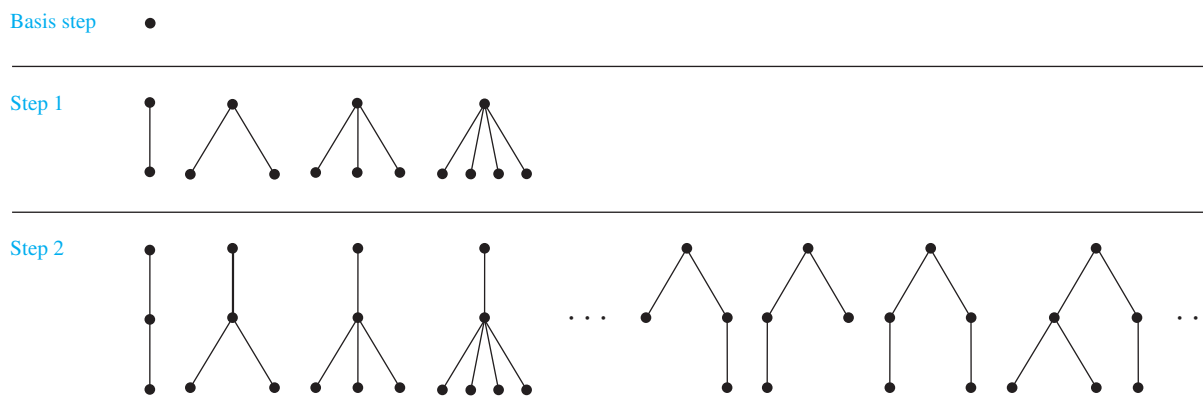


FIGURE 2 Building up rooted trees.

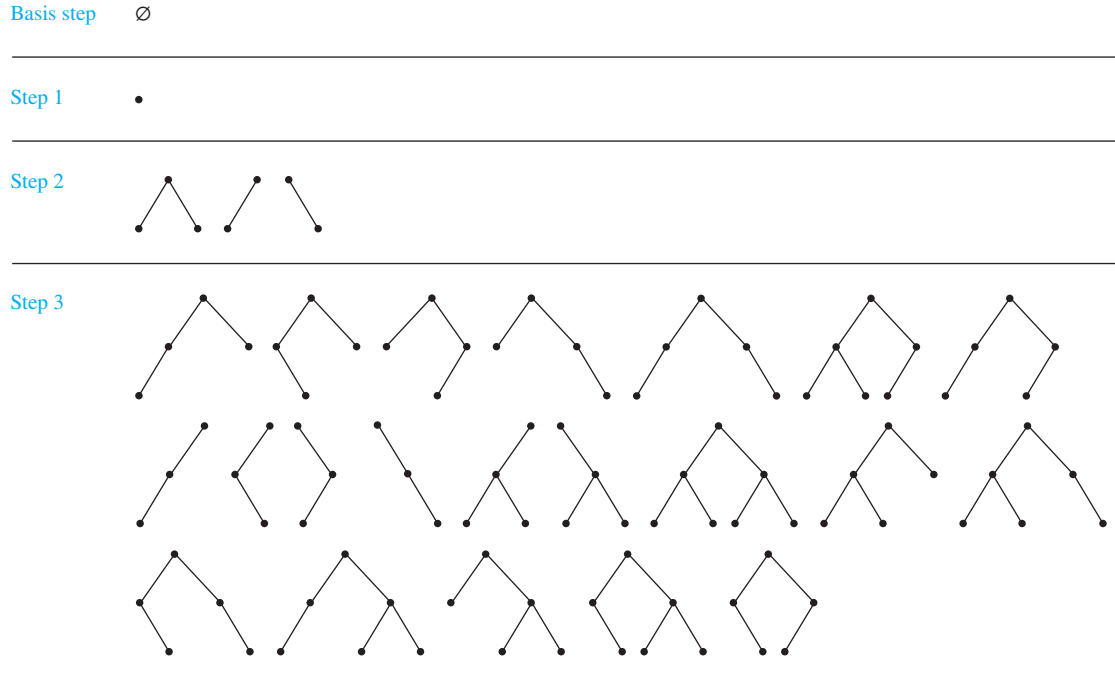


FIGURE 3 Building up extended binary trees.

Binary trees are a special type of rooted trees. We will provide recursive definitions of two types of binary trees—full binary trees and extended binary trees. In the recursive step of the definition of each type of binary tree, two binary trees are combined to form a new tree with one of these trees designated the left subtree and the other the right subtree. In extended binary trees, the left subtree or the right subtree can be empty, but in full binary trees this is not possible. Binary trees are one of the most important types of structures in computer science. In Chapter 11 we will see how they can be used in searching and sorting algorithms, in algorithms for compressing data, and in many other applications. We first define extended binary trees.

Definition 4

The set of *extended binary trees* can be defined recursively by these steps:

BASIS STEP: The empty set is an extended binary tree.

RECURSIVE STEP: If T_1 and T_2 are disjoint extended binary trees, there is an extended binary tree, denoted by $T_1 \cdot T_2$, consisting of a root r together with edges connecting the root to each of the roots of the left subtree T_1 and the right subtree T_2 when these trees are nonempty.

Figure 3 shows how extended binary trees are built up by applying the recursive step from one to three times.

We now show how to define the set of full binary trees. Note that the difference between this recursive definition and that of extended binary trees lies entirely in the basis step.

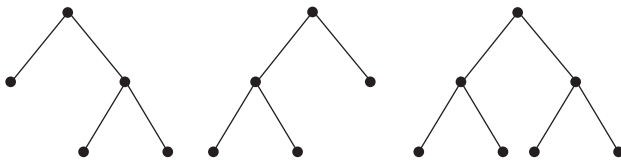
Basis step



Step 1



Step 2

**FIGURE 4** Building up full binary trees.**Definition 5**

The set of full binary trees can be defined recursively by these steps:

BASIS STEP: There is a full binary tree consisting only of a single vertex r .

RECURSIVE STEP: If T_1 and T_2 are disjoint full binary trees, there is a full binary tree, denoted by $T_1 \cdot T_2$, consisting of a root r together with edges connecting the root to each of the roots of the left subtree T_1 and the right subtree T_2 .

Figure 4 shows how full binary trees are built up by applying the recursive step one and two times.

5.3.4 Structural Induction

To prove results about recursively defined sets, we generally use some form of mathematical induction. Example 10 illustrates the connection between recursively defined sets and mathematical induction.

EXAMPLE 10 Show that the set S defined in Example 5 by specifying that $3 \in S$ and that if $x \in S$ and $y \in S$, then $x + y \in S$, is the set of all positive integers that are multiples of 3.

Solution: Let A be the set of all positive integers divisible by 3. To prove that $A = S$, we must show that A is a subset of S and that S is a subset of A . To prove that A is a subset of S , we must show that every positive integer divisible by 3 is in S . We will use mathematical induction to prove this.

Let $P(n)$ be the statement that $3n$ belongs to S . The basis step holds because by the first part of the recursive definition of S , $3 \cdot 1 = 3$ is in S . To establish the inductive step, assume that $P(k)$ is true, namely, that $3k$ is in S . Because $3k$ is in S and because 3 is in S , it follows from the second part of the recursive definition of S that $3k + 3 = 3(k + 1)$ is also in S .

To prove that S is a subset of A , we use the recursive definition of S . First, the basis step of the definition specifies that 3 is in S . Because $3 = 3 \cdot 1$, all elements specified to be in S in this step are divisible by 3 and are therefore in A . To finish the proof, we must show that all integers in S generated using the second part of the recursive definition are in A . This consists of showing that $x + y$ is in A whenever x and y are elements of S also assumed to be in A . Now if x and y are both in A , it follows that $3 \mid x$ and $3 \mid y$. By part (i) of Theorem 1 of Section 4.1, it follows that $3 \mid (x + y)$ completing the proof. ◀

In Example 10 we used mathematical induction over the set of positive integers and a recursive definition to prove a result about a recursively defined set. However, instead of using mathematical induction directly to prove results about recursively defined sets, we can use a more convenient form of induction known as **structural induction**. A proof by structural induction consists of two parts. These parts are

BASIS STEP: Show that the result holds for all elements specified in the basis step of the recursive definition to be in the set.

RECURSIVE STEP: Show that if the statement is true for each of the elements used to construct new elements in the recursive step of the definition, the result holds for these new elements.

The validity of structural induction follows from the principle of mathematical induction for the nonnegative integers. To see this, let $P(n)$ state that the claim is true for all elements of the set that are generated by n or fewer applications of the rules in the recursive step of a recursive definition. We will have established that the principle of mathematical induction implies the principle of structural induction if we can show that $P(n)$ is true whenever n is a positive integer. In the basis step of a proof by structural induction we show that $P(0)$ is true. That is, we show that the result is true of all elements specified to be in the set in the basis step of the definition. A consequence of the recursive step is that if we assume $P(k)$ is true, it follows that $P(k + 1)$ is true. When we have completed a proof using structural induction, we have shown that $P(0)$ is true and that $P(k)$ implies $P(k + 1)$. By mathematical induction it follows that $P(n)$ is true for all nonnegative integers n . This also shows that the result is true for all elements generated by the recursive definition, and shows that structural induction is a valid proof technique.

EXAMPLES OF PROOFS USING STRUCTURAL INDUCTION Structural induction can be used to prove that all members of a set constructed recursively have a particular property. We will illustrate this idea by using structural induction to prove results about well-formed formulae, strings, and binary trees. For each proof, we have to carry out the appropriate basis step and the appropriate recursive step. For example, to use structural induction to prove a result about the set of well-formed formulae defined in Example 8, where we specify that **T**, **F**, and every propositional variable s are well-formed formulae and where we specify that if E and F are well-formed formulae, then $(\neg E)$, $(E \wedge F)$, $(E \vee F)$, $(E \rightarrow F)$, and $(E \leftrightarrow F)$ are well-formed formulae, we need to complete this basis step and this recursive step.

BASIS STEP: Show that the result is true for **T**, **F**, and s whenever s is a propositional variable.

RECURSIVE STEP: Show that if the result is true for the compound propositions p and q , it is also true for $(\neg p)$, $(p \vee q)$, $(p \wedge q)$, $(p \rightarrow q)$, and $(p \leftrightarrow q)$.

Example 11 illustrates how we can prove results about well-formed formulae using structural induction.

EXAMPLE 11 Show that every well-formed formula for compound propositions, as defined in Example 8, contains an equal number of left and right parentheses.

Solution:

BASIS STEP: Each of the formula **T**, **F**, and s contains no parentheses, so clearly they contain an equal number of left and right parentheses.

RECURSIVE STEP: Assume p and q are well-formed formulae, each containing an equal number of left and right parentheses. That is, if l_p and l_q are the number of left parentheses in p and q , respectively, and r_p and r_q are the number of right parentheses in p and q , respectively, then $l_p = r_p$ and $l_q = r_q$. To complete the inductive step, we need to show that each of $(\neg p)$, $(p \vee q)$, $(p \wedge q)$, $(p \rightarrow q)$, and $(p \leftrightarrow q)$ also contains an equal number of left and right parentheses. The number of left parentheses in the first of these compound propositions equals

$l_p + 1$ and in each of the other compound propositions equals $l_p + l_q + 1$. Similarly, the number of right parentheses in the first of these compound propositions equals $r_p + 1$ and in each of the other compound propositions equals $r_p + r_q + 1$. Because $l_p = r_p$ and $l_q = r_q$, it follows that each of these compound expressions contains the same number of left and right parentheses. This completes the proof by structural induction. 

Suppose that $P(w)$ is a propositional function over the set of strings $w \in \Sigma^*$. To use structural induction to prove that $P(w)$ holds for all strings $w \in \Sigma^*$, we need to complete both a basis step and a recursive step. These steps are:

BASIS STEP: Show that $P(\lambda)$ is true.

RECURSIVE STEP: Assume that $P(w)$ is true, where $w \in \Sigma^*$. Show that if $x \in \Sigma$, then $P(wx)$ must also be true.

Example 12 illustrates how structural induction can be used in proofs about strings.

EXAMPLE 12 Use structural induction to prove that $l(xy) = l(x) + l(y)$, where x and y belong to Σ^* , the set of strings over the alphabet Σ .

Solution: We will base our proof on the recursive definition of the set Σ^* given in Definition 1 and the definition of the length of a string in Example 7, which specifies that $l(\lambda) = 0$ and $l(wx) = l(w) + 1$ when $w \in \Sigma^*$ and $x \in \Sigma$. Let $P(y)$ be the statement that $l(xy) = l(x) + l(y)$ whenever x belongs to Σ^* .

BASIS STEP: To complete the basis step, we must show that $P(\lambda)$ is true. That is, we must show that $l(x\lambda) = l(x) + l(\lambda)$ for all $x \in \Sigma^*$. Because $l(x\lambda) = l(x) = l(x) + 0 = l(x) + l(\lambda)$ for every string x , it follows that $P(\lambda)$ is true.

RECURSIVE STEP: To complete the inductive step, we assume that $P(y)$ is true and show that this implies that $P(ya)$ is true whenever $a \in \Sigma$. What we need to show is that $l(xya) = l(x) + l(ya)$ for every $a \in \Sigma$. To show this, note that by the recursive definition of $l(w)$ (given in Example 7), we have $l(xya) = l(xy) + 1$ and $l(ya) = l(y) + 1$. And, by the inductive hypothesis, $l(xy) = l(x) + l(y)$. We conclude that $l(xya) = l(x) + l(y) + 1 = l(x) + l(ya)$. 

We can prove results about trees or special classes of trees using structural induction. For example, to prove a result about full binary trees using structural induction we need to complete this basis step and this recursive step.

BASIS STEP: Show that the result is true for the tree consisting of a single vertex.

RECURSIVE STEP: Show that if the result is true for the full binary trees T_1 and T_2 , then it is true for tree $T_1 \cdot T_2$ consisting of a root r , which has T_1 as its left subtree and T_2 as its right subtree.

Before we provide an example showing how structural induction can be used to prove a result about full binary trees, we need some definitions. We will recursively define the height $h(T)$ and the number of vertices $n(T)$ of a full binary tree T . We begin by defining the height of a full binary tree.

Definition 6

We define the height $h(T)$ of a full binary tree T recursively.

BASIS STEP: The height of the full binary tree T consisting of only a root r is $h(T) = 0$.

RECURSIVE STEP: If T_1 and T_2 are full binary trees, then the full binary tree $T = T_1 \cdot T_2$ has height $h(T) = 1 + \max(h(T_1), h(T_2))$.

If we let $n(T)$ denote the number of vertices in a full binary tree, we observe that $n(T)$ satisfies the following recursive formula:

BASIS STEP: The number of vertices $n(T)$ of the full binary tree T consisting of only a root r is $n(T) = 1$.

RECURSIVE STEP: If T_1 and T_2 are full binary trees, then the number of vertices of the full binary tree $T = T_1 \cdot T_2$ is $n(T) = 1 + n(T_1) + n(T_2)$.

We now show how structural induction can be used to prove a result about full binary trees.

THEOREM 2

If T is a full binary tree T , then $n(T) \leq 2^{h(T)+1} - 1$.

Proof: We prove this inequality using structural induction.

BASIS STEP: For the full binary tree consisting of just the root r the result is true because $n(T) = 1$ and $h(T) = 0$, so that $n(T) = 1 \leq 2^{0+1} - 1 = 1$.

RECURSIVE STEP: For the inductive hypothesis we assume that $n(T_1) \leq 2^{h(T_1)+1} - 1$ and $n(T_2) \leq 2^{h(T_2)+1} - 1$ whenever T_1 and T_2 are full binary trees. By the recursive formulae for $n(T)$ and $h(T)$ we have $n(T) = 1 + n(T_1) + n(T_2)$ and $h(T) = 1 + \max(h(T_1), h(T_2))$.

We find that

$$\begin{aligned}
 n(T) &= 1 + n(T_1) + n(T_2) && \text{by the recursive formula for } n(T) \\
 &\stackrel{\text{IH}}{\leq} 1 + (2^{h(T_1)+1} - 1) + (2^{h(T_2)+1} - 1) && \text{by the inductive hypothesis} \\
 &\leq 2 \cdot \max(2^{h(T_1)+1}, 2^{h(T_2)+1}) - 1 && \text{because the sum of two terms is at most 2} \\
 &\quad \text{times the larger} \\
 &= 2 \cdot 2^{\max(h(T_1), h(T_2))+1} - 1 && \text{because } \max(2^x, 2^y) = 2^{\max(x,y)} \\
 &= 2 \cdot 2^{h(T)} - 1 && \text{by the recursive definition of } h(T) \\
 &= 2^{h(T)+1} - 1.
 \end{aligned}$$

This completes the recursive step. ◀

5.3.5 Generalized Induction

We can extend mathematical induction to prove results about other sets that have the well-ordering property besides the set of integers. Although we will discuss this concept in detail in Section 9.6, we provide an example here to illustrate the usefulness of such an approach.

As an example, note that we can define an ordering on $\mathbf{N} \times \mathbf{N}$, the ordered pairs of non-negative integers, by specifying that (x_1, y_1) is less than or equal to (x_2, y_2) if either $x_1 < x_2$, or $x_1 = x_2$ and $y_1 < y_2$; this is called the **lexicographic ordering**. The set $\mathbf{N} \times \mathbf{N}$ with this ordering has the property that every subset of $\mathbf{N} \times \mathbf{N}$ has a least element (see Exercise 53 in Section 9.6). This implies that we can recursively define the terms $a_{m,n}$, with $m \in \mathbf{N}$ and $n \in \mathbf{N}$, and prove results about them using a variant of mathematical induction, as illustrated in Example 13.

EXAMPLE 13 Suppose that $a_{m,n}$ is defined recursively for $(m, n) \in \mathbf{N} \times \mathbf{N}$ by $a_{0,0} = 0$ and

$$a_{m,n} = \begin{cases} a_{m-1,n} + 1 & \text{if } n = 0 \text{ and } m > 0 \\ a_{m,n-1} + n & \text{if } n > 0. \end{cases}$$

Show that $a_{m,n} = m + n(n+1)/2$ for all $(m, n) \in \mathbf{N} \times \mathbf{N}$, that is, for all pairs of nonnegative integers.

Solution: We can prove that $a_{m,n} = m + n(n+1)/2$ using a generalized version of mathematical induction. The basis step requires that we show that this formula is valid when $(m, n) = (0, 0)$. The induction step requires that we show that if the formula holds for all pairs smaller than (m, n) in the lexicographic ordering of $\mathbf{N} \times \mathbf{N}$, then it also holds for (m, n) .

BASIS STEP: Let $(m, n) = (0, 0)$. Then by the basis case of the recursive definition of $a_{m,n}$ we have $a_{0,0} = 0$. Furthermore, when $m = n = 0$, $m + n(n+1)/2 = 0 + (0 \cdot 1)/2 = 0$. This completes the basis step.

INDUCTIVE STEP: Suppose that $a_{m',n'} = m' + n'(n'+1)/2$ whenever (m', n') is less than (m, n) in the lexicographic ordering of $\mathbf{N} \times \mathbf{N}$. By the recursive definition, if $n = 0$, then $a_{m,n} = a_{m-1,n} + 1$. Because $(m-1, n)$ is smaller than (m, n) , the inductive hypothesis tells us that $a_{m-1,n} = m-1 + n(n+1)/2$, so that $a_{m,n} = m-1 + n(n+1)/2 + 1 = m + n(n+1)/2$, giving us the desired equality. Now suppose that $n > 0$, so $a_{m,n} = a_{m,n-1} + n$. Because $(m, n-1)$ is smaller than (m, n) , the inductive hypothesis tells us that $a_{m,n-1} = m + (n-1)n/2$, so $a_{m,n} = m + (n-1)n/2 + n = m + (n^2 - n + 2n)/2 = m + n(n+1)/2$. This finishes the inductive step. 

As mentioned, we will justify this proof technique in Section 9.6.

Exercises

- Find $f(1), f(2), f(3)$, and $f(4)$ if $f(n)$ is defined recursively by $f(0) = 1$ and for $n = 0, 1, 2, \dots$
 - $f(n+1) = f(n) + 2$.
 - $f(n+1) = 3f(n)$.
 - $f(n+1) = 2^{f(n)}$.
 - $f(n+1) = f(n)^2 + f(n) + 1$.
- Find $f(1), f(2), f(3), f(4)$, and $f(5)$ if $f(n)$ is defined recursively by $f(0) = 3$ and for $n = 0, 1, 2, \dots$
 - $f(n+1) = -2f(n)$.
 - $f(n+1) = 3f(n) + 7$.
 - $f(n+1) = f(n)^2 - 2f(n) - 2$.
 - $f(n+1) = 3^{f(n)/3}$.
- Find $f(2), f(3), f(4)$, and $f(5)$ if f is defined recursively by $f(0) = -1, f(1) = 2$, and for $n = 1, 2, \dots$
 - $f(n+1) = f(n) + 3f(n-1)$.
 - $f(n+1) = f(n)^2 f(n-1)$.
 - $f(n+1) = 3f(n)^2 - 4f(n-1)^2$.
 - $f(n+1) = f(n-1)/f(n)$.
- Find $f(2), f(3), f(4)$, and $f(5)$ if f is defined recursively by $f(0) = f(1) = 1$ and for $n = 1, 2, \dots$
 - $f(n+1) = f(n) - f(n-1)$.
 - $f(n+1) = f(n)f(n-1)$.
 - $f(n+1) = f(n)^2 + f(n-1)^3$.
 - $f(n+1) = f(n)/f(n-1)$.
- Determine whether each of these proposed definitions is a valid recursive definition of a function f from the set of nonnegative integers to the set of integers. If f is well defined, find a formula for $f(n)$ when n is a nonnegative integer and prove that your formula is valid.
 - $f(0) = 0, f(n) = 2f(n-2)$ for $n \geq 1$
 - $f(0) = 1, f(n) = f(n-1) - 1$ for $n \geq 1$
 - $f(0) = 2, f(1) = 3, f(n) = f(n-1) - 1$ for $n \geq 2$
 - $f(0) = 1, f(1) = 2, f(n) = 2f(n-2)$ for $n \geq 2$
 - $f(0) = 1, f(n) = 3f(n-1)$ if n is odd and $n \geq 1$ and $f(n) = 9f(n-2)$ if n is even and $n \geq 2$
- Determine whether each of these proposed definitions is a valid recursive definition of a function f from the set of nonnegative integers to the set of integers. If f is well defined, find a formula for $f(n)$ when n is a nonnegative integer and prove that your formula is valid.
 - $f(0) = 1, f(n) = -f(n-1)$ for $n \geq 1$
 - $f(0) = 1, f(1) = 0, f(2) = 2, f(n) = 2f(n-3)$ for $n \geq 3$
 - $f(0) = 0, f(1) = 1, f(n) = 2f(n+1)$ for $n \geq 2$
 - $f(0) = 0, f(1) = 1, f(n) = 2f(n-1)$ for $n \geq 1$
 - $f(0) = 2, f(n) = f(n-1)$ if n is odd and $n \geq 1$ and $f(n) = 2f(n-2)$ if $n \geq 2$

7. Give a recursive definition of the sequence $\{a_n\}$, $n = 1, 2, 3, \dots$ if
- a) $a_n = 6n$. b) $a_n = 2n + 1$.
 c) $a_n = 10^n$. d) $a_n = 5$.

8. Give a recursive definition of the sequence $\{a_n\}$, $n = 1, 2, 3, \dots$ if
- a) $a_n = 4n - 2$. b) $a_n = 1 + (-1)^n$.
 c) $a_n = n(n + 1)$. d) $a_n = n^2$.

9. Let F be the function such that $F(n)$ is the sum of the first n positive integers. Give a recursive definition of $F(n)$.

10. Give a recursive definition of $S_m(n)$, the sum of the integer m and the nonnegative integer n .

11. Give a recursive definition of $P_m(n)$, the product of the integer m and the nonnegative integer n .

In Exercises 12–19 f_n is the n th Fibonacci number.

12. Prove that $f_1^2 + f_2^2 + \dots + f_n^2 = f_n f_{n+1}$ when n is a positive integer.

13. Prove that $f_1 + f_3 + \dots + f_{2n-1} = f_{2n}$ when n is a positive integer.

- *14. Show that $f_{n+1}f_{n-1} - f_n^2 = (-1)^n$ when n is a positive integer.

- *15. Show that $f_0f_1 + f_1f_2 + \dots + f_{2n-1}f_{2n} = f_{2n}^2$ when n is a positive integer.

- *16. Show that $f_0 - f_1 + f_2 - \dots - f_{2n-1} + f_{2n} = f_{2n-1} - 1$ when n is a positive integer.

17. Determine the number of divisions used by the Euclidean algorithm to find the greatest common divisor of the Fibonacci numbers f_n and f_{n+1} , where n is a nonnegative integer. Verify your answer using mathematical induction.

18. Let

$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}.$$

Show that

$$\mathbf{A}^n = \begin{bmatrix} f_{n+1} & f_n \\ f_n & f_{n-1} \end{bmatrix}$$

when n is a positive integer.

19. By taking determinants of both sides of the equation in Exercise 18, prove the identity given in Exercise 14. (Recall that the determinant of the matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is $ad - bc$.)

- *20. Give a recursive definition of the functions $\max$ and $\min$ so that $\max(a_1, a_2, \dots, a_n)$ and $\min(a_1, a_2, \dots, a_n)$ are the maximum and minimum of the n numbers $a_1, a_2, \dots, a_n$, respectively.

- *21. Let $a_1, a_2, \dots, a_n$, and $b_1, b_2, \dots, b_n$ be real numbers. Use the recursive definitions that you gave in Exercise 20 to prove these.

- a) $\max(-a_1, -a_2, \dots, -a_n) = -\min(a_1, a_2, \dots, a_n)$
 b) $\max(a_1 + b_1, a_2 + b_2, \dots, a_n + b_n)$
 $\leq \max(a_1, a_2, \dots, a_n) + \max(b_1, b_2, \dots, b_n)$
 c) $\min(a_1 + b_1, a_2 + b_2, \dots, a_n + b_n)$
 $\geq \min(a_1, a_2, \dots, a_n) + \min(b_1, b_2, \dots, b_n)$

22. Show that the set S defined by $1 \in S$ and $s + t \in S$ whenever $s \in S$ and $t \in S$ is the set of positive integers.

23. Give a recursive definition of the set of positive integers that are multiples of 5.

24. Give a recursive definition of

- a) the set of odd positive integers.
 b) the set of positive integer powers of 3.
 c) the set of polynomials with integer coefficients.

25. Give a recursive definition of

- a) the set of even integers.
 b) the set of positive integers congruent to 2 modulo 3.
 c) the set of positive integers not divisible by 5.

26. Let S be the set of positive integers defined by

Basis step: $1 \in S$.

Recursive step: If $n \in S$, then $3n + 2 \in S$ and $n^2 \in S$.

- a) Show that if $n \in S$, then $n \equiv 1 \pmod{4}$.
 b) Show that there exists an integer $m \equiv 1 \pmod{4}$ that does not belong to S .

27. Let S be the set of positive integers defined by

Basis step: $5 \in S$.

Recursive step: If $n \in S$, then $3n \in S$ and $n^2 \in S$.

- a) Show that if $n \in S$, then $n \equiv 5 \pmod{10}$.
 b) Show that there exists an integer $m \equiv 5 \pmod{10}$ that does not belong to S .

28. Let S be the subset of the set of ordered pairs of integers defined recursively by

Basis step: $(0, 0) \in S$.

Recursive step: If $(a, b) \in S$, then $(a + 2, b + 3) \in S$ and $(a + 3, b + 2) \in S$.

- a) List the elements of S produced by the first five applications of the recursive definition.
 b) Use strong induction on the number of applications of the recursive step of the definition to show that $5 \mid a + b$ when $(a, b) \in S$.
 c) Use structural induction to show that $5 \mid a + b$ when $(a, b) \in S$.

29. Let S be the subset of the set of ordered pairs of integers defined recursively by

Basis step: $(0, 0) \in S$.

Recursive step: If $(a, b) \in S$, then $(a, b + 1) \in S$, $(a + 1, b + 1) \in S$, and $(a + 2, b + 1) \in S$.

- a) List the elements of S produced by the first four applications of the recursive definition.
 b) Use strong induction on the number of applications of the recursive step of the definition to show that $a \leq 2b$ whenever $(a, b) \in S$.
 c) Use structural induction to show that $a \leq 2b$ whenever $(a, b) \in S$.

30. Give a recursive definition of each of these sets of ordered pairs of positive integers. [Hint: Plot the points in the set in the plane and look for lines containing points in the set.]

- a) $S = \{(a, b) \mid a \in \mathbf{Z}^+, b \in \mathbf{Z}^+, \text{ and } a + b \text{ is odd}\}$
 b) $S = \{(a, b) \mid a \in \mathbf{Z}^+, b \in \mathbf{Z}^+, \text{ and } a \mid b\}$
 c) $S = \{(a, b) \mid a \in \mathbf{Z}^+, b \in \mathbf{Z}^+, \text{ and } 3 \mid a + b\}$

31. Give a recursive definition of each of these sets of ordered pairs of positive integers. Use structural induction

to prove that the recursive definition you found is correct. [Hint: To find a recursive definition, plot the points in the set in the plane and look for patterns.]

- a) $S = \{(a, b) \mid a \in \mathbf{Z}^+, b \in \mathbf{Z}^+, \text{ and } a + b \text{ is even}\}$
 - b) $S = \{(a, b) \mid a \in \mathbf{Z}^+, b \in \mathbf{Z}^+, \text{ and } a \text{ or } b \text{ is odd}\}$
 - c) $S = \{(a, b) \mid a \in \mathbf{Z}^+, b \in \mathbf{Z}^+, a + b \text{ is odd, and } 3 \mid b\}$
32. Prove that in a bit string, the string 01 occurs at most one more time than the string 10.
33. Define well-formed formulae of sets, variables representing sets, and operators from $\{\bar{}, \cup, \cap, -\}$.
34. a) Give a recursive definition of the function $\text{ones}(s)$, which counts the number of ones in a bit string s .
 b) Use structural induction to prove that $\text{ones}(st) = \text{ones}(s) + \text{ones}(t)$.
35. a) Give a recursive definition of the function $m(s)$, which equals the smallest digit in a nonempty string of decimal digits.
 b) Use structural induction to prove that $m(st) = \min(m(s), m(t))$.

The **reversal** of a string is the string consisting of the symbols of the string in reverse order. The reversal of the string w is denoted by w^R .

36. Find the reversal of the following bit strings.
 a) 0101 b) 11011 c) 100010010111
37. Give a recursive definition of the reversal of a string. [Hint: First define the reversal of the empty string. Then write a string w of length $n + 1$ as xy , where x is a string of length n , and express the reversal of w in terms of x^R and y .]
- *38. Use structural induction to prove that $(w_1 w_2)^R = w_2^R w_1^R$.
39. Give a recursive definition of w^i , where w is a string and i is a nonnegative integer. (Here w^i represents the concatenation of i copies of the string w .)
- *40. Give a recursive definition of the set of bit strings that are palindromes.
41. When does a string belong to the set A of bit strings defined recursively by
- $$\lambda \in A$$
- $$0x1 \in A \text{ if } x \in A,$$
- where λ is the empty string?
- *42. Recursively define the set of bit strings that have more zeros than ones.
43. Use Exercise 39 and mathematical induction to show that $l(w^i) = i \cdot l(w)$, where w is a string and i is a nonnegative integer.
- *44. Show that $(w^R)^i = (w^i)^R$ whenever w is a string and i is a nonnegative integer; that is, show that the i th power of the reversal of a string is the reversal of the i th power of the string.
45. Use structural induction to show that $n(T) \geq 2h(T) + 1$, where T is a full binary tree, $n(T)$ equals the number of vertices of T , and $h(T)$ is the height of T .

The set of leaves and the set of internal vertices of a full binary tree can be defined recursively.

Basis step: The root r is a leaf of the full binary tree with exactly one vertex r . This tree has no internal vertices.

Recursive step: The set of leaves of the tree $T = T_1 \cdot T_2$ is the union of the sets of leaves of T_1 and of T_2 . The internal vertices of T are the root r of T and the union of the set of internal vertices of T_1 and the set of internal vertices of T_2 .

46. Use structural induction to show that $l(T)$, the number of leaves of a full binary tree T , is 1 more than $i(T)$, the number of internal vertices of T .
47. Use generalized induction as was done in Example 13 to show that if $a_{m,n}$ is defined recursively by $a_{0,0} = 0$ and

$$a_{m,n} = \begin{cases} a_{m-1,n} + 1 & \text{if } n = 0 \text{ and } m > 0 \\ a_{m,n-1} + 1 & \text{if } n > 0, \end{cases}$$

then $a_{m,n} = m + n$ for all $(m, n) \in \mathbf{N} \times \mathbf{N}$.

48. Use generalized induction as was done in Example 13 to show that if $a_{m,n}$ is defined recursively by $a_{1,1} = 5$ and

$$a_{m,n} = \begin{cases} a_{m-1,n} + 2 & \text{if } n = 1 \text{ and } m > 1 \\ a_{m,n-1} + 2 & \text{if } n > 1, \end{cases}$$

then $a_{m,n} = 2(m + n) + 1$ for all $(m, n) \in \mathbf{Z}^+ \times \mathbf{Z}^+$.

- *49. A **partition** of a positive integer n is a way to write n as a sum of positive integers where the order of terms in the sum does not matter. For instance, $7 = 3 + 2 + 1 + 1$ is a partition of 7. Let P_m equal the number of different partitions of m , and let $P_{m,n}$ be the number of different ways to express m as the sum of positive integers not exceeding n .
- a) Show that $P_{m,m} = P_m$.
 b) Show that the following recursive definition for $P_{m,n}$ is correct:

$$P_{m,n} = \begin{cases} 1 & \text{if } m = 1 \\ 1 & \text{if } n = 1 \\ P_{m,m} & \text{if } m < n \\ 1 + P_{m,m-1} & \text{if } m = n > 1 \\ P_{m,n-1} + P_{m-n,n} & \text{if } m > n > 1. \end{cases}$$

- c) Find the number of partitions of 5 and of 6 using this recursive definition.



Consider the following inductive definition of a version of **Ackermann's function**. This function was named after Wilhelm Ackermann, a German mathematician who was a student of the great mathematician David Hilbert. Ackermann's function plays an important role in the theory of recursive functions and in the study of the complexity of certain algorithms involving set unions. (There are several different variants of this function. All are called Ackermann's function and have similar properties even though their values do not always agree.)

$$A(m, n) = \begin{cases} 2n & \text{if } m = 0 \\ 0 & \text{if } m \geq 1 \text{ and } n = 0 \\ 2 & \text{if } m \geq 1 \text{ and } n = 1 \\ A(m-1, A(m, n-1)) & \text{if } m \geq 1 \text{ and } n \geq 2 \end{cases}$$

Exercises 50–57 involve this version of Ackermann's function.

50. Find these values of Ackermann's function.

- a) $A(1, 0)$ b) $A(0, 1)$
c) $A(1, 1)$ d) $A(2, 2)$

51. Show that $A(m, 2) = 4$ whenever $m \geq 1$.

52. Show that $A(1, n) = 2^n$ whenever $n \geq 1$.

53. Find these values of Ackermann's function.

- a) $A(2, 3)$ *b) $A(3, 3)$

*54. Find $A(3, 4)$.

**55. Prove that $A(m, n+1) > A(m, n)$ whenever m and n are nonnegative integers.

*56. Prove that $A(m+1, n) \geq A(m, n)$ whenever m and n are nonnegative integers.

57. Prove that $A(i, j) \geq j$ whenever i and j are nonnegative integers.

58. Use mathematical induction to prove that a function F defined by specifying $F(0)$ and a rule for obtaining $F(n+1)$ from $F(n)$ is well defined.

59. Use strong induction to prove that a function F defined by specifying $F(0)$ and a rule for obtaining $F(n+1)$ from the values $F(k)$ for $k = 0, 1, 2, \dots, n$ is well defined.

60. Show that each of these proposed recursive definitions of a function on the set of positive integers does not produce a well-defined function.

- a) $F(n) = 1 + F(\lfloor n/2 \rfloor)$ for $n \geq 1$ and $F(1) = 1$.
b) $F(n) = 1 + F(n-3)$ for $n \geq 2$, $F(1) = 2$, and $F(2) = 3$.
c) $F(n) = 1 + F(n/2)$ for $n \geq 2$, $F(1) = 1$, and $F(2) = 2$.
d) $F(n) = 1 + F(n/2)$ if n is even and $n \geq 2$, $F(n) = 1 - F(n-1)$ if n is odd, and $F(1) = 1$.
e) $F(n) = 1 + F(n/2)$ if n is even and $n \geq 2$, $F(n) = F(3n-1)$ if n is odd and $n \geq 3$, and $F(1) = 1$.

61. Show that each of these proposed recursive definitions of a function on the set of positive integers does not produce a well-defined function.

- a) $F(n) = 1 + F(\lfloor (n+1)/2 \rfloor)$ for $n \geq 1$ and $F(1) = 1$.
b) $F(n) = 1 + F(n-2)$ for $n \geq 2$ and $F(1) = 0$.
c) $F(n) = 1 + F(n/3)$ for $n \geq 3$, $F(1) = 1$, $F(2) = 2$, and $F(3) = 3$.
d) $F(n) = 1 + F(n/2)$ if n is even and $n \geq 2$, $F(n) = 1 + F(n-2)$ if n is odd, and $F(1) = 1$.
e) $F(n) = 1 + F(F(n-1))$ if $n \geq 2$ and $F(1) = 2$.

Exercises 62–64 deal with iterations of the logarithm function. Let $\log n$ denote the logarithm of n to the base 2, as usual. The function $\log^{(k)} n$ is defined recursively by

$$\log^{(k)} n = \begin{cases} n & \text{if } k = 0 \\ \log(\log^{(k-1)} n) & \text{if } \log^{(k-1)} n \text{ is defined} \\ & \text{and positive} \\ \text{undefined} & \text{otherwise.} \end{cases}$$

The **iterated logarithm** is the function $\log^* n$ whose value at n is the smallest nonnegative integer k such that $\log^{(k)} n \leq 1$.

62. Find these values.

- a) $\log^{(2)} 16$ b) $\log^{(3)} 256$
c) $\log^{(3)} 2^{65536}$ d) $\log^{(4)} 2^{65536}$

63. Find the value of $\log^* n$ for these values of n .

- a) 2 b) 4 c) 8 d) 16
e) 256 f) 65536 g) 2^{2048}

64. Find the largest integer n such that $\log^* n = 5$. Determine the number of decimal digits in this number.

Exercises 65–67 deal with values of iterated functions. Suppose that $f(n)$ is a function from the set of real numbers, or positive real numbers, or some other set of real numbers, to the set of real numbers such that $f(n)$ is monotonically increasing [that is, $f(n) < f(m)$ when $n < m$] and $f(n) < n$ for all n in the domain of f . The function $f^{(k)}(n)$ is defined recursively by

$$f^{(k)}(n) = \begin{cases} n & \text{if } k = 0 \\ f(f^{(k-1)}(n)) & \text{if } k > 0. \end{cases}$$

Furthermore, let c be a positive real number. The **iterated function** f_c^* is the number of iterations of f required to reduce its argument to c or less, so $f_c^*(n)$ is the smallest nonnegative integer k such that $f^{(k)}(n) \leq c$.

65. Let $f(n) = n - a$, where a is a positive integer. Find a formula for $f^{(k)}(n)$. What is the value of $f_0^*(n)$ when n is a positive integer?

66. Let $f(n) = n/2$. Find a formula for $f^{(k)}(n)$. What is the value of $f_1^*(n)$ when n is a positive integer?

67. Let $f(n) = \sqrt{n}$. Find a formula for $f^{(k)}(n)$. What is the value of $f_2^*(n)$ when n is a positive integer?

5.4 Recursive Algorithms

5.4.1 Introduction

Sometimes we can reduce the solution to a problem with a particular set of input values to the solution of the same problem with smaller input values. For instance, the problem of finding the greatest common divisor of two positive integers a and b , where $b > a$, can be reduced to finding the greatest common divisor of a pair of smaller integers, namely, $b \bmod a$ and a , because $\gcd(b \bmod a, a) = \gcd(a, b)$.

When such a reduction can be done, the solution to the original problem can be found with a sequence of reductions, until the problem has been reduced to some initial case for which the

Here's a famous humorous quote: "To understand recursion, you must first understand recursion."

solution is known. For instance, for finding the greatest common divisor, the reduction continues until the smaller of the two numbers is zero, because $\gcd(a, 0) = a$ when $a > 0$.

We will see that algorithms that successively reduce a problem to the same problem with smaller input are used to solve a wide variety of problems.

Definition 1

An algorithm is called *recursive* if it solves a problem by reducing it to an instance of the same problem with smaller input.

Links

We will describe a variety of different recursive algorithms in this section.

EXAMPLE 1

Give a recursive algorithm for computing $n!$, where n is a nonnegative integer.

Extra Examples

Solution: We can build a recursive algorithm that finds $n!$, where n is a nonnegative integer, based on the recursive definition of $n!$, which specifies that $n! = n \cdot (n - 1)!$ when n is a positive integer, and that $0! = 1$. To find $n!$ for a particular integer n , we use the recursive step n times, each time replacing a value of the factorial function with the value of the factorial function at the next smaller integer. At this last step, we insert the value of $0!$. The recursive algorithm we obtain is displayed as Algorithm 1.

To help understand how this algorithm works, we trace the steps used by the algorithm to compute $4!$. First, we use the recursive step to write $4! = 4 \cdot 3!$. We then use the recursive step repeatedly to write $3! = 3 \cdot 2!$, $2! = 2 \cdot 1!$, and $1! = 1 \cdot 0!$. Inserting the value of $0! = 1$, and working back through the steps, we see that $1! = 1 \cdot 1 = 1$, $2! = 2 \cdot 1! = 2$, $3! = 3 \cdot 2! = 3 \cdot 2 = 6$, and $4! = 4 \cdot 3! = 4 \cdot 6 = 24$. 

ALGORITHM 1 A Recursive Algorithm for Computing $n!$.

```

procedure factorial( $n$ : nonnegative integer)
if  $n = 0$  then return 1
else return  $n \cdot \text{factorial}(n - 1)$ 
{output is  $n!$ }

```

Example 2 shows how a recursive algorithm can be constructed to evaluate a function from its recursive definition.

EXAMPLE 2

Give a recursive algorithm for computing a^n , where a is a nonzero real number and n is a nonnegative integer.

Solution: We can base a recursive algorithm on the recursive definition of a^n . This definition states that $a^{n+1} = a \cdot a^n$ for $n > 0$ and the initial condition $a^0 = 1$. To find a^n , successively use the recursive step to reduce the exponent until it becomes zero. We give this procedure in Algorithm 2. 

ALGORITHM 2 A Recursive Algorithm for Computing a^n .

```

procedure power( $a$ : nonzero real number,  $n$ : nonnegative integer)
if  $n = 0$  then return 1
else return  $a \cdot \text{power}(a, n - 1)$ 
{output is  $a^n$ }

```

Next we give a recursive algorithm for finding greatest common divisors.

EXAMPLE 3 Give a recursive algorithm for computing the greatest common divisor of two nonnegative integers a and b with $a < b$.

Solution: We can base a recursive algorithm on the reduction $\text{gcd}(a, b) = \text{gcd}(b \bmod a, a)$ and the condition $\text{gcd}(0, b) = b$ when $b > 0$. This produces the procedure in Algorithm 3, which is a recursive version of the Euclidean algorithm.

We illustrate the workings of Algorithm 3 with a trace when the input is $a = 5$, $b = 8$. With this input, the algorithm uses the “else” clause to find that $\text{gcd}(5, 8) = \text{gcd}(8 \bmod 5, 5) = \text{gcd}(3, 5)$. It uses this clause again to find that $\text{gcd}(3, 5) = \text{gcd}(5 \bmod 3, 3) = \text{gcd}(2, 3)$, then to get $\text{gcd}(2, 3) = \text{gcd}(3 \bmod 2, 2) = \text{gcd}(1, 2)$, then to get $\text{gcd}(1, 2) = \text{gcd}(2 \bmod 1, 1) = \text{gcd}(0, 1)$. Finally, to find $\text{gcd}(0, 1)$ it uses the first step with $a = 0$ to find that $\text{gcd}(0, 1) = 1$. Consequently, the algorithm finds that $\text{gcd}(5, 8) = 1$. ◀

ALGORITHM 3 A Recursive Algorithm for Computing $\text{gcd}(a, b)$.

```

procedure  $\text{gcd}(a, b$ : nonnegative integers with  $a < b$ )
if  $a = 0$  then return  $b$ 
else return  $\text{gcd}(b \bmod a, a)$ 
{output is  $\text{gcd}(a, b)$ }

```

EXAMPLE 4 Devise a recursive algorithm for computing $b^n \bmod m$, where b , n , and m are integers with $m \geq 2$, $n \geq 0$, and $1 \leq b < m$.

Solution: We can base a recursive algorithm on the fact that

$$b^n \bmod m = (b \cdot (b^{n-1} \bmod m)) \bmod m,$$

which follows by Corollary 2 in Section 4.1, and the initial condition $b^0 \bmod m = 1$. We leave this as Exercise 12 for the reader.

However, we can devise a much more efficient recursive algorithm based on the observation that

$$b^n \bmod m = (b^{n/2} \bmod m)^2 \bmod m$$

when n is even and

$$b^n \bmod m = ((b^{\lfloor n/2 \rfloor} \bmod m)^2 \bmod m \cdot b \bmod m) \bmod m$$

when n is odd, which we describe in pseudocode as Algorithm 4.

We trace the execution of Algorithm 4 with input $b = 2$, $n = 5$, and $m = 3$ to illustrate how it works. First, because $n = 5$ is odd we use the “else” clause to see that $\text{mpower}(2, 5, 3) = (\text{mpower}(2, 2, 3)^2 \bmod 3 \cdot 2 \bmod 3) \bmod 3$. We next use the “else if” clause to see that $\text{mpower}(2, 2, 3) = \text{mpower}(2, 1, 3)^2 \bmod 3$. Using the “else” clause again, we see that $\text{mpower}(2, 1, 3) = (\text{mpower}(2, 0, 3)^2 \bmod 3 \cdot 2 \bmod 3) \bmod 3$. Finally, using the “if” clause, we see that $\text{mpower}(2, 0, 3) = 1$. Working backwards, it follows that $\text{mpower}(2, 1, 3) = (1^2 \bmod 3 \cdot 2 \bmod 3) \bmod 3 = 2$, so $\text{mpower}(2, 2, 3) = 2^2 \bmod 3 = 1$, and finally $\text{mpower}(2, 5, 3) = (1^2 \bmod 3 \cdot 2 \bmod 3) \bmod 3 = 2$. ◀

ALGORITHM 4 Recursive Modular Exponentiation.

```

procedure mpower(b, n, m: integers with  $b > 0$  and  $m \geq 2$ ,  $n \geq 0$ )
if  $n = 0$  then
    return 1
else if  $n$  is even then
    return mpower(b,  $n/2$ , m)2 mod m
else
    return (mpower(b,  $\lfloor n/2 \rfloor$ , m)2 mod m · b mod m) mod m
{output is  $b^n \bmod m$ }

```

We will now give recursive versions of searching algorithms that were introduced in Section 3.1.

EXAMPLE 5 Express the linear search algorithm as a recursive procedure.

Solution: To *search* for the first occurrence of x in the sequence $a_1, a_2, \dots, a_n$, at the i th step of the algorithm, x and a_i are compared. If x equals a_i , then the algorithm returns i , the location of x in the sequence. Otherwise, the search for the first occurrence of x is reduced to a search in a sequence with one fewer element, namely, the sequence $a_{i+1}, \dots, a_n$. The algorithm returns 0 when x is never found in the sequence after all terms have been examined. We can now give a recursive procedure, which is displayed as pseudocode in Algorithm 5.

Let *search*(i, j, x) be the procedure that searches for the first occurrence of x in the sequence $a_i, a_{i+1}, \dots, a_j$. The input to the procedure consists of the triple (i, j, x) . The algorithm terminates at a step if the first term of the remaining sequence is x or if there is only one term of the sequence and this is not x . If x is not the first term and there are additional terms, the same procedure is carried out but with a search sequence of one fewer term, obtained by deleting the first term of the search sequence. If the algorithm terminates without x having been found, the algorithm returns the value 0. 

ALGORITHM 5 A Recursive Linear Search Algorithm.

```

procedure search(i, j, x: integers,  $1 \leq i \leq j \leq n$ )
if  $a_i = x$  then
    return i
else if  $i = j$  then
    return 0
else
    return search( $i + 1, j, x$ )
{output is the location of  $x$  in  $a_1, a_2, \dots, a_n$  if it appears; otherwise it is 0}

```

EXAMPLE 6 Construct a recursive version of a binary search algorithm.

Solution: Suppose we want to locate x in the sequence $a_1, a_2, \dots, a_n$ of integers in increasing order. To perform a binary search, we begin by comparing x with the middle term, $a_{\lfloor (n+1)/2 \rfloor}$. Our algorithm will terminate if x equals this term and return the location of this term in the sequence. Otherwise, we reduce the search to a smaller search sequence, namely, the first half of the sequence if x is smaller than the middle term of the original sequence, and the second

half otherwise. We have reduced the solution of the search problem to the solution of the same problem with a sequence at most half as long. If we have never encountered the search term x , our algorithm returns the value 0. We express this recursive version of a binary search algorithm as Algorithm 6. 

ALGORITHM 6 A Recursive Binary Search Algorithm.

```

procedure binary search( $i, j, x$ : integers,  $1 \leq i \leq j \leq n$ )
 $m := \lfloor (i + j) / 2 \rfloor$ 
if  $x = a_m$  then
    return  $m$ 
else if ( $x < a_m$  and  $i < m$ ) then
    return binary search( $i, m - 1, x$ )
else if ( $x > a_m$  and  $j > m$ ) then
    return binary search( $m + 1, j, x$ )
else return 0
{output is location of  $x$  in  $a_1, a_2, \dots, a_n$  if it appears; otherwise it is 0}

```

5.4.2 Proving Recursive Algorithms Correct

Mathematical induction, and its variant strong induction, can be used to prove that a recursive algorithm is correct, that is, that it produces the desired output for all possible input values. Examples 7 and 8 illustrate how mathematical induction or strong induction can be used to prove that recursive algorithms are correct. First, we will show that Algorithm 2 is correct.

EXAMPLE 7 Prove that Algorithm 2, which computes powers of real numbers, is correct.

Solution: We use mathematical induction on the exponent n .

BASIS STEP: If $n = 0$, the first step of the algorithm tells us that $\text{power}(a, 0) = 1$. This is correct because $a^0 = 1$ for every nonzero real number a . This completes the basis step.

INDUCTIVE STEP: The inductive hypothesis is the statement that $\text{power}(a, k) = a^k$ for all $a \neq 0$ for an arbitrary nonnegative integer k . That is, the inductive hypothesis is the statement that the algorithm correctly computes a^k . To complete the inductive step, we show that if the inductive hypothesis is true, then the algorithm correctly computes a^{k+1} . Because $k + 1$ is a positive integer, when the algorithm computes a^{k+1} , the algorithm sets $\text{power}(a, k + 1) = a \cdot \text{power}(a, k)$. By the inductive hypothesis, we have $\text{power}(a, k) = a^k$, so $\text{power}(a, k + 1) = a \cdot \text{power}(a, k) = a \cdot a^k = a^{k+1}$. This completes the inductive step.

We have completed the basis step and the inductive step, so we can conclude that Algorithm 2 always computes a^n correctly when $a \neq 0$ and n is a nonnegative integer. 

Generally, we need to use strong induction to prove that recursive algorithms are correct, rather than just mathematical induction. Example 8 illustrates this; it shows how strong induction can be used to prove that Algorithm 4 is correct.

EXAMPLE 8 Prove that Algorithm 4, which computes modular powers, is correct.

Solution: We use strong induction on the exponent n .

BASIS STEP: Let b be an integer and m an integer with $m \geq 2$. When $n = 0$, the algorithm sets $\text{mpower}(b, n, m)$ equal to 1. This is correct because $b^0 \bmod m = 1$. The basis step is complete.

INDUCTIVE STEP: For the inductive hypothesis we assume that $\text{mpower}(b, j, m) = b^j \bmod m$ for all integers $0 \leq j < k$ whenever b is a positive integer and m is an integer with $m \geq 2$. To complete the inductive step, we show that if the inductive hypothesis is correct, then $\text{mpower}(b, k, m) = b^k \bmod m$. Because the recursive algorithm handles odd and even values of k differently, we split the inductive step into two cases.

When k is even, we have

$$\text{mpower}(b, k, m) = (\text{mpower}(b, k/2, m))^2 \bmod m = (b^{k/2} \bmod m)^2 \bmod m = b^k \bmod m,$$

where we have used the inductive hypothesis to replace $\text{mpower}(b, k/2, m)$ by $b^{k/2} \bmod m$.

When k is odd, we have

$$\begin{aligned} \text{mpower}(b, k, m) &= ((\text{mpower}(b, \lfloor k/2 \rfloor, m))^2 \bmod m \cdot b \bmod m) \bmod m \\ &= ((b^{\lfloor k/2 \rfloor} \bmod m)^2 \bmod m \cdot b \bmod m) \bmod m \\ &= b^{2\lfloor k/2 \rfloor + 1} \bmod m = b^k \bmod m, \end{aligned}$$

using Corollary 2 in Section 4.1, because $2\lfloor k/2 \rfloor + 1 = 2(k-1)/2 + 1 = k$ when k is odd. Here we have used the inductive hypothesis to replace $\text{mpower}(b, \lfloor k/2 \rfloor, m)$ by $b^{\lfloor k/2 \rfloor} \bmod m$. This completes the inductive step.

We have completed the basis step and the inductive step, so by strong induction we know that Algorithm 4 is correct. 

5.4.3 Recursion and Iteration

A recursive definition expresses the value of a function at a positive integer in terms of the values of the function at smaller integers. This means that we can devise a recursive algorithm to evaluate a recursively defined function at a positive integer. Instead of successively reducing the computation to the evaluation of the function at smaller integers, we can start with the value of the function at one or more integers, the base cases, and successively apply the recursive definition to find the values of the function at successive larger integers. Such a procedure is called **iterative**. Often an iterative approach for the evaluation of a recursively defined sequence requires much less computation than a procedure using recursion (unless special-purpose recursive machines are used). This is illustrated by the iterative and recursive procedures for finding the n th Fibonacci number. The recursive procedure is given first.

ALGORITHM 7 A Recursive Algorithm for Fibonacci Numbers.

```

procedure fibonacci( $n$ : nonnegative integer)
if  $n = 0$  then return 0
else if  $n = 1$  then return 1
else return fibonacci( $n - 1$ ) + fibonacci( $n - 2$ )
{output is fibonacci( $n$ )}
```

When we use a recursive procedure to find f_n , we first express f_n as $f_{n-1} + f_{n-2}$. Then we replace both of these Fibonacci numbers by the sum of two previous Fibonacci numbers, and so on. When f_1 or f_0 arises, it is replaced by its value.

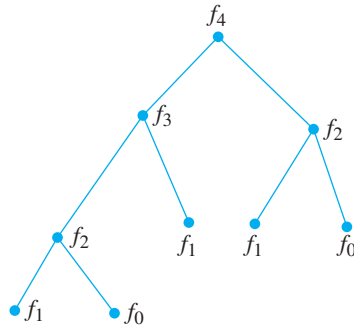


FIGURE 1 Evaluating f_4 recursively.

Note that at each stage of the recursion, until f_1 or f_0 is obtained, the number of Fibonacci numbers to be evaluated has doubled. For instance, when we find f_4 using this recursive algorithm, we must carry out all the computations illustrated in the tree diagram in Figure 1. This tree consists of a root labeled with f_4 , and branches from the root to vertices labeled with the two Fibonacci numbers f_3 and f_2 that occur in the reduction of the computation of f_4 . Each subsequent reduction produces two branches in the tree. This branching ends when f_0 and f_1 are reached. The reader can verify that this algorithm requires $f_{n+1} - 1$ additions to find f_n .

Now consider the amount of computation required to find f_n using the iterative approach in Algorithm 8.

ALGORITHM 8 An Iterative Algorithm for Computing Fibonacci Numbers.

```

procedure iterative fibonacci( $n$ : nonnegative integer)
if  $n = 0$  then return 0
else
     $x := 0$ 
     $y := 1$ 
    for  $i := 1$  to  $n - 1$ 
         $z := x + y$ 
         $x := y$ 
         $y := z$ 
    return  $y$ 
    {output is the  $n$ th Fibonacci number}

```

This procedure initializes x as $f_0 = 0$ and y as $f_1 = 1$. When the loop is traversed, the sum of x and y is assigned to the auxiliary variable z . Then x is assigned the value of y and y is assigned the value of the auxiliary variable z . Therefore, after going through the loop the first time, it follows that x equals f_1 and y equals $f_0 + f_1 = f_2$. Furthermore, after going through the loop $n - 1$ times, x equals f_{n-1} and y equals f_n (the reader should verify this statement). Only $n - 1$ additions have been used to find f_n with this iterative approach when $n > 1$. Consequently, this algorithm requires far less computation than does the recursive algorithm.

We have shown that a recursive algorithm may require far more computation than an iterative one when a recursively defined function is evaluated. It is sometimes preferable to use a recursive procedure even if it is less efficient than the iterative procedure. In particular, this is true when the recursive approach is easily implemented and the iterative approach is not. (Also, machines designed to handle recursion may be available that eliminate the advantage of using iteration.)

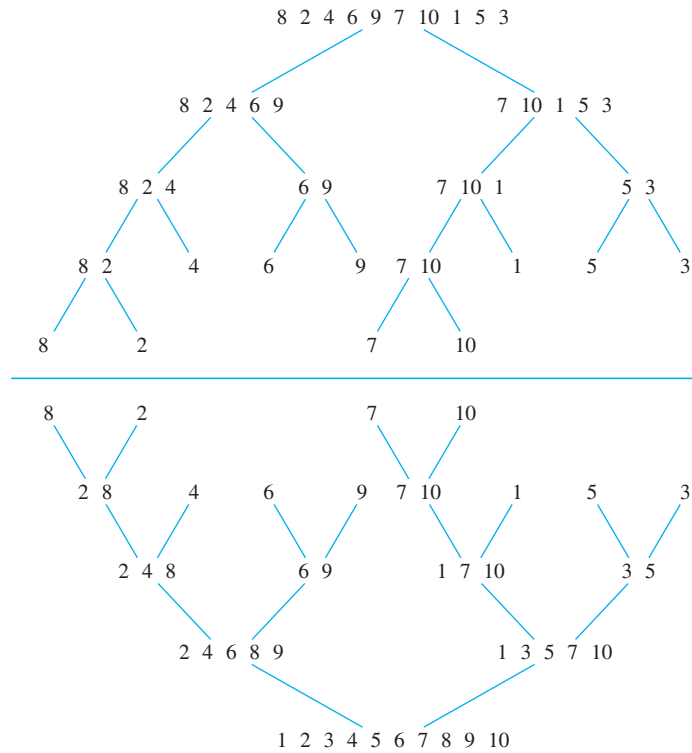


FIGURE 2 The merge sort of 8, 2, 4, 6, 9, 7, 10, 1, 5, 3.

5.4.4 The Merge Sort

Links

We now describe a recursive sorting algorithm called the **merge sort** algorithm. We will demonstrate how the merge sort algorithm works with an example before describing it in generality.

EXAMPLE 9 Use the merge sort to put the terms of the list 8, 2, 4, 6, 9, 7, 10, 1, 5, 3 in increasing order.

Solution: A merge sort begins by splitting the list into individual elements by successively splitting lists in two. The progression of sublists for this example is represented with the balanced binary tree of height 4 shown in the upper half of Figure 2.

Sorting is done by successively merging pairs of lists. At the first stage, pairs of individual elements are merged into lists of length two in increasing order. Then successive merges of pairs of lists are performed until the entire list is put into increasing order. The succession of merged lists in increasing order is represented by the balanced binary tree of height 4 shown in the lower half of Figure 2 (note that this tree is displayed “upside down”).

Demo

In general, a merge sort proceeds by iteratively splitting lists into two sublists of equal length (or where one sublist has one more element than the other) until each sublist contains one element. This succession of sublists can be represented by a balanced binary tree. The procedure continues by successively merging pairs of lists, where both lists are in increasing order, into a larger list with elements in increasing order, until the original list is put into increasing order. The succession of merged lists can be represented by a balanced binary tree.

We can also describe the merge sort recursively. To do a merge sort, we split a list into two sublists of equal, or approximately equal, size, sorting each sublist using the merge sort

algorithm, and then merging the two lists. The recursive version of the merge sort is given in Algorithm 9. This algorithm uses the subroutine *merge*, which is described in Algorithm 10.

ALGORITHM 9 A Recursive Merge Sort.

```

procedure mergesort( $L = a_1, \dots, a_n$ )
if  $n > 1$  then
     $m := \lfloor n/2 \rfloor$ 
     $L_1 := a_1, a_2, \dots, a_m$ 
     $L_2 := a_{m+1}, a_{m+2}, \dots, a_n$ 
     $L := \text{merge}(\text{mergesort}(L_1), \text{mergesort}(L_2))$ 
    { $L$  is now sorted into elements in nondecreasing order}

```

An efficient algorithm for merging two ordered lists into a larger ordered list is needed to implement the merge sort. We will now describe such a procedure.

EXAMPLE 10 Merge the two lists 2, 3, 5, 6 and 1, 4.

Solution: Table 1 illustrates the steps we use. First, compare the smallest elements in the two lists, 2 and 1, respectively. Because 1 is the smaller, put it at the beginning of the merged list and remove it from the second list. At this stage, the first list is 2, 3, 5, 6, the second is 4, and the combined list is 1.

Next, compare 2 and 4, the smallest elements of the two lists. Because 2 is the smaller, add it to the combined list and remove it from the first list. At this stage the first list is 3, 5, 6, the second is 4, and the combined list is 1, 2.

Continue by comparing 3 and 4, the smallest elements of their respective lists. Because 3 is the smaller of these two elements, add it to the combined list and remove it from the first list. At this stage the first list is 5, 6, and the second is 4. The combined list is 1, 2, 3.

Then compare 5 and 4, the smallest elements in the two lists. Because 4 is the smaller of these two elements, add it to the combined list and remove it from the second list. At this stage the first list is 5, 6, the second list is empty, and the combined list is 1, 2, 3, 4.

Finally, because the second list is empty, all elements of the first list can be appended to the end of the combined list in the order they occur in the first list. This produces the ordered list 1, 2, 3, 4, 5, 6. 

We will now consider the general problem of merging two ordered lists L_1 and L_2 into an ordered list L . We will describe an algorithm for solving this problem. Start with an empty list L . Compare the smallest elements of the two lists. Put the smaller of these two elements

TABLE 1 Merging the Two Sorted Lists 2, 3, 5, 6 and 1, 4.

First List	Second List	Merged List	Comparison
2 3 5 6	1 4		$1 < 2$
2 3 5 6	4	1	$2 < 4$
3 5 6	4	1 2	$3 < 4$
5 6	4	1 2 3	$4 < 5$
5 6		1 2 3 4	
		1 2 3 4 5 6	

at the right end of L , and remove it from the list it was in. Next, if one of L_1 and L_2 is empty, append the other (nonempty) list to L , which completes the merging. If neither L_1 nor L_2 is empty, repeat this process. Algorithm 10 gives a pseudocode description of this procedure.

We will need estimates for the number of comparisons used to merge two ordered lists in the analysis of the merge sort. We can easily obtain such an estimate for Algorithm 10. Each time a comparison of an element from L_1 and an element from L_2 is made, an additional element is added to the merged list L . However, when either L_1 or L_2 is empty, no more comparisons are needed. Hence, Algorithm 10 is least efficient when $m + n - 2$ comparisons are carried out, where m and n are the number of elements in L_1 and L_2 , respectively, leaving one element in each of L_1 and L_2 . The next comparison will be the last one needed, because it will make one of these lists empty. Hence, Algorithm 10 uses no more than $m + n - 1$ comparisons. Lemma 1 summarizes this estimate.

ALGORITHM 10 Merging Two Lists.

```
procedure merge( $L_1, L_2$ : sorted lists)
 $L :=$  empty list
while  $L_1$  and  $L_2$  are both nonempty
    remove smaller of first elements of  $L_1$  and  $L_2$  from its list; put it at the right end of  $L$ 
    if this removal makes one list empty then remove all elements from the other list and
        append them to  $L$ 
return  $L$  { $L$  is the merged list with elements in increasing order}
```

LEMMA 1

Two sorted lists with m elements and n elements can be merged into a sorted list using no more than $m + n - 1$ comparisons.

Sometimes two sorted lists of length m and n can be merged using far fewer than $m + n - 1$ comparisons. For instance, when $m = 1$, a binary search procedure can be applied to put the one element in the first list into the second list. This requires only $\lceil \log n \rceil$ comparisons, which is much smaller than $m + n - 1 = n$, for $m = 1$. On the other hand, for some values of m and n , Lemma 1 gives the best possible bound. That is, there are lists with m and n elements that cannot be merged using fewer than $m + n - 1$ comparisons. (See Exercise 47.)

We can now analyze the complexity of the merge sort. Instead of studying the general problem, we will assume that n , the number of elements in the list, is a power of 2, say 2^m . This will make the analysis less complicated, but when this is not the case, various modifications can be applied that will yield the same estimate.

At the first stage of the splitting procedure, the list is split into two sublists, of 2^{m-1} elements each, at level 1 of the tree generated by the splitting. This process continues, splitting the two sublists with 2^{m-1} elements into four sublists of 2^{m-2} elements each at level 2, and so on. In general, there are 2^{k-1} lists at level $k - 1$, each with 2^{m-k+1} elements. These lists at level $k - 1$ are split into 2^k lists at level k , each with 2^{m-k} elements. At the end of this process, we have 2^m lists each with one element at level m .

We start merging by combining pairs of the 2^m lists of one element into 2^{m-1} lists, at level $m - 1$, each with two elements. To do this, 2^{m-1} pairs of lists with one element each are merged. The merger of each pair requires exactly one comparison.

The procedure continues, so that at level k ($k = m, m - 1, m - 2, \dots, 3, 2, 1$), 2^k lists each with 2^{m-k} elements are merged into 2^{k-1} lists, each with 2^{m-k+1} elements, at level $k - 1$. To do this a total of 2^{k-1} mergers of two lists, each with 2^{m-k} elements, are needed. But, by Lemma 1,

each of these mergers can be carried out using at most $2^{m-k} + 2^{m-k} - 1 = 2^{m-k+1} - 1$ comparisons. Hence, going from level k to $k - 1$ can be accomplished using at most $2^{k-1}(2^{m-k+1} - 1)$ comparisons.

Summing all these estimates shows that the number of comparisons required for the merge sort is at most

$$\sum_{k=1}^m 2^{k-1}(2^{m-k+1} - 1) = \sum_{k=1}^m 2^m - \sum_{k=1}^m 2^{k-1} = m2^m - (2^m - 1) = n \log n - n + 1,$$

because $m = \log n$ and $n = 2^m$. (We evaluated $\sum_{k=1}^m 2^m$ by noting that it is the sum of m identical terms, each equal to 2^m . We evaluated $\sum_{k=1}^m 2^{k-1}$ using the formula for the sum of the terms of a geometric progression from Theorem 1 of Section 2.4.)

Theorem 1 summarizes what we have discovered about the worst-case complexity of the merge sort algorithm.

THEOREM 1 The number of comparisons needed to merge sort a list with n elements is $O(n \log n)$.

In Chapter 11 we will show that the fastest comparison-based sorting algorithm have $O(n \log n)$ time complexity. (A comparison-based sorting algorithm has the comparison of two elements as its basic operation.) Theorem 1 tells us that the merge sort achieves this best possible big- O estimate for the complexity of a sorting algorithm. We describe another efficient algorithm, the quick sort, in the preamble to Exercise 50.

Exercises

- Trace Algorithm 1 when it is given $n = 5$ as input. That is, show all steps used by Algorithm 1 to find $5!$, as is done in Example 1 to find $4!$.
- Trace Algorithm 1 when it is given $n = 6$ as input. That is, show all steps used by Algorithm 1 to find $6!$, as is done in Example 1 to find $4!$.
- Trace Algorithm 3 when it finds $\text{gcd}(8, 13)$. That is, show all the steps used by Algorithm 3 to find $\text{gcd}(8, 13)$.
- Trace Algorithm 3 when it finds $\text{gcd}(12, 17)$. That is, show all the steps used by Algorithm 3 to find $\text{gcd}(12, 17)$.
- Trace Algorithm 4 when it is given $m = 5$, $n = 11$, and $b = 3$ as input. That is, show all the steps Algorithm 4 uses to find $3^{11} \bmod 5$.
- Trace Algorithm 4 when it is given $m = 7$, $n = 10$, and $b = 2$ as input. That is, show all the steps Algorithm 4 uses to find $2^{10} \bmod 7$.
- Give a recursive algorithm for computing nx whenever n is a positive integer and x is an integer, using just addition.
- Give a recursive algorithm for finding the sum of the first n positive integers.
- Give a recursive algorithm for finding the sum of the first n odd positive integers.
- Give a recursive algorithm for finding the maximum of a finite set of integers, making use of the fact that the maximum of n integers is the larger of the last integer in the list and the maximum of the first $n - 1$ integers in the list.
- Give a recursive algorithm for finding the minimum of a finite set of integers, making use of the fact that the minimum of n integers is the smaller of the last integer in the list and the minimum of the first $n - 1$ integers in the list.
- Devise a recursive algorithm for finding $x^n \bmod m$ whenever n , x , and m are positive integers based on the fact that $x^n \bmod m = (x^{n-1} \bmod m \cdot x \bmod m) \bmod m$.
- Give a recursive algorithm for finding $n! \bmod m$ whenever n and m are positive integers.
- Give a recursive algorithm for finding a **mode** of a list of integers. (A **mode** is an element in the list that occurs at least as often as every other element.)
- Devise a recursive algorithm for computing the greatest common divisor of two nonnegative integers a and b with $a < b$ using the fact that $\text{gcd}(a, b) = \text{gcd}(a, b - a)$.
- Prove that the recursive algorithm for finding the sum of the first n positive integers you found in Exercise 8 is correct.

17. Describe a recursive algorithm for multiplying two non-negative integers x and y based on the fact that $xy = 2(x \cdot (y/2))$ when y is even and $xy = 2(x \cdot \lfloor y/2 \rfloor) + x$ when y is odd, together with the initial condition $xy = 0$ when $y = 0$.
 18. Prove that Algorithm 1 for computing $n!$ when n is a non-negative integer is correct.
 19. Prove that Algorithm 3 for computing $\gcd(a, b)$ when a and b are positive integers with $a < b$ is correct.
 20. Prove that the algorithm you devised in Exercise 17 is correct.
 21. Prove that the recursive algorithm that you found in Exercise 7 is correct.
 22. Prove that the recursive algorithm that you found in Exercise 10 is correct.
 23. Devise a recursive algorithm for computing n^2 where n is a nonnegative integer, using the fact that $(n+1)^2 = n^2 + 2n + 1$. Then prove that this algorithm is correct.
 24. Devise a recursive algorithm to find a^{2^n} , where a is a real number and n is a positive integer. [Hint: Use the equality $a^{2^{n+1}} = (a^{2^n})^2$.]
 25. How does the number of multiplications used by the algorithm in Exercise 24 compare to the number of multiplications used by Algorithm 2 to evaluate a^{2^n} ?
 - *26. Use the algorithm in Exercise 24 to devise an algorithm for evaluating a^n when n is a nonnegative integer. [Hint: Use the binary expansion of n .]
 - *27. How does the number of multiplications used by the algorithm in Exercise 26 compare to the number of multiplications used by Algorithm 2 to evaluate a^n ?
 28. How many additions are used by the recursive and iterative algorithms given in Algorithms 7 and 8, respectively, to find the Fibonacci number f_7 ?
 29. Devise a recursive algorithm to find the n th term of the sequence defined by $a_0 = 1$, $a_1 = 2$, and $a_n = a_{n-1} \cdot a_{n-2}$, for $n = 2, 3, 4, \dots$.
 30. Devise an iterative algorithm to find the n th term of the sequence defined in Exercise 29.
 31. Is the recursive or the iterative algorithm for finding the sequence in Exercise 29 more efficient?
 32. Devise a recursive algorithm to find the n th term of the sequence defined by $a_0 = 1$, $a_1 = 2$, $a_2 = 3$, and $a_n = a_{n-1} + a_{n-2} + a_{n-3}$, for $n = 3, 4, 5, \dots$.
 33. Devise an iterative algorithm to find the n th term of the sequence defined in Exercise 32.
 34. Is the recursive or the iterative algorithm for finding the sequence in Exercise 32 more efficient?
 35. Give iterative and recursive algorithms for finding the n th term of the sequence defined by $a_0 = 1$, $a_1 = 3$, $a_2 = 5$, and $a_n = a_{n-1} \cdot a_{n-2}^2 \cdot a_{n-3}^3$. Which is more efficient?
 36. Give a recursive algorithm to find the number of partitions of a positive integer based on the recursive definition given in Exercise 49 in Section 5.3.
 37. Give a recursive algorithm for finding the reversal of a bit string. (See the definition of the reversal of a bit string in the preamble of Exercise 36 in Section 5.3.)
 38. Give a recursive algorithm for finding the string w^i , the concatenation of i copies of w , when w is a bit string.
 39. Prove that the recursive algorithm for finding the reversal of a bit string that you gave in Exercise 37 is correct.
 40. Prove that the recursive algorithm for finding the concatenation of i copies of a bit string that you gave in Exercise 38 is correct.
 - *41. Give a recursive algorithm for tiling a $2^n \times 2^n$ checkerboard with one square missing using right triominoes.
 42. Give a recursive algorithm for triangulating a simple polygon with n sides, using Lemma 1 in Section 5.2.
 43. Give a recursive algorithm for computing values of the Ackermann function. [Hint: See the preamble to Exercise 48 in Section 5.3.]
 44. Use a merge sort to sort 4, 3, 2, 5, 1, 8, 7, 6 into increasing order. Show all the steps used by the algorithm.
 45. Use a merge sort to sort $b, d, a, f, g, h, z, p, o, k$ into alphabetic order. Show all the steps used by the algorithm.
 46. How many comparisons are required to merge these pairs of lists using Algorithm 10?
 - a) 1, 3, 5, 7, 9; 2, 4, 6, 8, 10
 - b) 1, 2, 3, 4, 5; 6, 7, 8, 9, 10
 - c) 1, 5, 6, 7, 8; 2, 3, 4, 9, 10
 47. Show that for all positive integers m and n there are sorted lists with m elements and n elements, respectively, such that Algorithm 10 uses $m + n - 1$ comparisons to merge them into one sorted list.
 - *48. What is the least number of comparisons needed to merge any two lists in increasing order into one list in increasing order when the number of elements in the two lists are
 - a) 1, 4? b) 2, 4? c) 3, 4? d) 4, 4?
 - *49. Prove that the merge sort algorithm is correct.
- The **quick sort** is an efficient algorithm. To sort $a_1, a_2, \dots, a_n$, this algorithm begins by taking the first element a_1 and forming two sublists, the first containing those elements that are less than a_1 , in the order they arise, and the second containing those elements greater than a_1 , in the order they arise. Then a_1 is put at the end of the first sublist. This procedure is repeated recursively for each sublist, until all sublists contain one item. The ordered list of n items is obtained by combining the sublists of one item in the order they occur.
50. Sort 3, 5, 7, 8, 1, 9, 2, 4, 6 using the quick sort.
 51. Let $a_1, a_2, \dots, a_n$ be a list of n distinct real numbers. How many comparisons are needed to form two sublists from this list, the first containing elements less than a_1 and the second containing elements greater than a_1 ?
 52. Describe the quick sort algorithm using pseudocode.
 53. What is the largest number of comparisons needed to order a list of four elements using the quick sort algorithm?
 54. What is the least number of comparisons needed to order a list of four elements using the quick sort algorithm?
 55. Determine the worst-case complexity of the quick sort algorithm in terms of the number of comparisons used.

5.5 Program Correctness

5.5.1 Introduction

Suppose that we have designed an algorithm to solve a problem and have written a program to implement it. How can we be sure that the program always produces the correct answer? After all the bugs have been removed so that the syntax is correct, we can test the program with sample input. It is not correct if an incorrect result is produced for any sample input. But even if the program gives the correct answer for all sample input, it may not always produce the correct answer (unless all possible input has been tested). We need a proof to show that the program *always* gives the correct output.

Program verification, the proof of correctness of programs, uses the rules of inference and proof techniques described in this chapter, including mathematical induction. Because an incorrect program can lead to disastrous results, a large amount of methodology has been constructed for verifying programs. Efforts have been devoted to automating program verification so that it can be carried out using a computer. However, only limited progress has been made toward this goal. Indeed, some mathematicians and theoretical computer scientists argue that it will never be realistic to mechanize the proof of correctness of complex programs.

Some of the concepts and methods used to prove that programs are correct will be introduced in this section. Many different methods have been devised for proving that programs are correct. We will discuss a widely used method for program verification introduced by Tony Hoare in this section; several other methods are also commonly used. Furthermore, we will not develop a complete methodology for program verification in this book. This section is meant to be a brief introduction to the area of program verification, which ties together the rules of logic, proof techniques, and the concept of an algorithm.

5.5.2 Program Verification

A program is said to be **correct** if it produces the correct output for every possible input. A proof that a program is correct consists of two parts. The first part shows that the correct answer is obtained if the program terminates. This part of the proof establishes the **partial correctness** of the program. The second part of the proof shows that the program always terminates.

To specify what it means for a program to produce the correct output, two propositions are used. The first is the **initial assertion**, which gives the properties that the input values must have. The second is the **final assertion**, which gives the properties that the output of the program should have, if the program did what was intended. The appropriate initial and final assertions must be provided when a program is checked.

Definition 1

A program, or program segment, S is said to be *partially correct with respect to* the initial assertion p and the final assertion q if whenever p is true for the input values of S and S terminates, then q is true for the output values of S . The notation $p\{S\}q$ indicates that the program, or program segment, S is partially correct with respect to the initial assertion p and the final assertion q .

Note: The notation $p\{S\}q$ is known as a *Hoare triple*. Tony Hoare introduced the concept of partial correctness.



Note that the notion of partial correctness has nothing to do with whether a program terminates; it focuses only on whether the program does what it is expected to do if it terminates.

A simple example illustrates the concepts of initial and final assertions.

EXAMPLE 1 Show that the program segment

$$\begin{aligned} y &:= 2 \\ z &:= x + y \end{aligned}$$

Extra Examples ➤

is correct with respect to the initial assertion $p: x = 1$ and the final assertion $q: z = 3$.

Solution: Suppose that p is true, so that $x = 1$ as the program begins. Then y is assigned the value 2, and z is assigned the sum of the values of x and y , which is 3. Hence, S is correct with respect to the initial assertion p and the final assertion q . Thus, $p\{S\}q$ is true. ◀

5.5.3 Rules of Inference

A useful rule of inference proves that a program is correct by splitting the program into a sequence of subprograms and then showing that each subprogram is correct.

Suppose that the program S is split into subprograms S_1 and S_2 . Write $S = S_1; S_2$ to indicate that S is made up of S_1 followed by S_2 . Suppose that the correctness of S_1 with respect to the initial assertion p and final assertion q , and the correctness of S_2 with respect to the initial assertion q and the final assertion r , have been established. It follows that if p is true and S_1 is executed and terminates, then q is true; and if q is true, and S_2 executes and terminates, then r is true. Thus, if p is true and $S = S_1; S_2$ is executed and terminates, then r is true. This rule of inference, called the **composition rule**, can be stated as

$$\frac{p\{S_1\}q \quad q\{S_2\}r}{\therefore p\{S_1; S_2\}r}.$$

This rule of inference will be used later in this section.

Next, some rules of inference for program segments involving conditional statements and loops will be given. Because programs can be split into segments for proofs of correctness, this will let us verify many different programs.

5.5.4 Conditional Statements

First, rules of inference for conditional statements will be given. Suppose that a program segment has the form

if *condition* **then**
 S

where S is a block of statements. Then S is executed if *condition* is true, and it is not executed when *condition* is false. To verify that this segment is correct with respect to the initial assertion p and final assertion q , two things must be done. First, it must be shown that when p is true and *condition* is also true, then q is true after S terminates. Second, it must be shown that when p is true and *condition* is false, then q is true (because in this case S does not execute).

This leads to the following rule of inference:

$$\frac{(p \wedge \text{condition})\{S\}q \quad (p \wedge \neg \text{condition}) \rightarrow q}{\therefore p\{\text{if } \text{condition} \text{ then } S\}q}.$$

Example 2 illustrates how this rule of inference is used.

EXAMPLE 2 Verify that the program segment

```

if  $x > y$  then
     $y := x$ 

```

is correct with respect to the initial assertion **T** and the final assertion $y \geq x$.

Solution: When the initial assertion is true and $x > y$, the assignment $y := x$ is carried out. Hence, the final assertion, which asserts that $y \geq x$, is true in this case. Moreover, when the initial assertion is true and $x > y$ is false, so that $x \leq y$, the final assertion is again true. Hence, using the rule of inference for program segments of this type, this program is correct with respect to the given initial and final assertions. ▶

Similarly, suppose that a program has a statement of the form

```

if condition then
     $S_1$ 
else
     $S_2$ 

```

If *condition* is true, then S_1 executes; if *condition* is false, then S_2 executes. To verify that this program segment is correct with respect to the initial assertion p and the final assertion q , two things must be done. First, it must be shown that when p is true and *condition* is true, then q is true after S_1 terminates. Second, it must be shown that when p is true and *condition* is false, then q is true after S_2 terminates. This leads to the following rule of inference:

$$\frac{(p \wedge \text{condition})\{S_1\}q \quad (p \wedge \neg \text{condition})\{S_2\}q}{\therefore p\{\text{if condition then } S_1 \text{ else } S_2\}q.}$$

Links ▶



Courtesy of Tony Hoare

C. ANTHONY R. HOARE (BORN 1934) Tony Hoare was born in Colombo, Ceylon (now known as Sri Lanka), where his father was a civil servant of the British Empire and his mother's father owned a plantation. He spent his early childhood in Ceylon, moving to England in 1945. Hoare studied philosophy, together with the classics, at the University of Oxford, where he became interested in computing as a result of his fascination with the power of mathematical logic and the certainty of mathematical truth. He received his bachelors degree from Oxford in 1956.

Hoare learned Russian during his service in the Royal Navy, and later studied the computer translation of natural languages at Moscow State University. He returned to England in 1960, taking a job at a small computer manufacturer, where he wrote a compiler for the programming language Algol. In 1968, he became Professor of Computing Science at the Queen's University, Belfast; in 1977, he moved to the

University of Oxford as Professor of Computing; he is now Professor Emeritus. He is a Fellow of the Royal Society and also holds a position at Microsoft Research in Cambridge.

Hoare has made many contributions to the theory of programming languages and to programming methodology. He was first to define a programming language based on how programs could be proved correct with respect to their specifications. Hoare also invented quick sort, one of the most commonly used sorting algorithms (see the preamble to Exercise 50 in Section 5.4). He received the ACM Turing Award in 1980 and in 2000 he was knighted for services to education and computer science. Hoare is a noted writer in the technical and social aspects of computer science.

Example 3 illustrates how this rule of inference is used.

EXAMPLE 3 Verify that the program segment

```

if  $x < 0$  then
     $abs := -x$ 
else
     $abs := x$ 

```

is correct with respect to the initial assertion **T** and the final assertion $abs = |x|$.

Solution: Two things must be demonstrated. First, it must be shown that if the initial assertion is true and $x < 0$, then $abs = |x|$. This is correct, because when $x < 0$ the assignment statement $abs := -x$ sets $abs = -x$, which is $|x|$ by definition when $x < 0$. Second, it must be shown that if the initial assertion is true and $x < 0$ is false, so that $x \geq 0$, then $abs = |x|$. This is also correct, because in this case the program uses the assignment statement $abs := x$, and x is $|x|$ by definition when $x \geq 0$, so $abs := x$. Hence, using the rule of inference for program segments of this type, this segment is correct with respect to the given initial and final assertions. ◀

5.5.5 Loop Invariants



Next, proofs of correctness of **while** loops will be described. To develop a rule of inference for program segments of the type

```

while condition
    S

```

note that *S* is repeatedly executed until *condition* becomes false. An assertion that remains true each time *S* is executed must be chosen. Such an assertion is called a **loop invariant**. In other words, *p* is a loop invariant if $(p \wedge \text{condition})\{S\}p$ is true.

Suppose that *p* is a loop invariant. It follows that if *p* is true before the program segment is executed, *p* and $\neg \text{condition}$ are true after termination, if it occurs. This rule of inference is

$$\frac{(p \wedge \text{condition})\{S\}p}{\therefore p\{\text{while } \text{condition } S\}(\neg \text{condition} \wedge p)}.$$

The use of a loop invariant is illustrated in Example 4.

EXAMPLE 4 A loop invariant is needed to verify that the program segment



```

i := 1
factorial := 1
while i < n
    i := i + 1
    factorial := factorial · i

```

terminates with $\text{factorial} = n!$ when *n* is a positive integer.

Let p be the assertion “ $factorial = i!$ and $i \leq n$.” We first prove that p is a loop invariant. Suppose that, at the beginning of one execution of the **while** loop, p is true and the condition of the **while** loop holds; in other words, assume that $factorial = i!$ and that $i < n$. The new values i_{new} and $factorial_{\text{new}}$ of i and $factorial$ are $i_{\text{new}} = i + 1$ and $factorial_{\text{new}} = factorial \cdot (i + 1) = (i + 1)! = i_{\text{new}}!$. Because $i < n$, we also have $i_{\text{new}} = i + 1 \leq n$. Thus, p is true at the end of the execution of the loop. This shows that p is a loop invariant.

Now we consider the program segment. Just before entering the loop, $i = 1 \leq n$ and $factorial = 1 = 1! = i!$ both hold, so p is true. Because p is a loop invariant, the rule of inference just introduced implied that if the **while** loop terminates, it terminates with p true and with $i < n$ false. In this case, at the end, $factorial = i!$ and $i \leq n$ are true, but $i < n$ is false; in other words, $i = n$ and $factorial = i! = n!$, as desired.

Finally, we need to check that the **while** loop actually terminates. At the beginning of the program i is assigned the value 1, so after $n - 1$ traversals of the loop, the new value of i will be n , and the loop terminates at that point. ◀

A final example will be given to show how the various rules of inference can be used to verify the correctness of a longer program.

EXAMPLE 5 We will outline how to verify the correctness of the program S for computing the product of two integers.

```

procedure multiply( $m, n$ : integers)

   $S_1 \left\{ \begin{array}{l} \text{if } n < 0 \text{ then } a := -n \\ \text{else } a := n \end{array} \right.$ 

   $S_2 \left\{ \begin{array}{l} k := 0 \\ x := 0 \end{array} \right.$ 

   $S_3 \left\{ \begin{array}{l} \text{while } k < a \\ \quad x := x + m \\ \quad k := k + 1 \end{array} \right.$ 

   $S_4 \left\{ \begin{array}{l} \text{if } n < 0 \text{ then } product := -x \\ \text{else } product := x \end{array} \right.$ 

  return product
  {product equals  $mn$ }

```

The goal is to prove that after S is executed, $product$ has the value mn . The proof of correctness can be carried out by splitting S into four segments, with $S = S_1; S_2; S_3; S_4$, as shown in the listing of S . The rule of composition can be used to build the correctness proof. Here is how the argument proceeds. The details will be left as an exercise for the reader.

Let p be the initial assertion “ m and n are integers.” Then, it can be shown that $p\{S_1\}q$ is true, when q is the proposition $p \wedge (a = |n|)$. Next, let r be the proposition $q \wedge (k = 0) \wedge (x = 0)$. It is easily verified that $q\{S_2\}r$ is true. It can be shown that “ $x = mk$ and $k \leq a$ ” is an invariant for the loop in S_3 . Furthermore, it is easy to see that the loop terminates after a iterations, with $k = a$, so $x = ma$ at this point. Because r implies that $x = m \cdot 0$ and $0 \leq a$, the loop invariant is true before the loop is entered. Because the loop terminates with $k = a$, it follows that $r\{S_3\}s$ is true, where s is the proposition “ $x = ma$ and $a = |n|$.” Finally, it can be shown that S_4 is correct with respect to the initial assertion s and final assertion t , where t is the proposition “ $product = mn$.”

Putting all this together, because $p\{S_1\}q$, $q\{S_2\}r$, $r\{S_3\}s$, and $s\{S_4\}t$ are all true, it follows from the rule of composition that $p\{S\}t$ is true. Furthermore, because all four segments terminate, S does terminate. This verifies the correctness of the program. ◀

Exercises

1. Prove that the program segment

```
y := 1
z := x + y
```

is correct with respect to the initial assertion $x = 0$ and the final assertion $z = 1$.

2. Verify that the program segment

```
if x < 0 then x := 0
```

is correct with respect to the initial assertion **T** and the final assertion $x \geq 0$.

3. Verify that the program segment

```
x := 2
z := x + y
if y > 0 then
  z := z + 1
else
  z := 0
```

is correct with respect to the initial assertion $y = 3$ and the final assertion $z = 6$.

4. Verify that the program segment

```
if x < y then
  min := x
else
  min := y
```

is correct with respect to the initial assertion **T** and the final assertion $(x \leq y \wedge \text{min} = x) \vee (x > y \wedge \text{min} = y)$.

- *5. Devise a rule of inference for verification of partial correctness of statements of the form

```
if condition 1 then
  S1
else if condition 2 then
  S2
  ⋮
else
  Sn
```

where $S_1, S_2, \dots, S_n$ are blocks.

6. Use the rule of inference developed in Exercise 5 to verify that the program

```
if x < 0 then
  y := -2|x|/x
else if x > 0 then
  y := 2|x|/x
else if x = 0 then
  y := 2
```

is correct with respect to the initial assertion **T** and the final assertion $y = 2$.

7. Use a loop invariant to prove that the following program segment for computing the n th power, where n is a positive integer, of a real number x is correct.

```
power := 1
i := 1
while i ≤ n
  power := power * x
  i := i + 1
```

- *8. Prove that the iterative program for finding f_n given in Section 5.4 is correct.

9. Provide all the details in the proof of correctness given in Example 5.

10. Suppose that both the conditional statement $p_0 \rightarrow p_1$ and the program assertion $p_1\{S\}q$ are true. Show that $p_0\{S\}q$ also must be true.

11. Suppose that both the program assertion $p\{S\}q_0$ and the conditional statement $q_0 \rightarrow q_1$ are true. Show that $p\{S\}q_1$ also must be true.

12. This program computes quotients and remainders.

```
r := a
q := 0
while r ≥ d
  r := r - d
  q := q + 1
```

Verify that it is partially correct with respect to the initial assertion " a and d are positive integers" and the final assertion " q and r are integers such that $a = dq + r$ and $0 \leq r < d$."

13. Use a loop invariant to verify that the Euclidean algorithm (Algorithm 1 in Section 4.3) is partially correct with respect to the initial assertion " a and b are positive integers" and the final assertion " $x = \text{gcd}(a, b)$."

Key Terms and Results

TERMS

sequence: a function with domain that is a subset of the set of integers

geometric progression: a sequence of the form $a, ar, ar^2, \dots$, where a and r are real numbers

arithmetic progression: a sequence of the form $a, a + d, a + 2d, \dots$, where a and d are real numbers

the principle of mathematical induction: The statement $\forall n P(n)$ is true if $P(1)$ is true and $\forall k[P(k) \rightarrow P(k + 1)]$ is true.

basis step: the proof of $P(1)$ in a proof by mathematical induction of $\forall nP(n)$

inductive step: the proof of $P(k) \rightarrow P(k+1)$ for all positive integers k in a proof by mathematical induction of $\forall nP(n)$

strong induction: The statement $\forall nP(n)$ is true if $P(1)$ is true and $\forall k[(P(1) \wedge \cdots \wedge P(k)) \rightarrow P(k+1)]$ is true.

well-ordering property: Every nonempty set of nonnegative integers has a least element.

recursive definition of a function: a definition of a function that specifies an initial set of values and a rule for obtaining values of this function at integers from its values at smaller integers

recursive definition of a set: a definition of a set that specifies an initial set of elements in the set and a rule for obtaining other elements from those in the set

structural induction: a technique for proving results about recursively defined sets

recursive algorithm: an algorithm that proceeds by reducing a problem to the same problem with smaller input

merge sort: a sorting algorithm that sorts a list by splitting it in two, sorting each of the two resulting lists, and merging the results into a sorted list

iteration: a procedure based on the repeated use of operations in a loop

program correctness: verification that a procedure always produces the correct result

loop invariant: a property that remains true during every traversal of a loop

initial assertion: the statement specifying the properties of the input values of a program

final assertion: the statement specifying the properties the output values should have if the program worked correctly

Review Questions

1. a) Can you use the principle of mathematical induction to find a formula for the sum of the first n terms of a sequence?
b) Can you use the principle of mathematical induction to determine whether a given formula for the sum of the first n terms of a sequence is correct?
c) Find a formula for the sum of the first n even positive integers, and prove it using mathematical induction.
2. a) For which positive integers n is $11n + 17 \leq 2^n$?
b) Prove the conjecture you made in part (a) using mathematical induction.
3. a) Which amounts of postage can be formed using only 5-cent and 9-cent stamps?
b) Prove the conjecture you made using mathematical induction.
c) Prove the conjecture you made using strong induction.
d) Find a proof of your conjecture different from the ones you gave in (b) and (c).
4. Give two different examples of proofs that use strong induction.
5. a) State the well-ordering property for the set of positive integers.
b) Use this property to show that every positive integer greater than one can be written as the product of primes.
6. a) Explain why a function f from the set of positive integers to the set of real numbers is well defined if it is defined recursively by specifying $f(1)$ and a rule for finding $f(n)$ from $f(n-1)$.
b) Provide a recursive definition of the function $f(n) = (n+1)!$.
7. a) Give a recursive definition of the Fibonacci numbers.
b) Show that $f_n > \alpha^{n-2}$ whenever $n \geq 3$, where f_n is the n th term of the Fibonacci sequence and $\alpha = (1 + \sqrt{5})/2$.
8. a) Explain why a sequence a_n is well defined if it is defined recursively by specifying a_1 and a_2 and a rule for finding a_n from $a_1, a_2, \dots, a_{n-1}$ for $n = 3, 4, 5, \dots$.
b) Find the value of a_n if $a_1 = 1, a_2 = 2$, and $a_n = a_{n-1} + a_{n-2} + \cdots + a_1$, for $n = 3, 4, 5, \dots$.
9. Give two examples of how well-formed formulae are defined recursively for different sets of elements and operators.
10. a) Give a recursive definition of the length of a string.
b) Use the recursive definition from part (a) and structural induction to prove that $l(xy) = l(x) + l(y)$.
11. a) What is a recursive algorithm?
b) Describe a recursive algorithm for computing the sum of n numbers in a sequence.
12. Describe a recursive algorithm for computing the greatest common divisor of two positive integers.
13. a) Describe the merge sort algorithm.
b) Use the merge sort algorithm to put the list 4, 10, 1, 5, 3, 8, 7, 2, 6, 9 in increasing order.
c) Give a big- O estimate for the number of comparisons used by the merge sort.
14. a) Does testing a computer program to see whether it produces the correct output for certain input values verify that the program always produces the correct output?
b) Does showing that a computer program is partially correct with respect to an initial assertion and a final assertion verify that the program always produces the correct output? If not, what else is needed?
15. What techniques can you use to show that a long computer program is partially correct with respect to an initial assertion and a final assertion?
16. What is a loop invariant? How is a loop invariant used?

Supplementary Exercises

- Use mathematical induction to show that $\frac{2}{3} + \frac{2}{9} + \frac{2}{27} + \dots + \frac{2}{3^n} = 1 - \frac{1}{3^n}$ whenever n is a positive integer.
- Use mathematical induction to show that $1^3 + 3^3 + 5^3 + \dots + (2n+1)^3 = (n+1)^2(2n^2 + 4n + 1)$ whenever n is a positive integer.
- Use mathematical induction to show that $1 \cdot 2^0 + 2 \cdot 2^1 + 3 \cdot 2^2 + \dots + n \cdot 2^{n-1} = (n-1) \cdot 2^n + 1$ whenever n is a positive integer.
- Use mathematical induction to show that
$$\frac{1}{1 \cdot 3} + \frac{1}{3 \cdot 5} + \dots + \frac{1}{(2n-1)(2n+1)} = \frac{n}{2n+1}$$
 whenever n is a positive integer.
- Show that
$$\frac{1}{1 \cdot 4} + \frac{1}{4 \cdot 7} + \dots + \frac{1}{(3n-2)(3n+1)} = \frac{n}{3n+1}$$
 whenever n is a positive integer.
- Use mathematical induction to show that $2^n > n^2 + n$ whenever n is an integer greater than 4.
- Use mathematical induction to show that $2^n > n^3$ whenever n is an integer greater than 9.
- Find an integer N such that $2^n > n^4$ whenever n is an integer greater than N . Prove that your result is correct using mathematical induction.
- Use mathematical induction to prove that $a - b$ is a factor of $a^n - b^n$ whenever n is a positive integer.
- Use mathematical induction to prove that 9 divides $n^3 + (n+1)^3 + (n+2)^3$ whenever n is a nonnegative integer.
- Use mathematical induction to prove that 43 divides $6^{n+1} + 7^{2n-1}$ for every positive integer n .
- Use mathematical induction to prove that 64 divides $3^{2n+2} + 56n + 55$ for every positive integer n .
- Use mathematical induction to prove this formula for the sum of the terms of an arithmetic progression.
$$a + (a+d) + \dots + (a+nd) = (n+1)(2a+nd)/2$$
- Suppose that $a_j \equiv b_j \pmod{m}$ for $j = 1, 2, \dots, n$. Use mathematical induction to prove that
 - $\sum_{j=1}^n a_j \equiv \sum_{j=1}^n b_j \pmod{m}$.
 - $\prod_{j=1}^n a_j \equiv \prod_{j=1}^n b_j \pmod{m}$.
- Show that if n is a positive integer, then
$$\sum_{k=1}^n \frac{k+4}{k(k+1)(k+2)} = \frac{n(3n+7)}{2(n+1)(n+2)}.$$
- For which positive integers n is $n+6 < (n^2-8n)/16$? Prove your answer using mathematical induction.
- (Requires calculus) Suppose that $f(x) = e^x$ and $g(x) = xe^x$. Use mathematical induction together with the product rule and the fact that $f'(x) = e^x$ to prove that $g^{(n)}(x) = (x+n)e^x$ whenever n is a positive integer.
- (Requires calculus) Suppose that $f(x) = e^x$ and $g(x) = e^{cx}$, where c is a constant. Use mathematical induction together with the chain rule and the fact that $f'(x) = e^x$ to prove that $g^{(n)} = c^n e^{cx}$ whenever n is a positive integer.
- * Formulate a conjecture about which Fibonacci numbers are even, and use a form of mathematical induction to prove your conjecture.
- * Determine which Fibonacci numbers are divisible by 3. Use a form of mathematical induction to prove your conjecture.
- * Prove that $f_k f_n + f_{k+1} f_{n+1} = f_{n+k+1}$ for all nonnegative integers n and k , where f_i denotes the i th Fibonacci number. Recall from Example 15 of Section 2.4 that the sequence of **Lucas numbers** is defined by $l_0 = 2$, $l_1 = 1$, and $l_n = l_{n-1} + l_{n-2}$ for $n = 2, 3, 4, \dots$.
- Show that $f_n + f_{n+2} = l_{n+1}$ whenever n is a positive integer, where f_i and l_i are the i th Fibonacci number and i th Lucas number, respectively.
- Show that $l_0^2 + l_1^2 + \dots + l_n^2 = l_n l_{n+1} + 2$ whenever n is a nonnegative integer and l_i is the i th Lucas number.
- * Use mathematical induction to show that the product of any n consecutive positive integers is divisible by $n!$. [Hint: Use the identity $m(m+1) \dots (m+n-1)/n! = (m-1)m(m+1) \dots (m+n-2)/n! + m(m+1) \dots (m+n-2)/(n-1)!]$
- Use mathematical induction to show that $(\cos x + i \sin x)^n = \cos nx + i \sin nx$ whenever n is a positive integer. (Here i is the square root of -1 .) [Hint: Use the identities $\cos(a+b) = \cos a \cos b - \sin a \sin b$ and $\sin(a+b) = \sin a \cos b + \cos a \sin b$.]
- * Use mathematical induction to show that $\sum_{j=1}^n \cos jx = \cos[(n+1)x/2] \sin(nx/2) / \sin(x/2)$ whenever n is a positive integer and $\sin(x/2) \neq 0$.
- Use mathematical induction to prove that $\sum_{j=1}^n j^2 2^j = n^2 2^{n+1} - n 2^{n+2} + 3 \cdot 2^{n+1} - 6$ for every positive integer n .
- (Requires calculus) Suppose that the sequence $x_1, x_2, \dots, x_n, \dots$ is recursively defined by $x_1 = 0$ and $x_{n+1} = \sqrt{x_n + 6}$.
 - Use mathematical induction to show that $x_1 < x_2 < \dots < x_n < \dots$, that is, the sequence $\{x_n\}$ is monotonically increasing.
 - Use mathematical induction to prove that $x_n < 3$ for $n = 1, 2, \dots$.
 - Show that $\lim_{n \rightarrow \infty} x_n = 3$.
- Show if n is a positive integer with $n \geq 2$, then
$$\sum_{j=2}^n \frac{1}{j^2 - 1} = \frac{(n-1)(3n+2)}{4n(n+1)}.$$

30. Use mathematical induction to prove Theorem 1 in Section 4.2, that is, show if b is an integer, where $b > 1$, and n is a positive integer, then n can be expressed uniquely in the form $n = a_k b^k + a_{k-1} b^{k-1} + \cdots + a_1 b + a_0$.
- *31. A **lattice point** in the plane is a point (x, y) where both x and y are integers. Use mathematical induction to show that at least $n + 1$ straight lines are needed to ensure that every lattice point (x, y) with $x \geq 0$, $y \geq 0$, and $x + y \leq n$ lies on one of these lines.
32. (Requires calculus) Use mathematical induction and the product rule to show that if n is a positive integer and $f_1(x), f_2(x), \dots, f_n(x)$, are all differentiable functions, then
- $$\frac{(f_1(x)f_2(x) \cdots f_n(x))'}{f_1(x)f_2(x) \cdots f_n(x)} = \frac{f_1'(x)}{f_1(x)} + \frac{f_2'(x)}{f_2(x)} + \cdots + \frac{f_n'(x)}{f_n(x)}.$$
33. (Requires material in Section 2.6) Suppose that $\mathbf{B} = \mathbf{M}\mathbf{A}\mathbf{M}^{-1}$, where $\mathbf{A}$ and $\mathbf{B}$ are $n \times n$ matrices and $\mathbf{M}$ is invertible. Show that $\mathbf{B}^k = \mathbf{M}\mathbf{A}^k\mathbf{M}^{-1}$ for all positive integers k . (Consult both the text of Section 2.6 and the preamble to Exercise 18 of Section 2.6.)
34. Use mathematical induction to show that if you draw lines in the plane you only need two colors to color the regions formed so that no two regions that have an edge in common have a common color.
35. Show that $n!$ can be represented as the sum of n of its distinct positive divisors whenever $n \geq 3$. [Hint: Use inductive loading. First try to prove this result using mathematical induction. By examining where your proof fails, find a stronger statement that you can easily prove using mathematical induction.]
- *36. Use mathematical induction to prove that if $x_1, x_2, \dots, x_n$ are positive real numbers with $n \geq 2$, then
- $$\left(x_1 + \frac{1}{x_1}\right)\left(x_2 + \frac{1}{x_2}\right) \cdots \left(x_n + \frac{1}{x_n}\right) \geq \left(x_1 + \frac{1}{x_2}\right)\left(x_2 + \frac{1}{x_3}\right) \cdots \left(x_{n-1} + \frac{1}{x_n}\right)\left(x_n + \frac{1}{x_1}\right)$$
37. Use mathematical induction to prove that if n people stand in a line, where n is a positive integer, and if the first person in the line is a woman and the last person in line is a man, then somewhere in the line there is a woman directly in front of a man.
- *38. Suppose that for every pair of cities in a country there is a direct one-way road connecting them in one direction or the other. Use mathematical induction to show that there is a city that can be reached from every other city either directly or via exactly one other city.
39. Use mathematical induction to show that when n circles divide the plane into regions, these regions can be colored with two different colors such that no regions with a common boundary are colored the same.
- *40. Suppose that among a group of cars on a circular track there is enough fuel for one car to complete a lap. Use mathematical induction to show that there is a car in the

group that can complete a lap by obtaining gas from other cars as it travels around the track.

41. Show that if n is a positive integer, then

$$\sum_{j=1}^n (2j-1) \left(\sum_{k=j}^n 1/k \right) = n(n+1)/2.$$

42. Use mathematical induction to show that if a , b , and c are the lengths of the sides of a right triangle, where c is the length of the hypotenuse, then $a^n + b^n < c^n$ for all integers n with $n \geq 3$.
- *43. Use mathematical induction to show that if n is a positive integer, the sequence $2 \bmod n, 2^2 \bmod n, 2^{2^2} \bmod n, \dots$ is eventually constant (that is, all terms after a finite number of terms are all the same).
44. A **unit** or **Egyptian fraction** is a fraction of the form $1/n$, where n is a positive integer. In this exercise, we will use strong induction to show that a greedy algorithm can be used to express every rational number p/q with $0 < p/q < 1$ as the sum of distinct unit fractions. At each step of the algorithm, we find the smallest positive integer n such that $1/n$ can be added to the sum without exceeding p/q . For example, to express $5/7$ we first start the sum with $1/2$. Because $5/7 - 1/2 = 3/14$ we add $1/5$ to the sum because 5 is the smallest positive integer k such that $1/k < 3/14$. Because $3/14 - 1/5 = 1/70$, the algorithm terminates, showing that $5/7 = 1/2 + 1/5 + 1/70$. Let $T(p)$ be the statement that this algorithm terminates for all rational numbers p/q with $0 < p/q < 1$. We will prove that the algorithm always terminates by showing that $T(p)$ holds for all positive integers p .
- a) Show that the basis step $T(1)$ holds.
- b) Suppose that $T(k)$ holds for positive integers k with $k < p$. That is, assume that the algorithm terminates for all rational numbers k/r , where $1 \leq k < p$. Show that if we start with p/q and the fraction $1/n$ is selected in the first step of the algorithm, then $p/q = p'/q' + 1/n$, where $p' = np - q$ and $q' = nq$. After considering the case where $p/q = 1/n$, use the inductive hypothesis to show that the greedy algorithm terminates when it begins with p'/q' and complete the inductive step.

The **McCarthy 91 function** (defined by John McCarthy, one of the founders of artificial intelligence) is defined using the rule

$$M(n) = \begin{cases} n - 10 & \text{if } n > 100 \\ M(M(n + 11)) & \text{if } n \leq 100 \end{cases}$$

for all positive integers n .

45. By successively using the defining rule for $M(n)$, find
- a) $M(102)$. b) $M(101)$. c) $M(99)$.
d) $M(97)$. e) $M(87)$. f) $M(76)$.
- **46. Show that the function $M(n)$ is a well-defined function from the set of positive integers to the set of positive integers. [Hint: Prove that $M(n) = 91$ for all positive integers n with $n \leq 101$.]

47. Is this proof that

$$\frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \cdots + \frac{1}{(n-1)n} = \frac{3}{2} - \frac{1}{n},$$

whenever n is a positive integer, correct? Justify your answer.

Basis step: The result is true when $n = 1$ because

$$\frac{1}{1 \cdot 2} = \frac{3}{2} - \frac{1}{1}.$$

Inductive step: Assume that the result is true for n . Then

$$\begin{aligned} \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \cdots + \frac{1}{(n-1)n} + \frac{1}{n(n+1)} \\ = \frac{3}{2} - \frac{1}{n} + \left(\frac{1}{n} - \frac{1}{n+1} \right) \\ = \frac{3}{2} - \frac{1}{n+1}. \end{aligned}$$

Hence, the result is true for $n + 1$ if it is true for n . This completes the proof.

48. Suppose that $A_1, A_2, \dots, A_n$ are a collection of sets. Suppose that $R_2 = A_1 \oplus A_2$ and $R_k = R_{k-1} \oplus A_k$ for $k = 3, 4, \dots, n$. Use mathematical induction to prove that $x \in R_n$ if and only if x belongs to an odd number of the sets $A_1, A_2, \dots, A_n$. (Recall that $S \oplus T$ is the symmetric difference of the sets S and T defined in the preamble to Exercise 38 of Section 2.2.)

*49. Show that n circles divide the plane into $n^2 - n + 2$ regions if every two circles intersect in exactly two points and no three circles contain a common point.

*50. Show that n planes divide three-dimensional space into $(n^3 + 5n + 6)/6$ regions if any three of these planes have exactly one point in common and no four contain a common point.

*51. Use the well-ordering property to show that $\sqrt{2}$ is irrational. [Hint: Assume that $\sqrt{2}$ is rational. Show that the set of positive integers of the form $b\sqrt{2}$ has a least

element a . Then show that $a\sqrt{2} - a$ is a smaller positive integer of this form.]

52. A set is **well ordered** if every nonempty subset of this set has a least element. Determine whether each of the following sets is well ordered.

- a) the set of integers
- b) the set of integers greater than -100
- c) the set of positive rationals
- d) the set of positive rationals with denominator less than 100

53. a) Show that if $a_1, a_2, \dots, a_n$ are positive integers, then $\gcd(a_1, a_2, \dots, a_{n-1}, a_n) = \gcd(a_1, a_2, \dots, a_{n-2}, \gcd(a_{n-1}, a_n))$.

b) Use part (a), together with the Euclidean algorithm, to develop a recursive algorithm for computing the greatest common divisor of a set of n positive integers.

*54. Describe a recursive algorithm for writing the greatest common divisor of n positive integers as a linear combination of these integers.

55. Find an explicit formula for $f(n)$ if $f(1) = 1$ and $f(n) = f(n-1) + 2n - 1$ for $n \geq 2$. Prove your result using mathematical induction.

**56. Give a recursive definition of the set of bit strings that contain twice as many 0s as 1s.

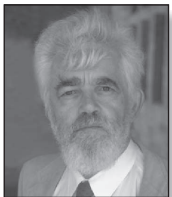
57. Let S be the set of bit strings defined recursively by $\lambda \in S$ and $0x \in S, x1 \in S$ if $x \in S$, where λ is the empty string.

- a) Find all strings in S of length not exceeding five.
- b) Give an explicit description of the elements of S .

58. Let S be the set of strings defined recursively by $abc \in S$, $bac \in S$, and $acb \in S$, where a, b , and c are fixed letters; and for all $x \in S$, $abxc \in S$, $axbc \in S$, and $xabc \in S$, where x is a variable representing a string of letters.

- a) Find all elements of S of length eight or less.
- b) Show that every element of S has a length divisible by three.

Links



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JOHN MCCARTHY (1927–2011) John McCarthy was born in Boston. He grew up in Boston and in Los Angeles. He studied mathematics as both an undergraduate and a graduate student, receiving his B.S. in 1948 from the California Institute of Technology and his Ph.D. in 1951 from Princeton. After graduating from Princeton, McCarthy held positions at Princeton, Stanford, Dartmouth, and M.I.T. He held a position at Stanford from 1962 until 1994, and was an emeritus professor there. At Stanford, he was the director of the Artificial Intelligence Laboratory, held a named chair in the School of Engineering, and was a senior fellow at the Hoover Institution.

McCarthy was a pioneer in the study of artificial intelligence, a term he coined in 1955. He worked on problems related to the reasoning and information needs required for intelligent computer behavior. McCarthy was among the first computer scientists to design time-sharing computer systems. He developed LISP, a programming language for computing using symbolic expressions. He played an important role in using logic to verify the correctness of computer programs. McCarthy has also worked on the social implications of computer technology. In his later years he worked on the problem of how people and computers make conjectures through assumptions that complications are absent from situations. McCarthy was an advocate of the sustainability of human progress and was optimistic about the future of humanity. In his later years he also wrote science fiction stories. Some of his last work explored the possibility that the world is a computer program written by some higher force.

Among the awards McCarthy won are the Turing Award from the Association for Computing Machinery, the Research Excellence Award of the International Conference on Artificial Intelligence, the Kyoto Prize, and the National Medal of Science.

The set B of all **balanced strings of parentheses** is defined recursively by $\lambda \in B$, where λ is the empty string; $(x) \in B$, $xy \in B$ if $x, y \in B$.

59. Show that $((()))$ is a balanced string of parentheses and $(()))$ is not a balanced string of parentheses.
60. Find all balanced strings of parentheses with exactly six symbols.
61. Find all balanced strings of parentheses with four or fewer symbols.
62. Use induction to show that if x is a balanced string of parentheses, then the number of left parentheses equals the number of right parentheses in x .

Define the function N on the set of strings of parentheses by

$$N(\lambda) = 0, N(()) = 1, N() = -1, \\ N(uv) = N(u) + N(v),$$

where λ is the empty string, and u and v are strings. It can be shown that N is well defined.

63. Find
 - a) $N(())$.
 - b) $N(()))((())$.
 - c) $N(((())((()))$.
 - d) $N()(((((())((()))$.
- **64. Show that a string w of parentheses is balanced if and only if $N(w) = 0$ and $N(u) \geq 0$ whenever u is a prefix of w , that is, $w = uv$.
- *65. Give a recursive algorithm for finding all balanced strings of parentheses containing n or fewer symbols.
66. Give a recursive algorithm for finding $\gcd(a, b)$, where a and b are nonnegative integers not both zero, based on these facts: $\gcd(a, b) = \gcd(b, a)$ if $a > b$, $\gcd(0, b) = b$, $\gcd(a, b) = 2 \gcd(a/2, b/2)$ if a and b are even, $\gcd(a, b) = \gcd(a/2, b)$ if a is even and b is odd, and $\gcd(a, b) = \gcd(a, b - a)$.
67. Verify the program segment


```
if  $x > y$  then
   $x := y$ 
```

 with respect to the initial assertion $\mathbf{T}$ and the final assertion $x \leq y$.
- *68. Develop a rule of inference for verifying recursive programs and use it to verify the recursive algorithm for computing factorials given as Algorithm 1 in Section 5.4.

69. Devise a recursive algorithm that counts the number of times the integer 0 occurs in a list of integers.

Exercises 70–77 deal with some unusual sequences, informally called **self-generating sequences**, produced by simple recurrence relations or rules. In particular, Exercises 70–75 deal with the sequence $\{a(n)\}$ defined by $a(n) = n - a(a(n-1))$ for $n \geq 1$ and $a(0) = 0$. (This sequence, as well as those in Exercises 74 and 75, are defined in Douglas Hofstadter's fascinating book *Gödel, Escher, Bach* ([Ho99]).

70. Find the first 10 terms of the sequence $\{a(n)\}$ defined in the preamble to this exercise.
 - *71. Prove that this sequence is well defined. That is, show that $a(n)$ is uniquely defined for all nonnegative integers n .
 - **72. Prove that $a(n) = \lfloor (n+1)\mu \rfloor$ where $\mu = (-1 + \sqrt{5})/2$. [Hint: First show for all $n > 0$ that $(\mu n - \lfloor \mu n \rfloor) + (\mu^2 n - \lfloor \mu^2 n \rfloor) = 1$. Then show for all real numbers α with $0 \leq \alpha < 1$ and $\alpha \neq 1 - \mu$ that $\lfloor (1 + \mu)(1 - \alpha) \rfloor + \lfloor \alpha + \mu \rfloor = 1$, considering the cases $0 \leq \alpha < 1 - \mu$ and $1 - \mu < \alpha < 1$ separately.]
 - *73. Use the formula from Exercise 72 to show that $a(n) = a(n-1)$ if $\mu n - \lfloor \mu n \rfloor < 1 - \mu$ and $a(n) = a(n-1) + 1$ otherwise.
 74. Find the first 10 terms of each of the following self-generating sequences:
 - a) $a(n) = n - a(a(a(n-1)))$ for $n \geq 1$, $a(0) = 0$
 - b) $a(n) = n - a(a(a(a(n-1))))$ for $n \geq 1$, $a(0) = 0$
 - c) $a(n) = a(n - a(n-1)) + a(n - a(n-2))$ for $n \geq 3$, $a(1) = 1$ and $a(2) = 1$
 75. Find the first 10 terms of both the sequences $m(n)$ and $f(n)$ defined by the following pair of interwoven recurrence relations: $m(n) = n - f(m(n-1))$, $f(n) = n - m(f(n-1))$ for $n \geq 1$, $f(0) = 1$ and $m(0) = 0$.
- Golomb's self-generating sequence** is the unique nondecreasing sequence of positive integers $a_1, a_2, a_3, \dots$ that has the property that it contains exactly a_k occurrences of k for each positive integer k .
76. Find the first 20 terms of Golomb's self-generating sequence.
 - *77. Show that if $f(n)$ is the largest integer m such that $a_m = n$, where a_m is the m th term of Golomb's self-generating sequence, then $f(n) = \sum_{k=1}^n a_k$ and $f(f(n)) = \sum_{k=1}^n ka_k$.

Computer Projects

Write programs with these input and output.

- **1. Given a $2^n \times 2^n$ checkerboard with one square missing, construct a tiling of this checkerboard using right triominoes.
- **2. Generate all well-formed formulae for expressions involving the variables x, y , and z and the operators $\{+, *, /, -\}$ with n or fewer symbols.
- **3. Generate all well-formed formulae for propositions with n or fewer symbols where each symbol is $\mathbf{T}, \mathbf{F}$, one of

the propositional variables p and q , or an operator from $\{\neg, \vee, \wedge, \rightarrow, \leftrightarrow\}$.

4. Given a string, find its reversal.
5. Given a real number a and a nonnegative integer n , find a^n using recursion.
6. Given a real number a and a nonnegative integer n , find a^{2^n} using recursion.

- *7. Given a real number a and a nonnegative integer n , find a^n using the binary expansion of n and a recursive algorithm for computing a^{2^k} .
- 8. Given two integers not both zero, find their greatest common divisor using recursion.
- 9. Given a list of integers and an element x , locate x in this list using a recursive implementation of a linear search.
- 10. Given a list of integers and an element x , locate x in this list using a recursive implementation of a binary search.
- 11. Given a nonnegative integer n , find the n th Fibonacci number using iteration.
- 12. Given a nonnegative integer n , find the n th Fibonacci number using recursion.
- 13. Given a positive integer, find the number of partitions of this integer. (See Exercise 49 of Section 5.3.)
- 14. Given positive integers m and n , find $A(m, n)$, the value of Ackermann's function at the pair (m, n) . (See the preamble to Exercise 50 of Section 5.3.)
- 15. Given a list of integers, sort these integers using the merge sort.

Computations and Explorations

Use a computational program or programs you have written to do these exercises.

- 1. What are the largest values of n for which $n!$ has fewer than 100 decimal digits and fewer than 1000 decimal digits?
- 2. Determine which Fibonacci numbers are divisible by 5, which are divisible by 7, and which are divisible by 11. Prove that your conjectures are correct.
- 3. Construct tilings using right triominoes of various 16×16 , 32×32 , and 64×64 checkerboards with one square missing.
- 4. Explore which $m \times n$ checkerboards can be completely covered by right triominoes. Can you make a conjecture that answers this question?
- **5. Implement an algorithm for determining whether a point is in the interior or exterior of a simple polygon.
- **6. Implement an algorithm for triangulating a simple polygon.
- 7. Which values of Ackermann's function are small enough that you are able to compute them?
- 8. Compare either the number of operations or the time needed to compute Fibonacci numbers recursively versus that needed to compute them iteratively.

Writing Projects

Respond to these with essays using outside sources.

- 1. Describe the origins of mathematical induction. Who were the first people to use it and to which problems did they apply it?
- 2. Explain how to prove the Jordan curve theorem for simple polygons and describe an algorithm for determining whether a point is in the interior or exterior of a simple polygon.
- 3. Describe how the triangulation of simple polygons is used in some key algorithms in computational geometry.
- 4. Describe a variety of different applications of the Fibonacci numbers to the biological and the physical sciences.
- 5. Discuss the uses of Ackermann's function both in the theory of recursive definitions and in the analysis of the complexity of algorithms for set unions.
- 6. Give the recursive definition of Knuth's up-arrow notation and illustrate its use with a variety of examples, including how it can be used to express values of the Ackermann function (defined in the preamble of Exercise 50 in Section 5.3).
- 7. Discuss some of the various methodologies used to establish the correctness of programs and compare them to Hoare's methods described in Section 5.5.
- 8. Explain how the ideas and concepts of program correctness can be extended to prove that operating systems are secure.